A Perturbative Approach to the Ground State of Interacting Fermions in the Mean Field Limit

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Contents

Chapter 1

Introduction

The rigorous analysis of many body systems poses notoriously hard challenges, even just tackling the conceptually simple case of a gas with pair-wise interactions proves to be mathematically extremely challenging. This difficulty forces us to develop more and more sophisticated tools to overcome the obstacles that appear along the way: an example of a tool that has proved itself invaluable in physics, and that allows for a close connection between theory and experiment, is perturbation theory. It allows for the analysis of complicated interactions, dominant contributions to specific processes, coarse properties of a system starting from more elementary interactions, and more. In the present work, we take a perturbative approach to analyze the behavior of the ground state energy of the interacting Fermi gas under specific assumptions, which we shall motivate here.

For a system of many interacting particles it can become a nightmare to discuss the effect of every interaction a single particle may have with the rest of the system, and it is therefore a good idea to first try looking for mechanisms that either make the system easier to study or the mathematics better behaved. An example that is very easy to picture is the screening of the Coulomb interaction between the nucleus of an atom and its orbiting electrons, be it in single atoms, molecules, crystal lattices, and so on. It is not at all a trivial endeavor trying to compute exactly even just the ground state energy of the atom, but one can make some useful observations before attempting to do so. If we consider electrons in the innermost shells we can expect their binding energy to be mainly due to the strong electrostatic interaction with the protons inside the nucleus. But if we move to higher and higher energy levels, the electrons, while still being attracted by the nucleus, will also be repelled by the presence of more and more lower-energy electrons, which effectively *screen* the total charge of the nucleus [7, p. 51]. Thus, electrons in higher shells will not feel the full long-range effect of the Coulomb interaction, but rather an effective interaction that is suppressed depending on their energy level, assuming lower energy levels are occupied.

The problem then becomes how this screened interaction behaves. This is where *mean field methods* come into play. To find the ground state energy without directly accounting for the very complicated correlations between each particle, one starts by assuming that electrons are found in independent single-electron atomic orbitals. Then, computing the overall effect of such a configuration on the energy of one of the electrons, one can look for a better configuration that minimizes this effect, and iterating this procedure leads to the Hartree-Fock approximation for electronic configurations [7, pp. 48–50]. Hartree-Fock theory can be formulated more generally and applied to other settings, furthermore it essentially corresponds to the first order contribution to the ground state energy in the case of homogeneous systems [5, p. 81], like a gas of uniform density.

This screening phenomenon is, of course, not limited to electrons in atomic orbitals. In any system with many interacting particles, one may ask whether the state of each particle can be described not by the true, *bare*, interaction that characterizes the system, but rather by a rescaled, *dressed*, interaction. If this is the case, it may also happen that ill-behaved quantities become well-behaved. As an example, in the case of an electron gas it is precisely the long range of the

Coulomb interaction that makes perturbation theory badly divergent in the IR [5, p. 38–39]. When computing the ground state energy, the Random Phase Approximation (RPA) comes as a way of carrying out such a computation once the screening of the interaction due to the high number of particles is accounted for; in fact, today we also understand the RPA as a partial resummation of perturbation theory at all orders. This resummation, coincidentally, also removes the divergence at low momentum of the Coulomb interaction [5, p. 293]).
Mean field theories are an example of *effective theories*, where the interaction potential that governs the dynamics of the system is not taken to be the bare potential, but a dressed potential whose strength depends, in some way or another, on the number of particles that partake in the interactions.

Our study of the Fermi gas in a mean field regime is structured as follows. We will cover the bread and butter of a many-body theorist: second quantization, a formalism that allows to adapt powerful concepts from statistical mechanics to the quantum world. By then we will have the tools necessary to understand why perturbation theory can be formulated using the clever diagrammatic approach first employed by Richard Feynman, which we introduce using complex Gaussian integrals and Grassmann Gaussian integrals. This approach is to be understood as a way of writing a formal perturbative series that is, in some way, identical to the one written in the operator formalism. Lastly we present the explicit computation at second order for the free energy F of the gas at arbitrary (inverse) temperature β, showing how to control the values of the diagrams that contribute to this order. The result is an expression that depends on the number of particles in the gas, and is to be tested against known exact results. This will lead to the main result of this thesis: the known scaling of this regime is rigorously proved at first order in perturbation theory. The second order is computed and discussed in some detail, but presents difficulties in controlling divergent quantities that require closer inspection. In particular, we isolate the problematic terms and explain where they come from.
Our role with this original work is to bring rigorous perturbation theory into the picture, and our hope is that, by laying the groundwork with the present results, higher orders can be treated systematically, ultimately moving us closer to a more complete and rigorous understanding of interacting fermions. For now, the challenges that arose in our calculations uncovered subtleties that our future efforts will need to address.

Chapter 2

Second Quantization

In this preliminary section we give an overview of the mathematical objects that appear in the description of quantum systems. The probabilistic nature of a quantum system, together with the superposition principle, are realized in the linear structure of a complex vector space of states. Such a vector space must be equipped with an inner product, which produces the probability amplitude for a system prepared in some state to be measured in another. Let us define these objects and give the results necessary to build a theory of interacting quantum particles. The majority of the definitions and theorems presented in this chapter are taken, and often reworked, from professor Solovej's lecture notes on many body quantum theory [8]. We will specify the page number for the proofs that are taken directly from those notes, while the proofs that do not have a page number are those that are not found there and that we came up with as useful exercises.

2.1 Principles of quantum mechanics

Definition 2.1.1 (Hilbert space)**.** A complex *Hilbert space* $\mathcal{H}$ is a vector space equipped with a sesquilinear map $\langle\cdot,\cdot\rangle : \mathcal{H}\times\mathcal{H}\to\mathbb{C}$ such that $\|\phi\| := \langle\phi,\phi\rangle^{1/2}$ defines a norm which makes $\mathcal{H}$ a complete metric space.

A note on the choice of notation. Sometimes, using Dirac's "bra-ket" notation can help avoid ambiguities. In this notation, a vector would be denoted by $|\psi\rangle \in \mathcal{H}$, a linear functional as $\langle\psi| \in \mathcal{H}^*$, and the inner product as $\langle\psi,\phi\rangle = \langle\psi|\phi\rangle$. An example of such a scenario where this notation might be preferred is that of the occupation number basis, which we discuss in this chapter and then make use of in Chapter 4.1.

Definition 2.1.2. A Hilbert space is said to be *separable* if it admits a countable dense subset.

The relevance of this definition stems from the fact that, if we can find a countable set $\mathcal{S} = \{u_n\}_{n\in\mathbb{N}}$ of vectors in the Hilbert space $\mathcal{H}$ such that the space of *finite* linear combinations of elements from $\mathcal{S}$ is dense, we have that any vector in $\mathcal{H}$ can be expressed as a limit of such finite linear combinations. That is, the set $\mathcal{S}$ is *complete* in the sense that any vector can be though of as a sequence of components $(a_n)_{n\in\mathbb{N}}$, with the condition that the series $\sum_{n\in\mathbb{N}} a_n u_n$ converges to said vector (in the norm topology of $\mathcal{H}$). When speaking of Hilbert spaces we will always assume that they are separable.

Lemma 2.1.3. *If X is a dense subspace of the Hilbert space $\mathcal{H}$, it is always possible to choose a complete orthonormal system (or orthonormal basis) $\{u_n\}_{n\in\mathbb{N}}$ for $\mathcal{H}$ composed only of elements in X.*

In quantum mechanics, observables are represented by a special type of linear map on the state space, so let us briefly recall the (purely formal) motivation behind this. If A is a physical observable, the result of an experiment is the collapse of the system onto a state associated to a well-defined value of A. A generic state ψ will be a superposition of such vectors, which we may call $\{\psi_i\}_{i\in\mathbb{N}}$, and let us call $\{\lambda_i\}_{i\in\mathbb{N}} \subset \mathbb{R}$ the well-defined values of A associated to such vectors. The generic state ψ then encodes the probability amplitudes $\langle\psi_i, \psi_i\rangle$ for the system to collapse in each ψ_i, which suggests that the statistical expectation for the measurement of A would read something like

$$\langle A\rangle_\psi = \sum_{i\in\mathbb{N}} \lambda_i \langle\psi_i, \psi_i\rangle = \langle\psi, A\psi\rangle$$

if we could represent the observable by a linear map $A : \mathcal{H} \to \mathcal{H}$. Let us make this precise and give meaning to what we just wrote.

Definition 2.1.4 (Operator)**.** A *linear operator* (or simply an *operator*) A on $\mathcal{H}$ is a linear map $A : D(A) \to \mathcal{H}$ defined on a subspace $D(A) \subseteq \mathcal{H}$. If $D(A)$ is dense in $\mathcal{H}$, A is said to be *densely defined*.

We will often write "operator" to mean a densely defined operator on a Hilbert space, unless stated otherwise. Moreover, note that the domain on which an operator acts is part of the definition, and will therefore need to be specified case by case.

Definition 2.1.5 (Extension of an operator)**.** If A and B are two operators such that $D(A) \subseteq D(B)$ and $A\phi = B\phi$ for all $\phi \in D(A)$, then we say that B *extends* (or is an extension of) A, and we write $A \subseteq B$.

Definition 2.1.6 (Bounded operator)**.** A densely defined operator A is said to be *bounded* if it is continuous with respect to the norm topology, and *unbounded* if it is not bounded.

Remark 1. By linearity, continuity is equivalent to the condition that

$$\|A\| := \sup_{\substack{\psi\in D(A)\\ \|\psi\|=1}} \|A\psi\| < \infty,$$

and the quantity $\|A\|$ is called the *operator norm* of A.

Proposition 2.1.7. *Given $A : D(A) \to \mathcal{H}$ a densely defined bounded operator on $\mathcal{H}$, there exists a unique extension to a bounded operator on the entire space $\tilde{A} : \mathcal{H} \to \mathcal{H}$.*

Proof. Let $(x_n)_{n\in\mathbb{N}} \subseteq D(A)$ be a Cauchy sequence in $D(A)$, then there exists a unique $x \in \mathcal{H}$ such that $x_n \to x$. Because A is bounded it holds that $\|Ax_n - Ax_m\| \leq \|A\|\|x_n - x_m\|$ and therefore the sequence $(Ax_n)_{n\in\mathbb{N}}$ is also a Cauchy sequence. As such, there exists a unique $y \in \mathcal{H}$ such that $Ax_n \to y$. Define the operator $\tilde{A}$ as

$$\begin{aligned} \tilde{A} : \mathcal{H} &\longrightarrow \mathcal{H} \\ x &\longmapsto \tilde{A}x := \lim_n Ax_n \end{aligned}$$

where $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence that converges to x. $\tilde{A}$ extends A and is defined on the entire space. Moreover, it is bounded as $\|A\| < \infty$ and

$$\|\tilde{A}x\| = \|\lim_n Ax_n\| = \lim_n \|Ax_n\| \leq \lim_n \|A\|\|x_n\| = \|A\|\|x\|.$$

□

It is worth pointing out that from Proposition 2.1.7 follows that for bounded operators it is possible to deal with dense subspaces without losing generality. In particular, since we are always assuming the Hilbert space to be separable, it will suffice to define the image of only the basis elements.

Definition 2.1.8 (Adjoint)**.** If A is an operator on $\mathcal{H}$ we define the *adjoint* A^* of A as the operator $A^* : D(A^*) \to \mathcal{H}$ defined on

$$D(A^*) = \{\phi \in \mathcal{H} : \sup_{\substack{\psi \in D(A) \\ \|\psi\|=1}} |\langle \phi, A\psi \rangle| < \infty\}$$

and with $A^*\phi$ defined for all $\phi \in D(A^*)$ as

$$\langle A^*\phi, \psi \rangle = \langle \phi, A\psi \rangle \quad \forall \psi \in D(A).$$

If $A^* = A$ then A is said to be *self-adjoint*.

Definition 2.1.9 (Observable)**.** A *physical observable* (or simply *observable*) is a self-adjoint operator A on the Hilbert space $\mathcal{H}$ of states. Given a normalized vector $\psi \in \mathcal{H}$ we define the *expectation of A on ψ* as

$$\langle A \rangle_\psi \coloneqq \langle \psi, A\psi \rangle.$$

Remark 2. From the self-adjointness of A we have that $\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle$ for all $\phi, \psi \in \mathcal{H}$, which implies that $\langle \psi, A\psi \rangle \in \mathbb{R}$ for all $\psi \in \mathcal{H}$, i.e., expectation values of A must be real numbers.

Proposition 2.1.10. *The adjoint of a bounded operator is a bounded operator.*

Proof. We are only considering densely defined operators, therefore, without loss of generality, let $A : \mathcal{H} \to \mathcal{H}$ be a bounded operator on the whole space $\mathcal{H}$. For all $\phi \in D(A^*)$, the map defined by $\varphi : \psi \mapsto \langle A^*\phi, \psi \rangle$ is a linear functional on $\mathcal{H}$, where

$$D(A^*) = \left\{ \phi \in \mathcal{H} : \sup_{\|\psi\|=1} \langle \phi, A\psi \rangle < \infty \right\}.$$

Then, by Schwarz's inequality, we have that for every $\phi \in D(A^*)$

$$\|A^*\phi\| = \sup_{\|\psi\|=1} |\langle A^*\phi, \psi \rangle| = \sup_{\|\psi\|=1} |\langle \phi, A\psi \rangle| \leq \sup_{\|\psi\|=1} \|A\psi\|\|\phi\|,$$

where the first step makes use of Theorem A.1.9, which implies that $\|A^*\phi\|_{\mathcal{H}} = \|\varphi\|_{\mathcal{H}^*}$. It follows from the boundedness of A that

$$\|A^*\phi\| \leq \|\phi\|\|A\| < \infty$$

which implies that A^* is bounded. □

Remark 3. We can introduce a partial ordering on self-adjoint operators. If A and B are self-adjoint and defined on a common domain $D \subseteq \mathcal{H}$, then we say that $A < B$ if $\langle \phi, A\phi \rangle < \langle \phi, B\phi \rangle$ for all $\phi \in D$.

We point to Appendix A.1 for a discussion on the most important definitions and properties of special types of operators. A key role in the discussion of quantum systems is played by the space of square integrable functions on $\mathbb{R}^n$ modulo the equivalence relation $\sim$ that identifies functions that are equal almost everywhere:

$$L^2(\mathbb{R}^n) = \left\{ f : \mathbb{R}^n \to \mathbb{C} \,\middle|\, \left(\int_{\mathbb{R}^n} \mathrm{d}x \, |f(x)|^2 \right)^{1/2} < \infty \right\} \Big/ \sim$$

$$\text{with} \quad \langle f, g \rangle \coloneqq \int_{\mathbb{R}^n} \mathrm{d}x \overline{f(x)} g(x),$$

whose elements are called wave functions. Its relevance stands from the fact that, in this space, the squared norm of a state $\psi \in L^2(\mathbb{R}^n)$ makes it possible to write probability amplitudes as integrals of a *probability density*, given as the point-wise modulus squared of the wave function: $|\psi(x)|^2$.

Going back to the definition of expectation values, we can now make an important generalization motivated by the following formal argument. Suppose the system has probability p_i of being prepared in some initial state ψ_i, of course given the normalization condition $\sum_i p_i = 1$, then the expectation of an observable A would be the weighted average of the expectations on each initial state. We call these statistical mixtures *mixed states*. If A is an observable, the expectation of A on a mixed state should be written as

$$\langle A \rangle = \sum_{i \in \mathbb{N}} p_i \langle \psi_i, A\psi_i \rangle.$$

We can take this further. Assuming we can write the observable A as a series of projectors onto its eigenspaces, and calling $\{\phi_\lambda : \lambda \in \sigma(A)\}$ the set of eigenvectors of A, then

$$A = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \cdot \rangle \phi_\lambda.$$

The expectation of A on the mixed state $\{(p_i, \psi_i)\}$ becomes

$$\langle A \rangle = \sum_{i \in \mathbb{N}} p_i \langle \psi_i, A\psi_i \rangle = \sum_{i \in \mathbb{N}, \lambda \in \sigma(A)} p_i \langle \psi_i, \lambda \langle \phi_\lambda, \psi_i \rangle \phi_\lambda \rangle = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \rho\phi_\lambda \rangle = \mathrm{Tr}(\rho A)$$

where in the last step we introduced the *density matrix* which uniquely identifies the mixed state

$$\rho \coloneqq \sum_i p_i \langle \psi_i, \cdot \rangle \psi_i.$$

This argument brings us to a general definition of a quantum state. Recall that $\mathfrak{B}(\mathcal{H})$ is the C^*-algebra of bounded operators on $\mathcal{H}$ (see Appendix A.1).

Definition 2.1.11. A general quantum state is an operator ρ on $\mathcal{H}$ such that

- $\mathrm{Tr}(\rho) = 1$ (probability normalization condition)
- $\rho^* = \rho$ (self-adjointness)
- $\rho \geq 0$ (positivity of probabilities)

and it defines the *expectation functional*

$$\begin{aligned} \langle \cdot \rangle : \mathfrak{B}(\mathcal{H}) &\longrightarrow \mathbb{C} \\ A &\longmapsto \langle A \rangle \coloneqq \mathrm{Tr}(\rho A). \end{aligned}$$

There is a correspondence between the set of mixed states (pure states are just a special case) and the set of quantum states (i.e., a subset of the dual of $\mathfrak{B}(\mathcal{H})$). We are interested in the state the system is found in when at thermal equilibrium, and from statistical mechanics we know this state is such that the energy spectrum is populated according to Boltzmann's distribution $p(\epsilon) = e^{-\beta\epsilon}/Z$. With the aforementioned correspondence we may identify such a state *at a given temperature* with the mixture $\{(e^{-\beta\epsilon_j}/Z, \psi_j)\}$ if ψ_j are energy eigenstates. Thermal equilibrium states are called Gibbs states, and we can define the quantum version for *stable* systems by presenting their associated expectation functional.

Definition 2.1.12 (Stability and ground state)**.** A physical system governed by a Hamiltonian, i.e., energy operator, H on $\mathcal{H}$ is said to be *stable* if

$$\inf_{\phi\in D(H),\|\phi\|=1}\langle\phi,H\phi\rangle > -\infty.$$

We define the *ground state energy* as

$$E = \inf_{\phi\in D(H),\|\phi\|=1}\langle\phi,H\phi\rangle,$$

and the *ground state*, **if it exists**, as a (normalized) minimizer for the energy expectation, i.e., a unit vector $\psi_0 \in D(H)$ such that $\langle\psi_0, H\psi_0\rangle = E$.

Remark 4. If the ground state ψ_0 exists and the ground state energy is E then it holds that $H\psi_0 = E\psi_0$, i.e., ψ_0 is an eigenvector of the energy operator with eigenvalue equal to the ground state energy.

Definition 2.1.13 (Gibbs state)**.** For a stable system with Hamiltonian H at temperature T, we define the *finite (inverse) temperature Gibbs state* as the quantum state given by

$$\langle\,\cdot\,\rangle_\beta = \frac{\mathrm{Tr}(\,\cdot\, e^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})}$$

where $\beta \coloneqq 1/T$ is the inverse temperature.

For $\beta < \infty$ they are generally mixed, while in the limit $\beta \to \infty$ we find the, usually pure, ground state. We will denote Gibbs states by $\langle\,\cdot\,\rangle_\beta$, specifying the temperature, and we will also specify the Hamiltonian, when needed, by writing $\langle\,\cdot\,\rangle_{\beta,H}$.

Remark 5. We could introduce Gibbs states following the standard thermodynamical approach. We would define the entropy as $S(\langle\,\cdot\,\rangle) = -\sum_n p_n \log(p_n)$, then the free energy as $F(T) = \inf_{\langle\,\cdot\,\rangle}(\langle H\rangle - TS(\langle\,\cdot\,\rangle)$, and say that a Gibbs state is a minimizer for the free energy, if it exists at finite temperature. We would find that such a minimizer is exactly the functional we wrote in the definition.

Remark 6. A free particle with state space $L^2(\mathbb{R}^n)$ and governed by the Hamiltonian $H = -\Delta$ is stable but does not admit a ground state. However, this Hamiltonian does admit a ground state when restricting the state space to wave functions on a compact box with periodic boundary conditions. The stability of free particles is essential, as it will be implicitly assumed in Chapter 4.

We refer to the appendix for the definition of trace class operators and point out that such operators are compact, which makes $e^{-\beta H}$ a very well-behaved operator in computations.

2.2 Canonical and grand canonical pictures

So far we talked about a generic (separable) Hilbert space, but for the most part we will deal with Hilbert spaces constructed in a specific way: tensor products, which are the natural spaces in which to set physical systems composed of multiple degrees of freedom. We are particularly interested in two special subspaces of a tensor product that encode the *statistics* of the particles considered, a property that describes indistinguishabilty between particles in a quantum system. Once again we point to Appendix A.2 for a discussion on the construction of tensor products (and direct sums) of Hilbert spaces.

Consider a system of N particles whose states belong to the spaces $\mathfrak{h}_1, \dots, \mathfrak{h}_N$. The possible states of this system are described by the tensor product

$$\mathcal{H}_N = \mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N.$$

Given the energy operators for each particle $h_1, \dots, h_N$, we lift these operators to the space $\mathcal{H}_N$ with the identification

$$h_i \longmapsto \mathbb{1} \otimes \cdots \otimes h_i \otimes \cdots \otimes \mathbb{1}.$$

The total energy of the system, disregarding any interaction, is naturally given by

$$H_N^{\text{in}} = h_1 + \cdots + h_N \tag{2.1}$$

with

$$D(H_N^{\text{in}}) = \operatorname{span}\{\phi_1 \otimes \cdots \otimes \phi_N : \phi_i \in D(h_i),\ i = 1, \dots, N\}$$

which is dense in $\mathcal{H}_N$ if $D(h_i)$ is dense in $\mathfrak{h}_i$ for all i.

Theorem 2.2.1 (Ground state of non-interacting (distinguishable) particles)**.** *[8, p. 15] If the ground state energies of the Hamiltonians $h_1, \dots, h_N$ are*

$$e_j = \inf_{\phi \in D(h_j), \|\phi\|=1} \langle \phi, h_j \phi \rangle, \quad j = 1, \dots, N$$

then the ground state energy of H_N^{in} is $\sum_{j=1}^N e_j$. Moreover, if $\phi_1, \dots, \phi_N$ are the ground states of these same Hamiltonians, then the ground state of H_N^{in} is $\phi_1 \otimes \cdots \otimes \phi_N$.

Proof. If $\Psi \in D(H_N^{\text{in}})$ is a unit vector, then we may write $\Psi = \psi_1 \otimes \Psi_1 + \cdots + \psi_K \otimes \Psi_k$, where $\psi_1, \dots, \psi_K \in D(h_1)$ and $\Psi_1, \dots, \Psi_K \in \mathfrak{h}_2 \otimes \cdots \otimes \mathfrak{h}_N$ are orthonormal. Since Ψ is normalized this implies that $\|\psi_1\|^2 + \cdots + \|\psi_K\|^2 = 1$, thus

$$\langle \Psi, h_1 \Psi \rangle = \sum_{i=1}^K \langle \psi_i, h_1 \psi_i \rangle \geq \sum_{i=1}^K \|\psi_i\|^2 e_1 = e_1.$$

This implies that $\langle \Psi, H_N^{\text{in}} \Psi \rangle \geq \sum_{j=1}^N e_j$. Conversely, for every $\varepsilon > 0$ we can choose unit vectors $\phi_j \in D(h_j)$ such that $\langle \phi_j, h_j \phi_j \rangle \leq e_j + \varepsilon$ with $j = 1, \dots, N$, which implies that, for $\Psi = \phi_1 \otimes \cdots \phi_N$ (still a unit vector),

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^N \langle \phi_j, h_j \phi_j \rangle \leq \sum_{j=1}^N e_j + N\varepsilon,$$

which holds if, and only if, $\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \sum_{j=1}^N e_j$, thus we proved the equality. Furthermore, it is clear that if $\phi_1, \dots, \phi_N$ are the respective ground state eigenvectors of the N one-body Hamiltonians, their tensor product is an eigenvector of H_N^{in} with eigenvalue given precisely by $\sum_{j=1}^N e_j$, the ground state energy. □

If we consider interactions we need to lift additional operators. For pair-wise interactions (but this procedure can be readily generalized to N-body interactions) to each pair (i, j) of distinct particles ($i \neq j$) we associate an operator W_{ij} on a suitable domain. To lift this operator, consider vectors $\phi_i \in \mathfrak{h}_i$ and identify W_{ij} with the operator on $\mathcal{H}_N$ that acts as

$$W_{ij}(\phi_1 \otimes \cdots \otimes \phi_i \otimes \cdots \otimes \phi_j \otimes \cdots \otimes \phi_N) = \phi_1 \otimes \cdots \otimes \psi_i \otimes \cdots \otimes \psi_j \otimes \cdots \otimes \phi_N$$

where $\psi_i \otimes \psi_j = W_{ij} \phi_i \otimes \phi_j$ (and express the image as a sum of simple tensors in the case where it is not factorizable as one). Without specifying the correct domain we arrive at the formal way of writing the Hamiltonian of a system of N pair-wise interacting particles:

$$H_N = H_N^{\text{in}} + \sum_{1 \leq i < j \leq N} W_{ij} = \sum_{j=1}^N h_j + \sum_{1 \leq i < j \leq N} W_{ij}. \tag{2.2}$$

Requiring that particles be indistinguishable from one another means that both their internal energy operators and the pair-wise interactions do not depend on the specific particle considered. This means that

$$\begin{aligned} \mathfrak{h}_1 &= \ldots = \mathfrak{h}_N =: \mathfrak{h}, \\ h_1 &= \ldots = h_N =: h, \\ W_{ij} &= W_{kl} =: W \quad \forall i \neq j, k \neq l. \end{aligned} \tag{2.3}$$

To make the definition of indistinguishability complete we also require an additional property that states must satisfy, so let us take the time to recall a formal explanation of what this property is, following the reasoning in [3, p. 302]. Consider the simplest case of a system of two indistinguishable particles in the states $\psi, \phi \in \mathfrak{h}$. Because quantum states can be told apart only through probability amplitudes, two vectors that differ only by an overall phase can be considered indistinguishable, as $|\langle \phi, e^{i\theta}\phi \rangle|^2 = \|\phi\|^2$ no matter the phase θ. Therefore we can consider an exchange operator that flips the order in a simple tensor in $\mathfrak{h} \otimes \mathfrak{h}$

$$\mathrm{Ex}(\phi \otimes \psi) = \psi \otimes \phi$$

and say that two indistinguishable particles only admit physical states that differ at most by an overall phase from their exchanged version, i.e., there must exist some α such that

$$\mathrm{Ex}(\phi \otimes \psi) = e^{i\alpha} \phi \otimes \psi.$$

But since Ex^2 should act as the identity, as flipping two times the order of the particle states results in the original state, we can conclude that

$$\mathrm{Ex}^2 = \mathbb{1} \implies e^{2i\alpha} = 1 \implies \alpha = \pm 1.$$

The sign $\alpha = \pm 1$ is what determines the statistics of the two identical particles: (+) for *bosons* (particles that follow Bose-Einstein statistics) and (-) for *fermions* (particles that follow Fermi-Dirac statistics).
Note, however, that this only works for two particles. We only use this line of reasoning to *postulate* that, given any number of particles greater or equal to two, the exchange of a pair of identical bosons leaves the state unaltered, while the exchange of a pair of identical fermions carries an overall minus sign.

We will now make precise the notion of bosonic and fermionic states. On the space $\mathcal{H}_N$ we can define a natural action for the group of permutations S_N. Indeed, we can build the unitary representation U_σ of $\sigma \in S_N$ on this space by specifying the action on the usual basis (see Appendix A.2):

$$U_\sigma(u_1 \otimes \cdots \otimes u_N) \coloneqq u_{\sigma^{-1}(1)} \otimes \cdots \otimes u_{\sigma^{-1}(N)}.$$

Define the operators

$$\begin{aligned} P_+ &\coloneqq \frac{1}{N!} \sum_{\sigma \in S_N} U_\sigma \\ P_- &\coloneqq \frac{1}{N!} \sum_{\sigma \in S_N} \mathrm{sgn}(\sigma) U_\sigma. \end{aligned} \tag{2.4}$$

These are unitary and self-adjoint by construction, and showing that they are also projections only involves rewriting the sum over two sets of permutations as a single one to get that $P_\pm^2 = P_\pm$.

Definition 2.2.2 (Symmetric and anti-symmetric tensor products)**.** Given $\mathfrak{h}$ the one-particle Hilbert space, the *symmetric tensor product* is the subspace

$$\bigotimes_{sym}^{N} \mathfrak{h} \coloneqq P_+ \left(\bigotimes^{N} \mathfrak{h} \right) \subset \mathcal{H}_N$$

while the *anti-symmetric tensor product* is the subspace

$$\bigwedge^{N} \mathfrak{h} \coloneqq P_- \left(\bigotimes^{N} \mathfrak{h} \right) \subset \mathcal{H}_N .$$

We also define the anti-symmetric tensor product of vectors as

$$u_1 \wedge \cdots \wedge u_N = \sqrt{N!}\, P_-(u_1 \otimes \cdots \otimes u_N).$$

Bosonic states belong in the symmetric tensor product, while fermionic states belong in the anti-symmetric tensor product.

Remark 7. In the special case where $\mathfrak{h} = L^2(\mathrm{R}^n)$, anti-symmetric tensor products between wave functions can be constructed as Slater determinants, i.e.

$$u_1 \wedge \cdots \wedge u_N(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} u_1(x_1) & \cdots & u_N(x_1) \\ \vdots & & \vdots \\ u_1(x_N) & \cdots & u_N(x_N) \end{pmatrix}.$$

Remark 8. If $\dim(\mathfrak{h}) < N$ then $\bigwedge^N \mathfrak{h} = \{0\}$, i.e., the anti-symmetrization of N vectors in a space of dimension less than N is always the null vector. Thus, if the Hilbert space of a single fermionic particle is finite-dimensional, the allowed number of particles in the canonical and grand canonical pictures will be finite.

With the conditions in Equation (2.3) that we impose when introducing indistinguishability, it is clear that H_N maps the spaces $\bigotimes_{\mathrm{sym}}^N \mathfrak{h}$ and $\bigwedge^N \mathfrak{h}$ into themselves, which means that we can restrict it to either of the two subspaces and get a well-defined operator (provided that a suitable domain for the starting Hamiltonian is chosen). The operators $P_+(H_N)$ and $P_-(H_N)$ are respectively called the *bosonic* and *fermionic* N-body Hamiltonians for pair-wise interacting indistinguishable particles. This way of defining the energy of the system characterizes the *canonical picture*, where the number of particles is fixed. But in statistical mechanics, allowing the number of particles to vary can actually make things easier. Allowing N to vary defines the *grand canonical picture*, and we shall now discuss its state space.

Definition 2.2.3 (Fock space)**.** For every $N \in \mathbb{N}$, let $\mathfrak{h}_1, \ldots, \mathfrak{h}_N$ be one-particle Hilbert spaces, the *Fock space* is defined as

$$\mathcal{F} \coloneqq \bigoplus_{N=0}^{\infty} \bigotimes_{i=1}^{N} \mathfrak{h}_i$$

where, as a convention, we interpret $\mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N$ for $N = 0$ to be just $\mathbb{C}$. The state $1 \in \mathcal{F}$ is called the *vacuum state* and often denoted $|0\rangle$, $|\Omega\rangle$ or Ω.

On this space, the energy operator is naturally given by the direct sum of all N-body Hamiltonians:

$$H = \bigoplus_{N=0}^{\infty} H_N, \quad \text{with } H \bigoplus_{N=0}^{\infty} \Psi_N = \bigoplus_{N=0}^{\infty} H_N \Psi_N,$$

with H_0 taken to be the null operator (which means we are viewing the vacuum as having zero energy), and its domain is

$$D(H) = \left\{ \Psi = \bigoplus_{N=0}^{\infty} \Psi_N \in \mathcal{F} : \Psi_N \in D(H_N),\ \sum_{N=0}^{\infty} \|H_N \Psi_N\|^2 < \infty \right\}.$$

Introducing indistinguishability in this picture means repeating the steps we took for the N-body case in each *N-particle sector*, which gets us to the definition of the *fermionic* and *bosonic* Fock spaces. Given $\mathfrak{h}$ the state space for a single particle, these spaces are, respectively

$$\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) := \bigoplus_{N=0}^{\infty} \bigwedge^{N} \mathfrak{h}$$
$$\mathcal{F}^{\mathrm{B}}(\mathfrak{h}) := \bigoplus_{N=0}^{\infty} \bigotimes_{\mathrm{sym}}^{N} \mathfrak{h}.$$

One of the most powerful aspects of the grand canonical picture is that all observables of the system can be written in the same way using special operators called *creation/annihilation operators.* Starting from the Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \bigotimes^{N} \mathfrak{h}$ we define operators that add or remove a single particle state $f \in \mathfrak{h}$ as

$$\begin{aligned} a(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N} \langle f, f_1 \rangle_{\mathfrak{h}} f_2 \otimes \cdots \otimes f_N, \\ a^*(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N+1} f \otimes f_1 \otimes \cdots \otimes f_N. \end{aligned} \tag{2.5}$$

Here, for every $f \in \mathfrak{h}$, $a(f)$ is called *annihilation operator* and $a^*(f)$ is called *creation operator.* Note the very important special case where an annihilation operator acts on the vacuum: $a(f)|0\rangle = 0$ for all $f \in \mathfrak{h}$, i.e., annihilation operators annihilate the vacuum. To indicate an operator that could be either of the two, we write $a^{\#}(f)$. With this definition the domain would be the set of pure tensors, but by linearity we can extend this to the space $\cup_{M=0}^{\infty} \bigoplus_{N=0}^{M} \mathcal{H}_N$, which is dense in $\mathcal{F}$.

Proposition 2.2.4. *Creation and annihilation operators are the adjoint of one another, i.e., for all $f \in \mathfrak{h}$ and for all $\Psi, \Phi \in \cup_{M=0}^{\infty} \bigoplus_{N=0}^{M} \mathcal{H}_N$ it holds that*

$$\langle a(f)\Psi, \Phi \rangle_{\mathcal{F}} = \langle \Psi, a^*(f)\Phi \rangle_{\mathcal{F}}.$$

Proof. It suffices to prove the equality for simple tensors, so let $\psi_1 \otimes \ldots \otimes \psi_N \in \bigotimes^N \mathfrak{h}$ and $\phi_0 \otimes \phi_1 \otimes \ldots \otimes \phi_N \in \bigotimes^N \mathfrak{h}$. Then, for all $f \in \mathfrak{h}$,

$$\begin{aligned} \langle \psi_1 \otimes \cdots \psi_N, a(f)\phi_0 \otimes \cdots \otimes \phi_N \rangle &= \sqrt{N+1} \langle f, \phi_0 \rangle \langle \psi_1 \otimes \cdots \otimes \psi_N, \phi_1 \otimes \cdots \otimes \phi_N \rangle \\ &= \sqrt{N+1} \langle f, \phi_0 \rangle \prod_{j=1}^{N} \langle \psi_j, \phi_j \rangle, \end{aligned}$$

and similarly

$$\langle a^*(f)\psi_1 \otimes \cdots \otimes \psi_N, \phi_0 \otimes \cdots \otimes \phi_N \rangle = \sqrt{N+1} \langle f, \phi_0 \rangle \prod_{j=1}^{N} \langle \psi_j, \phi_j \rangle.$$

□

Once again we introduce indistinguishability into the picture. Looking at the definition of these operators, it is clear that if a state $\Psi_N \in D(a^{\#}(f))$ is completely symmetric or completely anti-symmetric then applying an annihilation operator produces a state that is still completely symmetric or anti-symmetric (as Ψ_N can be written as a linear combination of all $N!$ permutations of N states and thus $a(f)$ removes every particle exactly once), but applying a creation operator does introduce a state that is not yet accounted for in the (anti-)symmetrization. We then define *bosonic (+)* and *fermionic (-)* creation/annihilation operators as the projection onto the correct subspaces:

$$a_\pm(f) \coloneqq a(f),\ a_\pm^*(f) \coloneqq P_\pm(a^*(f)) \quad \forall f \in \mathfrak{h}.$$

Remark 9. Bosonic and fermionic creation/annihilation operators are still densely defined and adjoints of one another (within their respective statistics).

Remark 10. The map $\mathfrak{h} \ni f \mapsto a_\pm^*(f)$ is linear but $\mathfrak{h} \ni f \mapsto a_\pm(f)$ is conjugate linear.

CHECKKKKKKKKK. The normalization we used in (2.5) changes slightly when using the (anti-)symmetric subspaces. This is because we want to choose a normalization such that canonical commutator from quantum mechanics still holds, and we show that the following normalization achieves this in the next proposition. The choice of normalization is

$$a_\pm(f)P_\pm(f_1 \otimes \cdots \otimes f_N) = \sqrt{N} \sum_{\sigma \in S_N} \langle f, f_{\sigma(1)} \rangle f_{\sigma(2)} \otimes \cdots \otimes f_{\sigma(N)},$$

$$a_\pm^*(f) f_1 \otimes \cdots \otimes f_N = \frac{1}{\sqrt{N+1}} \sum_{\sigma \in S_N} \sum_{j=1}^{N+1} f_{\sigma(1)} \otimes \cdots \otimes \underset{(j-\text{th})}{f} \otimes \cdots \otimes f_{\sigma(N)}.$$

Recall the definition of the commutator and anticommutator between two operators defined on a common domain: $\{A, B\} \coloneqq AB + BA$ and $[A, B] \coloneqq AB - BA$. The anticommutator may be denoted by $[\cdot,\cdot]_+$ while the commutator by $[\cdot,\cdot]_-$, to accommodate for the more compact notation $[\cdot,\cdot]_\mp$ to indicate either of the two.

Proposition 2.2.5 (CCR/CAR)**.** *The operators a_- and a_-^* satisfy the Canonical Commutation Relations (CCR), while a_+ and a_+^* the Canonical Anticommutation Relations (CAR), i.e., for all $f, g \in \mathfrak{h}$ it holds that*

$$[a_\pm(f), a_\pm^*(g)]_\mp = \langle f, g \rangle_{\mathfrak{h}} \mathbb{1}, \quad [a_\pm(f), a_\pm(g)]_\mp = [a_\pm^*(f), a_\pm^*(g)]_\mp = 0$$

Proof. We prove the claim on N-fold pure tensors. We compute the two terms in an (anti)commutator separately to highlight the terms that simplify. Let $\phi_1, \ldots, \phi_N \in \mathfrak{h}$, then

$$\begin{aligned}
&a_\pm(f)a_\pm^*(g)(\phi_1 \otimes \cdots \otimes \phi_N) = \\
&\quad = a_\pm(f)\frac{1}{\sqrt{N+1}} \sum_{\sigma \in S_N} \sum_{j=1}^{N+1} (\pm 1)^\sigma (\pm 1)^{j+1} \phi_{\sigma(1)} \otimes \cdots \otimes \underset{(j-\text{th})}{g} \otimes \cdots \otimes \phi_{\sigma(N)} \\
&\quad = \sum_{\sigma \in S_N} \Bigg((\pm 1)^\sigma \langle f, g \rangle \phi_{\sigma(1)} \otimes \cdots \otimes \phi_{\sigma(N)} \\
&\qquad + \sum_{j=2}^{N+1} (\pm 1)^\sigma (\pm 1)^{j+1} \langle f, \phi_{\sigma(1)} \rangle \phi_{\sigma(1)} \otimes \cdots \otimes \underset{(j-\text{th})}{g} \otimes \cdots \otimes \phi_{\sigma(N)} \Bigg) \\
&\quad = \langle f, g \rangle N! P_\pm(\phi_1 \otimes \cdots \otimes \phi_N) \\
&\qquad + \sum_{\sigma \in S_N} \sum_{j=2}^{N+1} (\pm 1)^\sigma (\pm 1)^{j+1} \langle f, \phi_{\sigma(1)} \rangle \phi_{\sigma(2)} \otimes \cdots \otimes \underset{(j-\text{th})}{g} \otimes \cdots \otimes \phi_{\sigma(N)}
\end{aligned}$$

and also

$$
\begin{aligned}
&a^*_\pm(g)a_\pm(f)P_\pm(\phi_1\otimes\cdots\otimes\phi_N)= \\
&\qquad = a^*_\pm(g)\sqrt{N}\sum_{\sigma\in S_N}(\pm 1)^\sigma\langle f,\phi_{\sigma(1)}\rangle\phi_{\sigma(2)}\otimes\cdots\otimes\phi_{\sigma(N)} \\
&\qquad = \sum_{\sigma\in S_N}\sum_{j=2}^{N+1}(\pm)^\sigma(\pm)^j\langle f,\phi_{\sigma(1)}\rangle\phi_{\sigma(2)}\otimes\cdots\otimes\underset{(j-\mathrm{th})}{g}\otimes\cdots\otimes\phi_{\sigma(N)}
\end{aligned}
$$

and note that in this last step, because the permutation that comes from summing over j starts as positive for $g\otimes\phi_{\sigma(2)}\otimes\cdots$, it carries an overall minus sign with respect to the permutation in the previous calculation, which started as positive for $g\otimes\phi_{\sigma(1)}\otimes\cdots$. Consider now $[a_\pm(f),a^*_\pm(g)]_\mp$, it is clear that both in the bosonic and fermionic cases, the only term that survives is $\langle f,g\rangle$ times the identity, as the other terms will always be equal but of opposite sign. Note that the factor of $N!$ in the first term is due to the fact that we didn't start from an (anti)symmetrized tensor, but now restricting ourselves to the correct domain leads us exactly to the thesis. □

Remark 11. Fermionic creation/annihilation operators are bounded. To see this, let $f\in\mathfrak{h}$, it is clear that for all normalized $\Psi\in D(a(f))$ it holds that

$$
\begin{aligned}
\|a_-(f)\Psi\|^2 &= \langle\Psi,a^*_-(f)a_-(f)\Psi\rangle = -\langle\Psi,a_-(f)a^*_-(f)\Psi\rangle+\langle\Psi,\{a_-(f),a^*_-(f)\}\Psi\rangle \\
&= -\langle a^*_-(f)\Psi,a^*_-(f)\Psi\rangle+1\le 1,
\end{aligned}
$$

which implies that $\|a_-(f)\|\le 1$, i.e., $a_-(f)$ is bounded for all $f\in\mathfrak{h}$. Similarly for $a^*_-(f)$.

An important quantity we will come back to shortly is the number operator, which is built from operators of the form $a^*_\pm(u)a_\pm(u)$, where $u\in\mathfrak{h}$ is a unit vector. Here, $\bigotimes^N u$ is an eigenvector with eigenvalue N for $a^*(u)a(u)$, which means this operator "counts" how many copies of a single-particle state compose a vector in $\mathcal{F}$. This is useful when considering the occupation number basis for a bosonic or fermionic Fock space, with the important distinction that bosonic modes can have arbitrary occupation number, while fermionic modes can only have occupation 0 or 1, since the CAR implies that $(a^*_-(f))^2=0$ for all f. This last statement is the content of Pauli's exclusion principle.

Proposition 2.2.6. *Let $u_1,\dots,u_r\in\mathfrak{h}$ be unit vectors and $\mathcal{H}_{n_1,\dots,n_r}$ the joint eigenspaces of the operators $a^*_\pm(u_1)a_\pm(u_1),\dots,a^*_\pm(u_r)a_\pm(u_r)$, then the fermionic and bosonic Fock spaces can be decomposed as direct sums of such eigenspaces:*

$$
\begin{aligned}
\mathcal{F}^{\mathrm{B}}(\mathfrak{h}) &= \bigoplus_{n_1=0}^{\infty}\cdots\bigoplus_{n_r=0}^{\infty}\mathcal{H}_{n_1,\dots,n_r} \\
\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) &= \bigoplus_{n_1=0}^{1}\cdots\bigoplus_{n_r=0}^{1}\mathcal{H}_{n_1,\dots,n_r}.
\end{aligned}
$$

Proof. We will only prove the fermionic case, with $\dim(\mathfrak{h})=M<\infty$ and $r=M$, as this provides the useful fermionic occupation number basis, which will be of great help to us later. Given $\{\phi_1,\dots,\phi_M\}$ an orthonormal basis for $\mathfrak{h}$, and calling $N_j=a^*_-(\phi_j)a_-(\phi_j)=a^*_ja_j$, we can prove that these number operators are commuting projections. These operators are indeed self-adjoint, as

$$N_j^*=(a^*_ja_j)^*=a^*_ja_j=N_j,$$

and idempotent, as the CAR imply

$$N_j^2=(a^*_ja_j)^2=-a^*_ja^*_ja_ja_j+a^*_ja_j=a^*_ja_j.$$

Moreover, for $j \neq k$, we have that

$$[N_j, N_k] = a_j^* a_j a_k^* a_k - a_k^* a_k a_j^* a_j = a_k^* a_k a_j^* a_j - a_k^* a_k a_j^* a_j = 0,$$

therefore, all the number operators are commuting projections, furthermore their spectra are all equal to $\{0, 1\}$. Note that these algebraic manipulations are well defined, since all the operators $a^{\#}$ (and their products) are defined on the entire space $\mathcal{F}^{\mathrm{F}}(\mathfrak{h})$, which is finite-dimensional if $\mathfrak{h}$ is finite-dimensional. We now turn to the direct sum decomposition. Let $\boldsymbol{n} = (n_1, \ldots, n_M) \in \{0, 1\}^M$ be an M-tuple of eigenvalues of the respective number operators, and define the joint projection as

$$P_{\boldsymbol{n}} \coloneqq \prod_{i=1}^{M} \left(n_i N_i + (1 - n_i)(\mathbb{1} - N_i) \right).$$

This is a product of commuting projections and is therefore still a projection, and still defined on the entire fermionic Fock space. The set $\{P_{\boldsymbol{n}} : \boldsymbol{n} \in \{0, 1\}^M\}$ is a complete set of projections, as shown by the completeness relation

$$\begin{aligned}
\sum_{\boldsymbol{n} \in \{0,1\}^M} P_{\boldsymbol{n}} &= \sum_{\boldsymbol{n}} \prod_{i=1}^{M} \left(n_i N_i + (1 - n_i)(\mathbb{1} - N_i) \right) \\
&= \prod_{i=1}^{M} \sum_{n_i=0}^{1} \left(n_i N_i + (1 - n_i)(\mathbb{1} - N_i) \right) \\
&= \prod_{i=1}^{M} \mathbb{1} = \mathbb{1},
\end{aligned}$$

again defined on the entire Fock space. We conclude by simply applying this completeness relation:

$$\mathcal{F}^{\mathrm{F}}(\mathfrak{h}) = \sum_{\boldsymbol{n} \in \{0,1\}^M} P_{\boldsymbol{n}} \mathcal{F}^{\mathrm{F}}(\mathfrak{h}) = \bigoplus_{\boldsymbol{n} \in \{0,1\}^M} P_{\boldsymbol{n}} \mathcal{F}^{\mathrm{F}}(\mathfrak{h}).$$

□

Remark 12. The case where $\mathfrak{h}$ is infinite-dimensional is not much different, and even then we can always choose a finite set of M independent number operators to obtain a decomposition into a direct sum of M (possibly infinite-dimensional) joint eigenspaces. But because we will put a UV cutoff on our system the allowed momenta will form a finite set, then the one-particle Hilbert space will be finite dimensional and the above decomposition will be realized by choosing the following basis for the fermionic Fock space:

$$\{|\boldsymbol{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0, 1\}\ \forall i = 1, \ldots, M\}$$

with eigenvalues n_i given by the occupation of the M independent modes.

We can now come back to a statement that we previously used as motivation for the introduction of creation/annihilation operators, and show that indeed observables in the grand canonical picture can be written in terms of such operators. We focus only on 1-body and 2-body observables because these are the types of operators that show up in the Hamiltonian we set out to study, but this procedure can be generalized to any N-body operator.

Theorem 2.2.7 (2nd quantization of 1-body operator)**.** *[8, p. 38] Let h be an observable on $\mathfrak{h}$ and $\{u_\alpha\}_{\alpha=1}^{\infty}$ be an orthonormal basis for $\mathfrak{h}$ with elements in the dense domain $D(h)$. Following the notation from the beginning of this section, we can write*

$$\bigoplus_{N=1}^{\infty} \sum_{j=1}^{N} h_j = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, h u_\beta \rangle_{\mathfrak{h}} a_{\pm}^*(u_\alpha) a_{\pm}(u_\beta)$$

on the domain $\cup_{M=0}^{\infty}\bigoplus_{N=1}^{M} P_\pm(D(\sum_{j=1}^{N} h_j))$ *(for* $M = 0$ *the domain is taken to be* $\mathbb{C}$*). This operator is called the second quantization of* h*.*

Proof. First we observe that the claim holds in the sense of quadratic forms on $D(h)$. Indeed, let $f, g \in D(h)$ and recall the definition of creation/annihilation operators in 2.5

$$\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle\langle g, a_\pm^*(u_\alpha)a_\pm(u_\beta)f\rangle = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle\langle g, u_\alpha\rangle\langle u_\beta, f\rangle = \sum_{\beta=1}^{\infty}\langle g, hu_\beta\rangle\langle u_\beta, f\rangle$$
$$= \sum_{\beta=1}^{\infty}\langle hg, u_\beta\rangle\langle u_\beta, f\rangle = \langle hg, f\rangle = \langle g, hf\rangle$$

where the indices α and β are removed by using the completeness of the basis $\{u_\alpha\}_{\alpha=1}^{\infty}$, and h can be moved on either side of the inner products as it is self-adjoint with $f, g \in D(h)$ and $u_\alpha \in D(h)$ for all α. Moving onto N-fold pure tensors, let $f_1, \dots, f_N \in D(h)$ for some $N \geq 1$ and consider the action of $a^*(u_\alpha)a(u_\beta)$ on their tensor product

$$a^*(u_\alpha)a(u_\beta)f_1 \otimes \cdots \otimes f_N = N\langle u_\beta, f_1\rangle u_\alpha \otimes f_2 \otimes \cdots \otimes f_N$$

then it follows that

$$\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle a^*(u_\alpha)a(u_\beta) = Nh_1$$

as quadratic form on finite linear combinations of pure tensor products of elements in $D(h)$. Recall the definition of the projections $P_\pm$ from Equation (2.4), then it is clear that

$$P_\pm Nh_1P_\pm = P_\pm\sum_{j=1}^{N} h_jP_\pm = \sum_{j=1}^{N} h_jP_\pm$$

on $\bigotimes^N \mathfrak{h}$, as each h_j acts in the same way as h on the respective factor in the tensor product, which means that it maps the (anti-)symmetric subspace into itself. Then it follows that

$$\sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle a_\pm^*(u_\alpha)a_\pm(u_\beta)P_\pm = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, hu_\beta\rangle P_\pm a^*(u_\alpha)a(u_\beta)P_\pm = \bigoplus_{N=1}^{\infty}\sum_{j=1}^{N} h_jP_\pm$$

from which the thesis follows immediately. □

Remark 13. Another way to lift an operator U to the Fock space is multiplicatively as $\bigoplus_{N=0}^{\infty}\bigotimes^N U$, and this plays an important role in the discussion of symmetries. Given a Lie group and a unitary representation γ on some Hilbert space, its push-forward $\mathrm{d}\gamma$ defines a representation of its associated Lie algebra as Hermitian operators. One can then define a map Γ that lifts the representation γ multiplicatively to the Fock space, and this induces yet another map $\mathrm{d}\Gamma$ which lifts the representation $\mathrm{d}\gamma$ additively to the Fock space. In general, given a unitary operator U on $\mathfrak{h}$ we may call $\Gamma(U) = \bigoplus_{N=0}^{\infty}\bigotimes_{j=1}^{N} U_j$ its multiplicative representation on $\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$, and given a self-adjoint operator A we may call $\mathrm{d}\Gamma(A) = \bigoplus_{N=0}^{\infty}\sum_{j=1}^{N} A_j$ its additive representation on $\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$.

Remark 14. The second quantization of the identity operator on $\mathfrak{h}$ is the number operator

$$\mathcal{N} = \bigoplus_{N=0}^{\infty}\sum_{j=0}^{N} \mathbb{1}_j = \bigoplus_{N=0}^{\infty} N = \sum_{\alpha=1}^{\infty} a_\pm^*(u_\alpha)a_\pm(u_\beta).$$

Now it is clear why we introduced the operator $a^*(f)a(f)$. The number operator is the sum of operators that count the occupation number of a given state over a complete set of states. In a much more delicate way one could define creation/annihilation *distributions* as objects that formally create/annihilate states completely localized at one point in space. If we called $|x\rangle$ an "eigenstate" (in a generalized sense that we will not discuss here) of the position operator with eigenvalue x, then the number density would be $\rho(x) = a_\pm^*(|x\rangle)a_\pm(|x\rangle)$ and the total number of particles would be written in a familiar way: $\mathcal{N} = \int_{\mathbb{R}} \mathrm{d}x\, \rho(x)$, but to make sense of these objects one needs tools that go beyond the scope of our work.

Theorem 2.2.8 (2nd quantization of 2-body operator)**.** *[8, p. 40] Let W be a 2-body potential for identical particles, i.e., an observable on $\mathfrak{h} \otimes \mathfrak{h}$ such that* $\mathrm{Ex}W\mathrm{Ex} = W$. *Let $\{u_\alpha\}_{\alpha=1}^\infty$ be an orthonormal basis for $\mathfrak{h}$ such that $u_\alpha \otimes u_\beta \in D(W)$ for all α and β. Then as a quadratic form on the domain $\cup_{M=2}^\infty \bigoplus_{N=2}^M P_\pm(D(\sum_{1\le i<j\le N} W_{ij}))$ we have*

$$\bigoplus_{N=2}^\infty \sum_{1\le i<j\le N} W_{ij} = \frac{1}{2}\sum_{\alpha,\beta,\nu,\mu=1}^\infty \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle a_\pm^*(u_\alpha)a_\pm^*(u_\beta)a_\pm(u_\nu)a_\pm(u_\mu).$$

Again, this operator is called the second quantization of W. Note that in the fermionic case, the order in which the indices appear in the creation-creation and annihilation-annihilation products matters, as these operators anticommute rather than commute.

Proof. We proceed in a similar fashion to the second quantization of one-body operators. We have that

$$W u_1 \otimes u_2 = \sum_{\alpha,\beta,\mu,\nu=1}^\infty \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle \langle u_\mu, u_1\rangle\langle u_\nu, u_2\rangle u_\alpha \otimes u_\beta,$$

using the resolution of the identity. But now notice that the product of creation/annihilation acts on pure tensors as

$$a^*(u_\alpha)a^*(u_\beta)a(u_\nu)a(u_\mu)u_1 \otimes \cdots \otimes u_N = N(N-1)\langle u_\mu, u_1\rangle\langle u_\nu, u_2\rangle u_\alpha \otimes u_\beta \otimes u_3 \otimes \cdots \otimes u_N,$$

and that, like in the one-body case, the projectors $P_\pm$ act on W as

$$\frac{N(N-1)}{2} P_\pm W_{12} P_\pm = \sum_{1\le i<j\le N} W_{ij} P_\pm,$$

where we used that $\mathrm{Ex}W\mathrm{Ex} = W$. Recall, finally, that $P_\pm a^*(f)P_\pm = P_\pm a^*(f)$ as it doesn't matter if the starting state is (anti-)symmetrized since the image gets (anti-)symmetrized. With this in mind, we may write

$$\begin{aligned}
\bigoplus_{N=2}^\infty \sum_{1\le i<j\le N} W_{ij} P_\pm &= \bigoplus_{N=2}^\infty \frac{N(N-1)}{2} P_\pm W_{12} P_\pm \\
&= P_\pm \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^\infty \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle a^*(u_\alpha)a^*(u_\beta)a(u_\nu)a(u_\mu)P_\pm \\
&= \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^\infty \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle P_\pm a^*(u_\alpha)P_\pm a^*(u_\beta)a(u_\nu)a(u_\mu)P_\pm \\
&= \frac{1}{2} \sum_{\alpha,\beta,\nu,\mu=1}^\infty \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu\rangle a_\pm^*(u_\alpha)a_\pm^*(u_\beta)a_\pm(u_\nu)a_\pm(u_\mu)P_\pm,
\end{aligned}$$

which concludes the proof. $\square$

2.3 One-particle density matrices

It is clear from our discussion of the second quantization of an operator, that expectation values of observables will inevitably reduce to expectation values of products of creation and annihilation operators. An important result that we will discuss in the next section states that – under appropriate conditions which are discussed in Theorem 2.4.5 – the expectation of an observable can be written in terms of expectations of operators of the form $a_\pm^\# a_\pm^\#$. We therefore begin the study of these kinds of expectations, starting from the following definition.

Definition 2.3.1 (One-particle density matrix (1-pdm))**.** Let $\Psi = \bigoplus_{N=0}^\infty \Psi_N$ be a normalized vector on the bosonic or fermionic Fock space $\mathcal{F}^{\mathrm{B,F}}(\mathfrak{h})$ with finite particle number expectation

$$\langle \Psi, \mathcal{N}\Psi\rangle = \sum_{N=0}^\infty N\|\Psi_N\|^2 < \infty.$$

We define the *1-particle density matrix* (or *2-point function*, or *propagator*) of Ψ as the operator γ_Ψ on $\mathfrak{h}$ given, as a quadratic form, by

$$\langle f, \gamma_\Psi g\rangle_{\mathfrak{h}} = \langle \Psi, a_\pm^*(g) a_\pm(f)\Psi\rangle$$

for all $f, g \in \mathfrak{h}$.

Proposition 2.3.2. *The operator γ_Ψ is positive semi-definite, i.e., $\gamma_\Psi \geq 0$, and trace class, with*

$$\mathrm{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N}\Psi\rangle.$$

Lemma 2.3.3. *Let $\mathfrak{h}$ be a Hilbert space. If $\Psi \in \mathfrak{h}$ is a finite linear combination of pure tensor products of elements from a subspace $X \subseteq \mathfrak{h}$ then γ_Ψ is a finite rank operator whose range is a subspace of X.*

Lemma 2.3.4. *[8, p. 42] If $\Psi \in \mathcal{F}^{\mathrm{F}}(\mathfrak{h})$ is a normalized vector then γ_Ψ satisfies the inequality*

$$0 \leq \gamma_\Psi \leq \mathbb{1}.$$

In particular, the eigenvalues of γ_Ψ lie in the interval $[0, 1]$.

Proof. We need to show that for every $f \in \mathfrak{h}$ it holds that $0 \leq \langle f, \gamma_\Psi f\rangle \leq \|f\|^2$. Using that $a(f)$ and $a^*(f)$ are adjoints of each other, we can see that

$$\langle f, \gamma_\Psi f\rangle = \langle \Psi, a_-^*(f) a_-(f)\Psi\rangle = \langle a_-(f)\Psi, a_-(f)\Psi\rangle = \|a_-(f)\Psi\|^2 \geq 0.$$

On the other hand, the CAR imply

$$\begin{aligned}\langle f, \gamma_\Psi f\rangle &= \langle \Psi, a_-^*(f) a_-(f)\Psi\rangle \leq \langle \Psi, a_-^*(f) a_-(f)\Psi\rangle + \langle a_-^*(f)\Psi, a_-^*(f)\Psi\rangle \\ &= \langle \Psi, (a_-^*(f)a_-(f) + a_-(f)a_-^*(f))\Psi\rangle = \langle \Psi, \{a_-^*(f), a_-(f)\}\Psi\rangle = \|f\|^2.\end{aligned}$$

This concludes the proof. □

Lemma 2.3.5. *If $u_1, \ldots, u_N$ are orthonormal vectors in $\mathfrak{h}$, then the 1-pdm of $\Psi = u_1 \wedge \cdots \wedge u_N$ is the projection onto the N-dimensional space* $\mathrm{span}\{u_1, \ldots, u_N\}$.

We are now ready to prove a fundamental theorem on which the treatment of any fermionic system is built, especially if perturbation theory is involved: the ground state energy and eigenvector of non-interacting fermions.

Theorem 2.3.6 (Ground state of non-interacting fermions)**.** *[8, p. 42] Consider the Hamiltonian $H_N^{\text{in}} = \sum_{j=1}^N h_j$ for N particles, introduced in Equation* (2.1)*, restricted to the anti-symmetric subspace $\bigwedge^N \mathfrak{h}$, i.e., with domain*

$$D_-(H_N^{\text{in}}) = P_- D(H_N^{\text{in}}).$$

Assume that the one-body operator h is bounded below, i.e., there exists some constant $E_0 \in \mathbb{R}$ such that $h \geq E_0$. The ground state energy of this fermionic system is

$$\inf \{\text{Tr}(Ph) : P \text{ is a projection onto an } N\text{-dimensional subspace of } D(h)\}.$$

If the infimum is attained for an N-dimensional projection P then H_N^{in} has a ground state eigenvector $f_1 \wedge \cdots \wedge f_N$, where $f_1, \ldots, f_N$ is an orthonormal basis for $P(\mathfrak{h})$. In particular, these vectors can be chosen to be eigenvectors of h, and the expectations of h restricted to $P(\mathfrak{h})^\perp \cap D(h)$ will be greater than the expectations of h restricted to $P(\mathfrak{h})$.

Proof. Since $D(h)$ is assumed to be dense, we can choose an orthonormal basis $\{u_\alpha\}_{\alpha=1}^\infty$ for $\mathfrak{h}$ of elements in $D(h)$. Let $\Psi \in D_-(H_N^{\text{in}})$ be a normalized vector, then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{\alpha,\beta=1}^{\infty} \langle u_\alpha, h u_\beta \rangle \langle \Psi, a_-^*(u_\alpha) a_-(u_\beta) \Psi \rangle = \sum_{\alpha,\beta=1}^{\infty} h_{\alpha\beta} \langle u_\beta, \gamma_\Psi u_\alpha \rangle.$$

Since, from the second quantization of an observable, Ψ is assumed to be a finite linear combination of pure tensor products of elements in $D(h)$, it follows from Lemma 2.3.3 that γ_Ψ has finite rank. Let $\{v_1, \ldots, v_n\}$ be the set of eigenvectors of γ_Ψ with eigenvalues $\{\lambda_1, \ldots, \lambda_n\}$, then it follows from the same lemma that $v_1, \ldots, v_n \in D(h)$. Then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^{n} \sum_{\alpha,\beta=1}^{\infty} \lambda_j h_{\alpha\beta} \langle u_\beta, v_j \rangle \langle v_j, u_\alpha \rangle = \sum_{j=1}^{n} \lambda_j \langle v_j, h v_j \rangle.$$

Because $\Psi \in D_-(H_N^{\text{in}})$, it must have fixed particle number N, then it follows that

$$\sum_{j=1}^{n} \lambda_j = \text{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N} \Psi \rangle = N,$$

but $0 \leq \lambda_j \leq 1$, and therefore it must be that $n \geq N$. We may assume the eigenvectors v_j to be ordered such that $\langle v_1, h v_1 \rangle \leq \ldots \leq \langle v_n, h v_n \rangle$ so that, by defining P as the projection onto $\text{span}\{v_1, \ldots, v_n\}$, we can provide the bound

$$\text{Tr}(Ph) = \sum_{j=1}^{N} \langle v_j, h v_j \rangle \leq \sum_{j=1}^{n} \lambda_j \langle v_j, h v_j \rangle = \langle \Psi, H_N^{\text{in}} \Psi \rangle.$$

Thus

$$\begin{aligned} &\inf\{\text{Tr}(Ph) : P \text{ is a projection onto an } N\text{-dimensional subspace of } D(h)\} \\ &\qquad \leq \inf\{\langle \Psi, H_N^{\text{in}} \Psi \rangle : \Psi \in D(H_N^{\text{in}}), \|\Psi\| = 1\}. \end{aligned}$$

It remains to show the opposite inequality. Let $\Psi = f_1 \wedge \cdots \wedge f_N$ with $f_1, \ldots, f_N \in D(h)$ orthonormal vectors. Again $\gamma_\Psi =: P$ is a projection onto $\text{span}\{f_1, \ldots, f_N\}$, and as above we find that

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \text{Tr}(\gamma_\Psi h)$$

and the inequality carries to the infimum

$$\inf_{\substack{\Psi \in D(H_N^{\text{in}}) \\ \|\Psi\|=1}} \langle \Psi, H_N^{\text{in}} \Psi \rangle \le \operatorname{Tr}(\gamma_\Psi h)$$

which finally proves the equality and the first part of the claim. Assume now that P is a minimizing projection for the variational problem. It is clear that the state $\Psi = f_1 \wedge \cdots \wedge f_N$ whose 1-pdm coincides with P is a ground state for H_N^{in} if $f_1, \ldots, f_N$ form an orthonormal basis for $P(\mathfrak{h})$.
We may now prove the stated properties for this minimizing projection. Let $\phi \in P(\mathfrak{h})$ and $\psi \in P(\mathfrak{h})^\perp \cap D(h)$ be two normalized states, then

$$\langle \phi, h\phi \rangle \le \langle \psi, h\psi \rangle.$$

Furthermore, consider the projection Q onto the N-dimensional space where ψ and ϕ are swapped, i.e., onto

$$\operatorname{span}(\{\psi\} \cup (P(\mathfrak{h}) \cap \{\phi\}^\perp),$$

and since P is a minimizer it is clear that

$$0 \le \operatorname{Tr}(Qh) - \operatorname{Tr}(Ph) = \langle \psi, h\psi \rangle - \langle \phi, h\phi \rangle.$$

We can use this same argument to show that if $\psi \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$ then $\langle \phi, h\phi \rangle \le \langle \psi, h\psi \rangle$, and we now use this fact to show that h maps $P(\mathfrak{h})$ which leads to the conclusion that $P(\mathfrak{h})$ is indeed spanned by eigenvectors of h.
We proceed by contradiction, so assume that there exists a vector $\phi \in P(\mathfrak{h})$ such that $h\phi \notin P(\mathfrak{h})$. Since $D(h)$ is dense there is a $g \in D(h)$ such that $\langle (\mathbb{1} - P)g, h\phi \rangle = \langle g, (\mathbb{1} - P)h\phi \rangle \ne 0$, and, without loss of generality, we may assume $\operatorname{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \ne 0$. We have that $(\mathbb{1} - P)g \in D(h)$. Consider $t \in \mathbb{R}$ (close to 0) and define

$$\phi_t = \frac{\phi + t(\mathbb{1} - P)g}{\|\phi + t(\mathbb{1} - P)g\|}.$$

Note that $\phi_t \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$, and therefore $\langle \phi, h\phi \rangle \le \langle \phi_t, h\phi_t \rangle$. But this is a contradiction, since

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \langle \phi_t, h\phi_t \rangle = \langle (\mathbb{1} - P)g, h\phi \rangle + \langle \phi, h(\mathbb{1} - P)g \rangle = 2\operatorname{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \ne 0.$$

□

Before going over the next section there is an important generalization to be made. The second quantization of the one-body operator of Theorem 2.2.7 contained only the term $a_\pm^*(f)a_\pm(g)$, and it is easy to see that such an operator preserves the number of particles, as a^*a maps an N-particle sector of the Fock space to itself. It comes quite natural, however, to ask whether we can still say something about terms of the form $a_\pm^*(f)a_\pm^*(g)$ and $a_\pm(f)a_\pm(g)$, since they all belong to the family of operators $a_\pm^\# a_\pm^\#$ and they might as well share some nice properties. This, and other questions, can be answered by appropriately generalizing the one-particle density matrix, and the aim of this section is precisely to introduce this generalization.

Definition 2.3.7 (Generalized 1-pdm)**.** Given a normalized vector $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ with finite particle number expectation, i.e., $\langle \Psi, \mathcal{N}\Psi \rangle < \infty$, we define the *generalized one-particle density matrix* as the positive semi-definite operator $\Gamma_\Psi : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ defined by

$$\langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1))\Psi \rangle_{\mathcal{F}^{\text{F,B}}}$$

where J is the canonical isomorphism between $\mathfrak{h}$ and $\mathfrak{h}^*$ (see Appendix A.1.

Given the normalized vector Ψ we may write Γ_Ψ in block-matrix form by using an auxiliary operator, $\alpha_\Psi : \mathfrak{h}^* \to \mathfrak{h}$, that "measures" how far the state Ψ is from having a fixed particle number. The operator is defined by

$$\langle f, \alpha_\Psi J g\rangle = \langle \Psi, a_\pm(g) a_\pm(f)\Psi\rangle$$

for all $f, g \in \mathfrak{h}$. Note that this map has a useful property, which essentially relates its adjoint to the corresponding operator $a_\pm^*(f)a_\pm^*(g)$, that reads

$$\alpha_\Psi^* = \pm J\alpha_\Psi J,$$

as can be shown by noting that, using the CAR or the CCR,

$$\langle \alpha_\Psi^* f, Jg\rangle = \langle f, \alpha_\Psi J g\rangle = \pm\langle g, \alpha_\Psi J f\rangle = \pm\langle J\alpha_\Psi J f, Jg\rangle.$$

With this operator we come to the following lemma.

Lemma 2.3.8. *Given a normalized state $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ with finite particle number expectation, its generalized one-particle density matrix can be written in the block-matrix form as*

$$\Gamma_\Psi = \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix}$$

Proof. It suffices to prove that, as a quadratic form on $\mathfrak{h} \oplus \mathfrak{h}^*$, this matrix reproduces exactly Definition 2.3.7. Let $f_1, g_1, f_2, g_2 \in \mathfrak{h}$, then it holds that

$$\begin{aligned}
&\begin{pmatrix} f_1 & Jg_1 \end{pmatrix} \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix} \begin{pmatrix} f_2 \\ Jg_2 \end{pmatrix} = \begin{pmatrix} f_1 & Jg_1 \end{pmatrix} \begin{pmatrix} \gamma_\Psi f_2 + \alpha_\Psi J g_2 \\ \pm J\alpha_\Psi J f_2 + (\mathbb{1} \pm J\gamma_\Psi J^*) J g_2 \end{pmatrix} \\
&= \langle f_1, \gamma_\Psi f_2\rangle + \langle f_1, \alpha_\Psi J g_2\rangle + \langle + \langle Jg_1, \alpha_\Psi^* f_2\rangle + \langle Jg_1, (\mathbb{1} \pm J\gamma_\Psi J^*) J g_2\rangle \\
&= \langle a_\pm^*(f_2) a_\pm(f_1)\rangle + \langle a_\pm(g_2) a_\pm(f_1)\rangle + \langle a_\pm^*(f_2) a_\pm^*(g_1)\rangle + \langle a_\pm(g_2) a_\pm^*(g_1)\rangle \\
&= \langle (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1))\rangle \\
&= \langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2\rangle.
\end{aligned}$$

□

The usefulness of this new object is that it allows for a very simple way to write this quantity in a way that closely resembles the standard one-particle density matrix. We just need to define the generalized creation and annihilation operators to see this.

Definition 2.3.9. Let $f, g \in \mathfrak{h}$ be one-particle states. The generalized annihilation and creation operators are defined, respectively, by the maps

$$\begin{aligned}
A_\pm(f \oplus Jg) &\coloneqq a_\pm(f) + a_\pm^*(g) \\
A_\pm^*(f \oplus Jg) &\coloneqq a_\pm^*(f) + a_\pm(g).
\end{aligned}$$

Remark 15. Once again we have that the generalized annihilation operators are conjugate linear from $\mathfrak{h} \oplus \mathfrak{h}^*$ to operators on $\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$. Furthermore, we have the following relation between these generalized operators. For all $F \in \mathfrak{h} \oplus \mathfrak{h}^*$ it holds that

$$A_\pm^*(F) = A_\pm(\mathcal{J}F), \quad \text{where } \mathcal{J} = \begin{pmatrix} 0 & J^* \\ J & 0 \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$$

Proposition 2.3.10 (CCR/CAR for generalized creation and annihilation operators)**.** *For all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have the following the canonical (anti-)commutation relations. For bosons*

$$[A_+(F_1), A_+^*(F_2)] = \langle F_1, \mathcal{S}F_2\rangle_{\mathfrak{h}\oplus\mathfrak{h}^*} \quad \text{where } \mathcal{S} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*,$$

and for fermions

$$\{A_-(F_1), A_-^*(F_2)\} = \langle F_1, F_2 \rangle_{\mathfrak{h}\oplus\mathfrak{h}^*}.$$

More compactly we could introduce the natural notation $[A_\pm(F_1), A_\pm^*(F_2)]_\mp = \langle F_1, \mathcal{S}_\pm F_2\rangle$, *where we define* $\mathcal{S}_-$ *as simply being the identity, and the sign on the square brackets indicates a commutator (-) or an anticommutator (+).*

Proof. The proof is trivial, as it only requires to expand the expression for the (anti-)commutator and to use explicitly the CCR/CAR. □

Remark 16. Note that, in general, the (anti-)commutators $[A_\pm(F), A_\pm(G)]_\mp$ and $[A_\pm^*(F), A_\pm^*(G)]_\mp$ need not be zero.

The special case where $F = f \oplus 0$ reduces to the usual creation/annihilation operators because $A_\pm^\#(f\oplus 0) = a_\pm^\#(f)$, which means that this is by all means an extension of the theory we developed so far. With these operators, the generalized one-particle density matrix can be written in a familiar form. Given, as usual, $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ normalized and for all $F_1, F_2 \in \mathfrak{h}\oplus\mathfrak{h}^*$ we have

$$\langle F_1, \Gamma_\Psi F_2\rangle_{\mathfrak{h}\oplus\mathfrak{h}^*} = \langle \Psi, A_\pm^*(F_1) A_\pm(F_2)\Psi\rangle$$

2.4 Bogolubov transformations and quasi-free pure states

We can finally realize the notion of a "change of modes" for creation/annihilation operators on the space $\mathfrak{h}\oplus\mathfrak{h}^*$. The following definition characterizes a class of isomorphisms on this space as those that, in some sense, preserve the structure of the generalized creation and annihilation operators.

Definition 2.4.1 (Bogolubov transformations)**.** A linear bounded isomorphism $\mathcal{V} : \mathfrak{h}\oplus\mathfrak{h}^* \to \mathfrak{h}\oplus\mathfrak{h}^*$ is called a (bosonic or fermionic) *Bogolubov map* if, for all $F, G \in \mathfrak{h}\oplus\mathfrak{h}^*$

- $A^*(\mathcal{V}F) = A(\mathcal{V}\mathcal{J}F)$;
- $[A_\pm(\mathcal{V}F), A_\pm^*(\mathcal{V}G)]_\mp = \langle F, \mathcal{S}_\pm G\rangle$.

One also refers to the map $A_\pm(F) \longmapsto A_\pm(\mathcal{V}F)$ as a *Bogolubov transformation.*

We can characterize Bogolubov maps by writing them in block-matrix form and noting that each block must satisfy specific properties that come directly from the definition. We first notice that the conditions in the definition can be rewritten as

$$\langle \mathcal{V}F_1, \mathcal{S}_\pm \mathcal{V}F_2\rangle = \langle F_1, \mathcal{S}_\pm F_2\rangle \quad \text{and} \quad \mathcal{J}\mathcal{V}F = \mathcal{V}\mathcal{J}F$$

for all $F, F_1, F_2 \in \mathfrak{h}\oplus\mathfrak{h}^*$. Moreover, since we are assuming that $\mathcal{V}$ is invertible, we can also write

$$\mathcal{V}^{-1} = \mathcal{S}_\pm \mathcal{V}^* \mathcal{S}_\pm.$$

In particular, $\mathcal{V}^{-1} = \mathcal{V}^*$ in the fermionic case. With these properties we immediately get to the following proposition, for which the proof can be read through what we just said:

Proposition 2.4.2. *A linear map* $\mathcal{V}$ *on* $\mathfrak{h}\oplus\mathfrak{h}^*$ *is a Bogolubov map if, and only if, the following conditions hold:*

$$\mathcal{V}^*\mathcal{S}_\pm\mathcal{V} = \mathcal{S}_\pm, \quad \mathcal{V}\mathcal{S}_\pm\mathcal{V}^* = \mathcal{S}_\pm, \quad \mathcal{J}\mathcal{V}\mathcal{J} = \mathcal{V}. \tag{2.6}$$

If we take bounded linear maps $U : \mathfrak{h} \to \mathfrak{h}$ and $V : \mathfrak{h} \to \mathfrak{h}^*$ we can finally write a Bogolubov map in block-matrix form

$$\mathcal{V} = \begin{pmatrix} U & J^*VJ^* \\ V & JUJ^* \end{pmatrix} \tag{2.7}$$

and the conditions (2.6) can then be read as

$$\begin{cases} U^*U = \mathbb{1} \pm V^*V \\ UU^* = \mathbb{1} \pm J^*VV^*J \\ JV^*JU = \pm JU^*J^*V \end{cases} .$$

Defining Bogolubov maps is not sufficient though, as such a map need not, in general, correspond to a unitary *on the Fock space*. Thus we come to the following theorem which tells us when this is possible, and which we will use to define an important class of states in a Fock space for which Wick's theorem, which we will come to shortly, will hold in its most general form.

Theorem 2.4.3 (Unitary implementability of Bogolubov maps)**.** *If $\mathcal{V} : \mathfrak{h}\oplus\mathfrak{h}^* \to \mathfrak{h}\oplus\mathfrak{h}^*$ is a Bogolubov map of the form (2.7) then there exists a unitary map*

$$\mathbb{U}_{\mathcal{V}} : \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h}) \to \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$$

such that, for $F \in \mathfrak{h} \oplus \mathfrak{h}^*$,

$$\mathbb{U}_{\mathcal{V}}^* A_{\pm}(F)\mathbb{U}_{\mathcal{V}} = A_{\pm}(\mathcal{V}F)$$

if, and only if, V^*V is trace-class. This trace-class condition is also referred to as the Shale-Stinespring condition.

What this means is that, under suitable conditions on the block-matrix form of a Bogolubov map, there exists a "change of basis" on the Fock space that sends generalized annihilation and - by taking the adjoint - creation operators to operators that annihilate/create a superposition of states determined by the linear map $\mathcal{V}$. We will not prove this theorem, as it is quite a long endeavor that ultimately doesn't add to our discussion of the physical system at hand.

We are finally ready to discuss the most important result of this chapter: Wick's theorem. This is what, among other things, allows us to write perturbative series in the way described in the next chapter. The content of this important theorem is, concretely, that all the expectations we are interested in reduce to products of the 2-point function. We first look at the case of pure states and then move onto Gibbs states.

Definition 2.4.4 (Quasi-free pure state)**.** A vector $\Psi \in \mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$ is called a *quasi-free pure state* if there exists a unitarily implementable Bogolubov map $\mathcal{V}$ such that $\Psi = \mathbb{U}_{\mathcal{V}}|0\rangle$, where $\mathbb{U}_{\mathcal{V}}$ is the unitary implementation of $\mathcal{V}$.

That is, a pure state is quasi-free if it is the vacuum state under an appropriate choice of physical modes.

Theorem 2.4.5 (Wick's theorem)**.** *If Ψ is a quasi-free pure state and $F_1, \dots, F_{2m} \in \mathfrak{h} \oplus \mathfrak{h}^*$ for some $m \geq 1$, then it holds that*

$$\begin{aligned} &\langle \Psi, A_{\pm}(F_1) \cdots A_{\pm}(F_2 m)\Psi\rangle = \\ &\qquad \sum_{\sigma\in P_{2m}} (\pm 1)^{\sigma} \langle \Psi, A_{\pm}(F_{\sigma(1)})A_{\pm}(F_{\sigma(2)})\Psi\rangle \cdots \langle \Psi, A_{\pm}(F_{\sigma(2m-1)})A_{\pm}(F_{\sigma(2m)})\Psi\rangle \end{aligned} \tag{2.8}$$

and

$$\langle \Psi, A_\pm(F_1)\cdots A_\pm(F_{2m-1})\Psi\rangle = 0$$

where P_{2m} is the set of pairings of the indices $\{1,\dots,2m\}$, i.e.

$$\begin{aligned} P_{2m} = \{\sigma \in S_{2m} : \sigma(2j-1) < \sigma(2j+1),\ j = 1,\dots,m-1 \text{ and}\\ \sigma(2j-1) < \sigma(2j),\ j = 1,\dots,m\}. \end{aligned}$$

Proof. From Ψ being quasi-free we know there exists some Bogolubov map $\mathcal{V}$ with unitary implementation $\mathbb{U}_\mathcal{V}$ such that $\mathbb{U}_\mathcal{V}^* A_\pm(F_i)\mathbb{U}_\mathcal{V} = A_\pm(\mathcal{V}F_i)$, let us call $G_i = \mathcal{V}F_i$. It follows that we may restrict our proof to the case of the vacuum state, as

$$\begin{aligned} \langle \Psi, A_\pm(F_1)\cdots A_\pm(F_{2m})\Psi\rangle &= \langle \Omega, \mathbb{U}_\mathcal{V}^* A_\pm(F_1)\mathbb{U}_\mathcal{V}\mathbb{U}_\mathcal{V}^*\cdots \mathbb{U}_\mathcal{V}\mathbb{U}_\mathcal{V}^* A_\pm(F_{2m})\mathbb{U}_\mathcal{V}\Omega\rangle \\ &= \langle \Omega, A_\pm(G_1)\cdots A_\pm(G_{2m})\Omega\rangle. \end{aligned}$$

Because $G_i \in \mathfrak{h}\oplus\mathfrak{h}^*$ there exist $f_i, g_i \in \mathfrak{h}$ such that

$$A_\pm(G_i) = a_\pm(f_i) + a_\pm^*(g_i) = a_i + a_i^*,$$

where we called $a_i = a_\pm(f_i)$ and $a_i^* = a_\pm(g_i)$. The odd case is straightforward: because we have an odd number of operators $A_\pm(G_i)$, expanding the expression of each one of them will always produce a product with an odd number of either creation or annihilation operators, therefore the product will take the state Ω to an $(N>0)$-particle sector, which is orthogonal to Ω, or directly annihilate it. Thus

$$\langle \Omega, A_\pm(G_1)\cdots A_\pm(G_{2m-1})\Omega\rangle = 0.$$

The even case is more involved and we proceed by induction. Assume that the claim holds for $2m-2$, with the base case $m=2$ being trivially true. Note that the (anti)commutator between an operator a_j and an operator A_k is a scalar times the identity operator:

$$[a_j, A_k]_\mp = [a_j, a_k + a_k^*] = \langle f_j, g_k\rangle \mathbb{1} = \langle \Omega, a_j A_k \Omega\rangle \mathbb{1} =: C_{jk}\mathbb{1}.$$

We expand the expression for the first operator in the product, calling $A_i = A_\pm(G_i)$, to get

$$\langle \Omega, (a_1 + a_1^*)A_2\cdots A_{2m}\Omega\rangle = \langle \Omega, A_2 a_1 A_3\cdots A_{2m}\Omega\rangle \pm C_{12}\langle \Omega, A_2A_3\cdots A_{2m}\Omega\rangle,$$

where the a_1^* term vanishes as it annihilates the vacuum on the left side of the inner product. Repeating this procedure we get all $2m-1$ terms where the k-th operator has been replaced with the scalar C_{1k}, i.e., all pairings that involve the operator A_1:

$$\langle \Omega, A_1\cdots A_{2m}\Omega\rangle = \sum_{k=2}^{2m} (\pm)^k C_{1k}\langle \Omega, A_2\cdots \widehat{A_k}\cdots A_{2m}\Omega\rangle,$$

where the hat symbol means omission. By the inductive hypothesis, the claim holds for the product of operators from A_2 to A_{2m}, and whenever the j-th operator is “taken out” from the product, all the pairings between A_j and an $A_{k>j}$ are produced. Thus we arrive at the thesis

$$\langle \Omega, A_1\cdots A_{2m}\Omega\rangle = \sum_{\sigma\in P_{2m}} (\pm)^\sigma \langle \Omega, A_{\sigma(1)}A_{\sigma(2)}\Omega\rangle\cdots\langle \Omega, A_{\sigma(2m-1)}A_{\sigma(2m)}\Omega\rangle.$$

□

We may now extend the definition of quasi-free pure states to include non-pure ones, and we may as well use Wick's theorem as the defining property of quasi-free expectations. Surprisingly enough, Gibbs states are quasi-free, as illustrated by the next theorem.

Definition 2.4.6 (General quasi-free states)**.** A *quasi-free state* is a state $\rho \in \mathfrak{B}^*(\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h}))$ that satisfies Wick's theorem, i.e., such that Equation (2.8) holds when replacing pure expectations $\langle \Psi, \cdot\, \Psi \rangle$ with mixed expectations $\langle \cdot \rangle_\rho$.

Theorem 2.4.7. *Let $\{F_n\}_{n\in\mathbb{N}}$ be an orthonormal basis for $\mathfrak{h} \oplus \mathfrak{h}^*$ and define the Hamiltonian $H = \sum_{n=0}^{\infty} \epsilon_n A_\pm^*(F_n) A_\pm(F_n)$, where $\mathrm{Tr}(e^{-\beta H}) = \sum_n e^{-\beta\epsilon_n} < \infty$. The Gibbs state $\langle \cdot \rangle_{\beta,H}$ is quasi-free.*

Sketch of the Proof. We give a sketch of the proof, omitting details about why some algebraic manipulations are well defined in our case. Because $F_i \in \mathfrak{h} \oplus \mathfrak{h}^*$ there exist $f_i, g_i \in \mathfrak{h}$ such that $A_\pm(F_i) = a_\pm(f_i) + a_\pm^*(g_i)$, so let us call $a_i = a_\pm(f_i)$ and $a_i^* = a_\pm^*(g_i)$. We first note the following identities. By making use of the commutator $[H, a_k] = -\epsilon_k a_k$, we can write (we actually derive this fact in a special case in Lemma 4.1.2)

$$e^{\beta H} a_k e^{-\beta H} = e^{-\beta\epsilon_k} a_k.$$

Then, by the cyclic property of the trace we have, for all bounded operators X,

$$\langle a_k X \rangle_{\beta,H} = \frac{\mathrm{Tr}(a_k X e^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})} = \frac{\mathrm{Tr}(X(e^{-\beta H} e^{\beta H}) e^{-\beta H} a_k)}{\mathrm{Tr}(e^{-\beta H})} = e^{-\beta\epsilon_k} \frac{\mathrm{Tr}(X a_k e^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})} = e^{-\beta\epsilon_k} \langle X a_k \rangle_{\beta,H}.$$

Using this cyclic property of Gibbs expectations, we can move the first operator all the way on the right and then "commute" it all the way back to the left. Calling $A_i = A_\pm(F_i)$, we proceed by induction. Let $n \in \mathbb{N}$ and assume that the claim holds for $n-1$. Then

$$\begin{aligned}\langle A_1 \cdots A_n \rangle_{\beta,H} &= e^{-\beta\epsilon_1} \langle A_2 \cdots A_n A_1 \rangle_{\beta,H} \\ &= e^{-\beta\epsilon_1} \left\langle (\pm)^{n-1} + \sum_{j=2}^{n} (\pm)^{n-j} [A_j, A_1]_\mp A_2 \cdots \widehat{A_j} \cdots A_n \right\rangle_{\beta,H}.\end{aligned}$$

We may factor the term on the left side with the last term of the right side to get that, recalling that we may treat $[A_j, A_k]$ as a scalar,

$$(1 - (\pm)^{k-1} e^{-\beta\epsilon_1}) \langle A_1 \cdots A_n \rangle_{\beta,H} = e^{-\beta\epsilon_1} \sum_{j=2}^{n} (\pm)^{n-j} [A_j, A_1]_\mp \langle A_2 \cdots \widehat{A_j} \cdots A_n \rangle_{\beta,H}.$$

In the special case $n = 2$ we have that

$$(1 \mp e^{-\beta\epsilon_1}) \langle A_1 A_2 \rangle_{\beta,H} = e^{-\beta\epsilon_1} [A_j, A_1]_\mp,$$

which implies, substituting it back into the equation, that

$$\langle A_1 \cdots A_n \rangle_{\beta,H} = e^{-\beta\epsilon_1} \sum_{j=2}^{n} (\pm)^{n-j} \langle A_j A_1 \rangle_{\beta,H} \langle A_2 \cdots \widehat{A_j} \cdots A_n \rangle_{\beta,H}.$$

By the inductive hypothesis, the expectation of $A_2 \cdots A_n$ produces all the pairings that do not involve A_1, and thus we arrive at the thesis. If n is odd, every term will contain an expectation of a single operator, which is zero as can be seen by the relation

$$\begin{aligned}&\langle A_1 \rangle_{\beta,H} = \langle A_1 \mathbb{1} \rangle_{\beta,H} = e^{-\beta\epsilon_1} \langle A_1 \rangle_{\beta,H} \\ &\implies (1 - e^{-\beta\epsilon_1}) \langle A_1 \rangle_{\beta,H} = 0.\end{aligned}$$

If n is even, we instead produce all possible pairings, just as we did in Theorem 2.4.5:

$$\langle A_1 \cdots A_n \rangle = \sum_{\sigma \in P_n} (\pm)^\sigma \langle A_{\sigma(1)} A_{\sigma(2)} \rangle_{\beta,H} \cdots \langle A_{\sigma(n-1)} A_{\sigma(n)} \rangle_{\beta,H}.$$

This concludes the proof. □

2.5 Quadratic Hamiltonians

Having introduced all the necessary results regarding Bogolubov maps, we can now move onto the study of generic quadratic Hamiltonians, which can be easily expressed in terms of generalized creation and annihilation operators. They play a crucial role in perturbative calculations, which are all about dealing with complicated interactions on top of an easy and well-defined "free theory", i.e., a theory whose n-point functions satisfy Wick's theorem.
We will not provide proofs for the following results, as they serve the purpose of giving a more complete picture of the tools at hand rather than being of fundamental importance to us.

Definition 2.5.1 (Quadratic Hamiltonian)**.** Let $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ be a self-adjoint, trace-class operator and assume moreover in the bosonic case that it is positive semi-definite. Given an orthonormal basis $\{F_1, \ldots, F_{2M}\}$ for $\mathfrak{h} \oplus \mathfrak{h}^*$, the operator

$$H^\pm_{\mathcal{A}} = \sum_{i,j=1}^{2M} \langle F_i, \mathcal{A} F_j \rangle A^*_\pm(F_i) A_\pm(F_j),$$

is called the *quadratic Hamiltonian associated to* $\mathcal{A}$.

Proposition 2.5.2. *If $H^\pm_{\mathcal{A}}$ is self-adjoint, than the above definition is independent of the choice of basis for $\mathfrak{h} \oplus \mathfrak{h}^*$.*

Theorem 2.5.3 (Variational principle for quadratic Hamiltonians)**.** *If $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ is self-adjoint, trace class (and positive semi-definite in the bosonic case) and $E^{H^\pm_{\mathcal{A}}}$ denotes the true ground state energy of $H^\pm_{\mathcal{A}}$, then it holds that*

$$E^{H^\pm_{\mathcal{A}}} = \inf \left\{ \langle \Psi, H^\pm_{\mathcal{A}} \Psi \rangle : \Psi \in \mathcal{F}^{\mathrm{F,B}} \textit{ is a quasi-free pure state} \right\}.$$

Theorem 2.5.4 (Ground state eigenvector for quadratic Hamiltonians)**.** *Let $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ be self-adjoint. Assume moreover, in the bosonic case, that it is positive definite. Then the Hamiltonian $H^\pm_{\mathcal{A}}$ admits a ground state eigenvector that is a quasi-free pure state. In the bosonic case, this eigenvector is also unique in the class of quasi-free pure states.*

Theorem 2.5.5 (Bogolubov transformations of quadratic Hamiltonians)**.** *If $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map and $\mathbb{U}_{\mathcal{V}}$ its unitary implementation on $\mathcal{F}^{\mathrm{F,B}}(\mathfrak{h})$, then for all self-adjoint $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \to \mathfrak{h} \oplus \mathfrak{h}^*$ it holds that*

$$\mathbb{U}^*_{\mathcal{V}} H^\pm_{\mathcal{A}} \mathbb{U}_{\mathcal{V}} = H^\pm_{\mathcal{V}\mathcal{A}\mathcal{V}^*}$$

The definition of quadratic Hamiltonians naturally includes kinetic terms that contain the so-called *anomalous* correlators $a_i a_j$ and $a^*_i a^*_j$, which are number-non-preserving terms. Usually, if a Hamiltonian contains such terms, one could try to find a Bogolubov transformation that takes it to the above form.

Example 1. An important example of such a procedure is the particle-hole transformation, which also represents an alternative approach to the calculations we carry out in our work. The transformation is as follows: consider a set of momentum states inside a ball of fixed radius k_{F}, the Fermi ball

$B_{\mathrm{F}} = \{k \in \mathbb{Z}^3 : |k| < k_{\mathrm{F}}\}$, then define a *hole* as the annihilation of a state inside B_{F}, and a *particle* as the creation of a state outside B_{F}. This is the result of the Bogolubov transformation R which acts as (calling $a_k^* = a_-^*(\psi_k)$ the creation of a particle in the state ψ_k of definite momentum k)

$$R^* a_k^* R = \begin{cases} a_k^* & \text{if } k \notin B_{\mathrm{F}} \\ a_k & \text{if } k \in B_{\mathrm{F}} \end{cases}$$

and

$$R|\Omega\rangle = \bigwedge_{k \in B_{\mathrm{F}}} \psi_k,$$

i.e., the new vacuum under this Bogolubov transformation is the Fermi sea itself. A particle is then just the excitation of a fermion outside the Fermi ball, and this excitation can recombine – and also exchange momentum – with the hole left inside the Fermi ball.

Chapter 3

Perturbation Theory

In the previous chapter we built the second quantization formalism to be able to introduce, in full generality, this key property that is Wick's theorem. Looking at the Gibbs state functional

$$\langle \cdot \rangle_\beta = \frac{\mathrm{Tr}(\,\cdot\, e^{-\beta H})}{\mathrm{Tr}(e^{-\beta H})}$$

we can imagine expanding the exponential in the numerator as (something similar to) a power series, and consider each term as a contribution at a specific order. Such terms will contain powers of the interaction term, i.e., polynomials of creation/annihilation operators. Then Wick's theorem tells us that at all orders we can write the correct contribution using more and more complicated combinations of the 2-point function. These combinations can be easily found by means of Feynman diagrams, and the aim of this chapter is to outline the inner workings of this approach in the specific case of a fermionic system at thermal equilibrium.

Remark 17. The idea behind the tools we will introduce and the way we will write the perturbative series are the same as the ones used for scattering processes in any QFT. In fact, there is a beautiful connection between the calculations done for such processes and the theory we build below, and it has to do with the link between the time evolution operator of quantum mechanics e^{-iHt} and its imaginary time version $e^{-\tau H}$, which defines a Gibbs state.

The following definitions and theorems are taken, and partially reworked, from lecture notes on quantum statistical mechanics [[4]].

3.1 A toy model: bosonic expectations

We build the theory of Feynman diagrams by first analyzing at a simple model: a classical Hamiltonian given by polynomials of complex variables. We will introduce expectations as Gaussian integrals over such variables and show that they exactly reproduce Wick's theorem, in a sense that will be made clear below. After that, we will interpret this tool as a formal way of writing expectations of bosonic (i.e., *commuting*) operators, and finally introduce an algebraic structure capable of carrying the results to the fermionic (i.e., *anticommuting*) case.

Suppose we are given a classical system of N coupled harmonic oscillators: a linear chain model, with Hamiltonian

$$H_0 = \sum_{j\in\mathbb{Z}_N} \left(\frac{p_j^2}{2} + (q_{j+1} - q_j)^2 \right).$$

It can be shown that there exists a canonical transformation that sends H_0 into

$$H_0 = \sum_{k\in\mathcal{D}} \omega_k z_k^* z_k$$

where $\omega_k = 2\sin(\frac{\pi k}{N})$ and $\mathcal{D} = \{k \in \mathbb{Z} : -\frac{N}{2} \leq k \leq \frac{N+1}{2}\}$. Following the classical theory, the partition function for this system is

$$Z_0(\beta) \propto \int \left(\prod_{k\in\mathcal{D}} \mathrm{d}z_k^* \,\mathrm{d}z_k\right) e^{-\beta H_0} = \prod_{k\in\mathcal{D}} \int \mathrm{d}z_k^* \,\mathrm{d}z_k e^{-\beta\omega_k z_k^* z_k} \propto \prod_{k\in\mathcal{D}} \frac{1}{\beta\omega_k}$$

where, recall, the integral is given by

$$\int \mathrm{d}z^* \,\mathrm{d}z f(z,z^*) := 2i \int_{\mathbb{R}^2} \mathrm{d}x \,\mathrm{d}y f(x+iy, x-iy).$$

Suppose now we want to add anharmonic corrections to the model. We start with a quartic term, and already we notice that exact diagonalization is not possible anymore. Performing the canonical transformation into the variables z_k and z_k^* leads to

$$\begin{aligned} H = H_0 + V &= \sum_{j\in\mathbb{Z}_N} \left(\frac{p_j^2}{2} + (q_{j+1}-q_j)^2 + \frac{\lambda}{4}(q_{j+1}-q_j)^4\right) \\ &= \sum_{k\in\mathcal{D}} \omega_k z_k^* z_k + \lambda \sum_{k\in\mathcal{D}} \Omega_{p,q,k} z_{q+k}^* z_{p-k}^* z_p z_q. \end{aligned}$$

Now the partition function reads as

$$Z(\beta,\lambda) = \int \left(\prod_{k\in\mathcal{D}} \mathrm{d}z_k^* \,\mathrm{d}z_k\right) e^{-\beta H_0} e^{-\beta\lambda V}, \tag{3.1}$$

and this is not exactly computable. If we suppose that $\lambda \ll 1$, we can expand the exponential of the correction term as a power series and truncate $Z(\beta,\lambda)$ to the desired order. For example

$$\begin{aligned} Z(\beta,\lambda) &= \int \left(\prod_{k\in\mathcal{D}} \mathrm{d}z_k^* \,\mathrm{d}z_k\right) e^{-\beta H_0} \left(1 - \lambda\beta V + \mathcal{O}(\lambda^2)\right) \\ &= Z_0(\beta) \left(1 - \lambda\beta \frac{\int \left(\prod_{k\in\mathcal{D}} \mathrm{d}z_k^* \,\mathrm{d}z_k\right) e^{-\beta H_0} V}{Z_0(\beta)} + \mathcal{O}(\lambda^2)\right). \end{aligned}$$

Looking at this first order correction, we could interpret the expression

$$\frac{\left(\prod_{k\in\mathcal{D}} \mathrm{d}z_k^* \,\mathrm{d}z_k\right) e^{-\beta H_0}}{Z_0(\beta)} =: P_0(\mathrm{d}z)$$

as some Gaussian probability measure, and correspondingly interpret $\int P_0(\mathrm{d}z)V$ as a kind of average of V. Higher order corrections can also be interpreted as higher moments of V with respect to this Gaussian probability measure.

We now want to show that this kind of expectation satisfies Wick's theorem, or Isserlis' theorem as it is known in the field of probability theory. After that it will be clear that the variables z_k^* and z_k may just as well be regarded to as a formal representation of creation/annihilation operators, and higher order corrections to the free partition function $Z_0(\beta)$ are going to be expectations of powers of the interaction potential V. The only obstacle is then the fact that complex Gaussian integrals actually satisfy the *bosonic* version of Wick's theorem, and this is a consequence of z_k^* and z_k being commuting variables rather than anticommuting. The missing fermionic signs will be the subject of Section 3.2.

We start our discussion of bosonic expectations by recalling some properties of complex Gaussian integrals.

Definition 3.1.1 (Complex covariance)**.** A matrix $G \in \mathrm{Mat}_{N\times N}$ is said to be a *complex covariance* if

(i) there exists a matrix U that diagonalizes G;

(ii) all eigenvalues of G have positive real part.

The space of complex covariances on will be denoted by $\mathcal{C}_N$.

To ease the notation, if $\boldsymbol{z}, \boldsymbol{w} \in \mathbb{C}^N$ we will write $\boldsymbol{z}\cdot\boldsymbol{w} = \sum_{j=1}^N z_j w_j$.

Lemma 3.1.2. *Let $G \in \mathcal{C}_N$ and $\boldsymbol{z} \in \mathbb{C}^N$, then*

$$\int \left(\prod_{j=1}^N \frac{\mathrm{d}z_j^*\,\mathrm{d}z_j}{2\pi i} \right) e^{-\boldsymbol{z}\cdot G^{-1}\boldsymbol{z}} = \det(G).$$

Proof. Because $G \in \mathcal{C}_N$ there exists U that diagonalizes G, i.e., there exist ξ such that $\boldsymbol{z} = U\boldsymbol{\xi}$. Then

$$\begin{gathered} \boldsymbol{z}^* \cdot G^{-1}\boldsymbol{z} = \boldsymbol{z}^* \cdot U^\dagger G^{-1} U \boldsymbol{z} =: \boldsymbol{\xi}^* \cdot D^{-1}\boldsymbol{\xi} \\ \mathrm{d}\boldsymbol{z}^*\,\mathrm{d}\boldsymbol{z} = \det(U^\dagger)\det(U)\mathrm{d}\boldsymbol{\xi}^*\,\mathrm{d}\boldsymbol{\xi} = \mathrm{d}\boldsymbol{\xi}^*\,\mathrm{d}\boldsymbol{\xi} \end{gathered}$$

where $D = \mathrm{diag}(\lambda_1, \dots, \lambda_N)$ is the diagonalization of G. Then, the left hand side becomes

$$\text{l.h.s.} = \prod_{j=1}^N \int \frac{\mathrm{d}\xi_j^*\,\mathrm{d}\xi_j}{2\pi i} e^{-\xi^*{}_j \frac{1}{\lambda_j}\xi_j} = \prod_{j=1}^N \int_{\mathbb{R}^2} \frac{\mathrm{d}x\,\mathrm{d}y}{\pi} e^{-\frac{x^2+y^2}{\lambda_j}} = \det(G).$$

□

Definition 3.1.3 (Complex Gaussian measure)**.** Let $G \in \mathcal{C}_N$. The complex integration measure

$$P_G(\mathrm{d}z) := \frac{1}{\det(G)} \int \left(\prod_{j=1}^N \frac{\mathrm{d}z_j^*\,\mathrm{d}z_j}{2\pi i} \right) e^{-\boldsymbol{z}^*\cdot G^{-1}\boldsymbol{z}}$$

is called *complex Gaussian measure* of covariance G.

Remark 18. The matrix G is also called a propagator because, as will be made clear by the theorems we prove, its matrix elements form exactly the 2-point function for a given system.

Note that a complex Gaussian measure is "normalized", in the sense that $\int P_G(\mathrm{d}z) = 1$. We are now ready to define expectation values and prove their fundamental properties.

Definition 3.1.4 (Complex/bosonic expectation)**.** Given $P_G(\mathrm{d}z)$ and a measurable function f, the *expectation of f* with respect to $P_G(\mathrm{d}z)$ is defined as

$$\mathbb{E}_G(f) := \int P_G(\mathrm{d}z) f(\boldsymbol{z}, \boldsymbol{z}^*).$$

Remark 19. In our toy model, we actually have $G = \mathrm{diag}\left(\frac{1}{\beta\omega_1}, \dots, \frac{1}{\beta\omega_N}\right)$ and the first order correction to the partition function was indeed $\mathbb{E}_G(V)$.

Lemma 3.1.5. *Let $G \in \mathcal{C}_N$ and let $\boldsymbol{J} \in \mathbb{C}^N$ be a fixed complex vector, often called a source term. Then*

$$\int P_G(\mathrm{d}z) e^{\boldsymbol{J}^* \cdot \boldsymbol{z} + \boldsymbol{J} \cdot \boldsymbol{z}^*} = e^{\boldsymbol{J}^* \cdot G \boldsymbol{J}}$$

Proof. Let U be the diagonalization matrix for G, let $\boldsymbol{z} = U\boldsymbol{\xi}$ and $\boldsymbol{J} = U\boldsymbol{\eta}$. Then we can rewrite the exponent of the integrand and complete the square:

$$\begin{aligned} -\boldsymbol{z}^* \cdot G^{-1}\boldsymbol{z} + \boldsymbol{z}^* \cdot \boldsymbol{J} + \boldsymbol{z} \cdot \boldsymbol{J}^* &= -\boldsymbol{\xi}^* \cdot D^{-1}\boldsymbol{\xi} + \boldsymbol{\xi}^* \cdot \boldsymbol{\eta} + \boldsymbol{\xi} \cdot \boldsymbol{\eta}^* \\ &= -(\boldsymbol{\xi}^* - D\boldsymbol{\eta}^*) \cdot D^{-1}(\boldsymbol{\xi} - D\boldsymbol{\eta}) + \boldsymbol{\eta}^* \cdot D\boldsymbol{\eta} \\ &= -(\boldsymbol{\xi}^* - D\boldsymbol{\eta}^*) \cdot D^{-1}(\boldsymbol{\xi} - D\boldsymbol{\eta}) + \boldsymbol{J}^* \cdot G\boldsymbol{J} \end{aligned}$$

where, once again, $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ is the diagonalization of G. We can apply the change of variables $\xi_j \mapsto \xi_j + \frac{1}{\lambda_j}\eta_j$ and take the term $J^* \cdot GJ$ out of the integral, but the remaining integral is just $\int P_G(\mathrm{d}\xi) = 1$, and the thesis follows immediately. □

To prove the next theorem we will need the two following identities:

$$\begin{aligned} P_G(\mathrm{d}z) z_j &= \frac{\partial}{\partial J_j^*} \left(P_G(\mathrm{d}z) e^{\boldsymbol{J}^* \cdot \boldsymbol{z} + \boldsymbol{J} \cdot \boldsymbol{z}^*} \right) \Big|_{\boldsymbol{J} = \boldsymbol{J}^* = 0} \\ P_G(\mathrm{d}z) z_j^* &= \frac{\partial}{\partial J_j} \left(P_G(\mathrm{d}z) e^{\boldsymbol{J}^* \cdot \boldsymbol{z} + \boldsymbol{J} \cdot \boldsymbol{z}^*} \right) \Big|_{\boldsymbol{J} = \boldsymbol{J}^* = 0} \end{aligned} \tag{3.2}$$

Theorem 3.1.6 (Isserlis' theorem)**.** *Given $G \in \mathcal{C}_N$, let M be a positive integer and consider two sets of indices: $\alpha_1, \ldots, \alpha_M$ and $\beta_1, \ldots, \beta_M$. Then it holds that*

$$\mathbb{E}_G(z_{\alpha_1} \cdots z_{\alpha_M} z_{\beta_1}^* \cdots z_{\beta_M}^*) = \sum_{\sigma \in S_N} \prod_{j=1}^M G_{\alpha_j, \beta_{\sigma(j)}}$$

where S_M is the set of permutations of M elements.

Proof. Using the identities in (3.2) and Lemma (3.1.5) we can write

$$\mathbb{E}_G(z_{\alpha_1} \cdots z_{\alpha_M} z_{\beta_1}^* \cdots z_{\beta_M}^*) = \frac{\partial^{2M}}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^* \partial J_{\beta_1} \cdots \partial J_{\beta_M}} \left(e^{\boldsymbol{J}^* \cdot G \boldsymbol{J}} \right) \Big|_{\boldsymbol{J} = \boldsymbol{J}^* = 0} = \star$$

Now we can expand the exponential as a power series and derive term by term. In fact, when deriving $2M$ times the n-th term of this series we have the following cases: if $n < M$ we are taking derivatives of order higher than the degree of the polynomial so it gives zero, if $n > M$ deriving will lead to a homogeneous polynomial evaluated at zero so it gives zero, if $n = M$ some terms survive. Indeed, if we look at the derivatives with respect to the J^* we see that

$$\begin{aligned} \frac{\partial^M}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^*} \frac{(\boldsymbol{J}^* \cdot G\boldsymbol{J})^M}{M!} &= \frac{1}{M!} \frac{\partial^M}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^*} \left(\sum_{j,k=1}^M J_{\rho_j}^* G_{\rho_j, \tau_k} J_{\tau_k} \right)^M \\ &= \frac{\partial^{M-1}}{\partial J_{\alpha_2}^* \cdots \partial J_{\alpha_M}^*} \frac{M}{M!} \left(\sum_{j,k=1}^M \delta_{\rho_j, \alpha_1} G_{\rho_j, \tau_k} J_{\tau_k} \right) \left(\sum_{j,k=1}^M J_{\rho_j}^* G_{\rho_j, \tau_k} J_{\tau_k} \right)^{M-1} \\ &= \prod_{\ell=1}^M \sum_{j,k=1}^M \delta_{\rho_j, \alpha_\ell} G_{\alpha_\ell, \tau_k} J_{\tau_k}. \end{aligned}$$

Using this in $\star$ we get that

$$\star = \frac{\partial^M}{\partial J_{\beta_1} \cdots \partial J_{\beta_M}} \left(\prod_{j=1}^{M} \sum_{k=1}^{M} G_{\alpha_j,\tau_k} J_{\tau_k} \right) \Bigg|_{J=J^*=0}$$

and applying Leibniz rule we arrive at the thesis. □

3.2 Grassmann integrals and fermionic expectations

It is time to introduce the anticipated algebraic structure that carries the results found so far to the fermionic case: Grassmann variables. Consider the unital complex Grassmann algebra generated by a set of $2N$ anticommuting vectors:

$$\begin{aligned} &\mathcal{G}_N = \mathrm{span}\{1, \psi_1, \bar{\psi}_1, \ldots, \psi_N, \bar{\psi}_N\} \\ &\{\psi_i, \psi_j\} = 0 \quad \forall i, j = 1, ..., N \\ &\{\psi_i, \bar{\psi}_j\} = 0 \quad \forall i, j = 1, ..., N \\ &\{\bar{\psi}_i, \bar{\psi}_j\} = 0 \quad \forall i, j = 1, ..., N \end{aligned} \tag{3.3}$$

where $\{\psi, \phi\} := \psi\phi + \phi\psi$ is the anticommutator. We need to define Gaussian integrals and expectations of Grassmann variables.
Given $\mathcal{G}_N$ the Grassmann algebra generated by $\{\psi_1, ..., \psi_N, \bar{\psi}_1, ..., \bar{\psi}_N\}$, denote by

$$\begin{aligned} \mathcal{G}_N^- &= \{\phi \in \mathcal{G}_N : [\phi, \psi] = 0 \; \forall \psi \in \mathcal{G}_N\} \\ \mathcal{G}_N^+ &= \{\phi \in \mathcal{G}_N : \{\phi, \psi\} = 0 \; \forall \psi \in \mathcal{G}_N\} \end{aligned}$$

the even (commuting) and odd (anticommuting) subsets of the algebra. It is easy to show that these are subspaces (by linearity of commutators and anticommutators), and one can also show that $\mathcal{G}_N$ can be decomposed as

$$\mathcal{G}_N \cong \mathcal{G}_N^+ \oplus \mathcal{G}_N^-.$$

Odd elements satisfy a really important property. If $\psi \in \mathcal{G}_N^+$ it follows that $\{\psi, \psi\} = 2\psi^2 = 0$. That is, all powers greater than one of an odd element vanish.

Definition 3.2.1 (Functions of Grassmann variables)**.** Given $\mathcal{G}$ a Grassmann algebra, as defined above, and $f : \mathbb{C}^k \to \mathbb{C}$ an analytic function given by its Taylor series centered at zero

$$f(\boldsymbol{z}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{z}^i$$

we define the associated *function of Grassmann variables* as

$$f(\boldsymbol{\psi}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{\psi}^i$$

where $\boldsymbol{\psi} \in \mathcal{G}_N \times ... \times \mathcal{G}_N$ is the list of k Grassmann variables.

Remark 20. If $\psi \in \mathcal{G}_N^+$ it follows that any function of Grassmann variables evaluated in ψ has at most first order terms, as all higher orders will always contain a factor of $\psi^2 = 0$. The most important example to us is the exponential of an odd element:

$$e^\psi = 1 + \psi + \frac{1}{2}\psi^2 + ... = 1 + \psi$$

We now introduce integrals of functions of Grassmann variables. Note that the definition is going to be purely algebraic and, in particular, motivated by our need to reproduce the fermionic version of Wick's theorem. We ask, first of all, for the integral to be a linear operator on a Grassmann algebra. Then, specifying the image of a basis for the algebra will uniquely determine the operator. The specific definition of the image of basis elements is such that a Grassmann integral over all such basis elements extracts the coefficient of the highest degree term of the integrand.

Definition 3.2.2 (Grassmann integral)**.** Given $\{1, \psi_1, ..., \psi_N, \bar{\psi}_1, ..., \bar{\psi}_N\}$ the anticommuting basis for a Grassmann algebra, the *Grassmann integral* is the linear operator that acts on this basis as

$$\int \mathrm{d}\psi_i \, \psi_j = \int \mathrm{d}\bar{\psi}_i \, \bar{\psi}_j := \delta_{ij},$$
$$\int \mathrm{d}\psi_i = \int \mathrm{d}\bar{\psi}_i := 0.$$

Repeated integration is denoted by a single integral symbol and multiple "differentials".

Example 2. In $\mathcal{G}_2$, consider the exponential function. Using the definition, we can evaluate the following integrals to see how this operator behaves.

- $\int \mathrm{d}\psi_1 \, e^{\psi_1 \bar{\psi}_2} = \int \mathrm{d}\psi_1 \, (1 + \psi_1 \bar{\psi}_2) = \bar{\psi}_2$;
- $\int \mathrm{d}\bar{\psi}_2 \, e^{\psi_1 \bar{\psi}_2} = \int \mathrm{d}\bar{\psi}_2 \, (1 - \bar{\psi}_2 \psi_1) = -\psi_1$;
- $\int \mathrm{d}\psi_1 \mathrm{d}\bar{\psi}_2 \, e^{\psi_1 \bar{\psi}_2} = -1$.

Remark 21. Repeated integration, i.e., the composition of multiple integral operators, inherits some form of anticommutativity when acting on the basis elements. Indeed in $\mathcal{G}_1$, for example, we have

$$\int \mathrm{d}\psi \mathrm{d}\bar{\psi} \, \psi\bar{\psi} = -\int \mathrm{d}\psi \mathrm{d}\bar{\psi} \, \bar{\psi}\psi = -1 = -\int \mathrm{d}\bar{\psi} \mathrm{d}\psi \, \psi\bar{\psi}. \tag{3.4}$$

And given the definition of Grassmann functions, this property will be quite useful to us.

Having defined integration, we can now introduce a *Gaussian measure*, again in a purely algebraic sense. To make the notation lighter, we define a "dot product", without any geometric meaning, as $\boldsymbol{\psi} \cdot \boldsymbol{\phi} := \sum_{i=1}^{k} \psi_i \phi_i$.

Definition 3.2.3 (Grassmann covariance)**.** A *Grassmann covariance* is an invertible complex matrix $G \in \mathrm{GL}(N, \mathbb{C})$.

Lemma 3.2.4. *Let G be a Grassmann covariance, and let $\boldsymbol{\psi} = (\psi_1, \dots, \psi_N)$ and $\bar{\boldsymbol{\psi}} = (\bar{\psi}_1, \dots, \bar{\psi}_N)$. Then*

$$\int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} = \det(G^{-1})$$

Proof. We carry out the calculation explicitly.

$$\begin{aligned}
\int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} &= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{(-1)^N}{N!} \left(\sum_{i,j=1}^{N} \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\
&= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{1}{N!} \left(\sum_{i,j=1}^{N} \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\
&= \int \left(\prod_{i=1}^{N} \mathrm{d}\bar{\psi}_i \mathrm{d}\psi_i \right) \frac{1}{N!} \sum_{\sigma, \tau \in \mathrm{S}_N} \prod_{j=1}^{N} (G^{-1})_{\sigma(j), \tau(j)} \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = \star
\end{aligned}$$

Where S_N is the set of permutations of N elements. We can split the product into a product of matrix elements times a product of Grassmann variables, then we use that

$$\prod_{j=1}^{N} \bar{\psi}_{\sigma(j)}\psi_{\tau(j)} = (-1)^{\mathrm{sgn}(\sigma)+\mathrm{sgn}(\tau)} \prod_{j=1}^{N} \bar{\psi}_j\psi_j$$

to find that

$$\begin{aligned}\star &= \frac{1}{N!} \sum_{\sigma,\tau\in \mathrm{S}_N} (-1)^{\mathrm{sgn}(\sigma)+\mathrm{sgn}(\tau)} \prod_{j=1}^{N} (G^{-1})_{\sigma(j),\tau(j)} \\ &= \frac{1}{N!} \sum_{\sigma\in \mathrm{S}_N} (-1)^{\mathrm{sgn}(\sigma)} \prod_{j=1}^{N} (G^{-1})_{j,\sigma j} \\ &= \det(G^{-1})\end{aligned}$$

□

Definition 3.2.5 (Grassmann Gaussian measure)**.** Let G be a Gaussian covariance. A *Grassmann Gaussian measure* is the Grassmann integral measure defined by

$$P_G(\mathrm{d}\psi) := \frac{1}{\det(G^{-1})} \left(\prod_{j=1}^{N} \mathrm{d}\bar{\psi}_j \mathrm{d}\psi_j \right) e^{-\bar{\boldsymbol{\psi}}\cdot G^{-1}\boldsymbol{\psi}}$$

where ψ_i and $\bar{\psi}_i$ are the usual anticommuting Grassmann variables.

Remark 22. The factor in front of the differentials guarantees a “normalized” measure, i.e., $\int P_G(\mathrm{d}\psi) = (1/\det(G^{-1}))\det(G^{-1}) = 1$.

We will prove the following statement later.

Lemma 3.2.6. *With the same notation used so far,*

$$\int P_G(\mathrm{d}\psi)\psi_i\bar{\psi}_j = G_{ij}$$

The following are useful facts about Grassmann integrals that we give without proof, we just point out that proving these only involves taking good care of the combinatorics that come from the definition of Grassmann integrals, and nothing more.

Lemma 3.2.7. *Let $D \in \mathrm{GL}(2,\mathbb{C})$, $\boldsymbol{\varphi}$ and $\boldsymbol{\eta}$ two vectors of Grassmann variables. Then, under the following change of variables*

$$\begin{pmatrix} \boldsymbol{\psi} \\ \bar{\boldsymbol{\psi}} \end{pmatrix} = D \begin{pmatrix} \boldsymbol{\varphi} \\ \bar{\boldsymbol{\varphi}} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\eta} \\ \bar{\boldsymbol{\eta}} \end{pmatrix}$$

the integral of a function of Grassmann variables f changes as follows

$$\int \left(\prod_i \mathrm{d}\bar{\psi}_i\psi_i \right) f(\boldsymbol{\psi}, \bar{\boldsymbol{\psi}}) = \det(G^{-1}) \int \left(\prod_i \mathrm{d}\bar{\varphi}_i\varphi_i \right) f(D(\boldsymbol{\varphi}, \bar{\boldsymbol{\varphi}}) + (\boldsymbol{\eta}, \bar{\boldsymbol{\eta}}))$$

Lemma 3.2.8. *Let $G \in \mathrm{GL}(N,\mathbb{C})$, then*

$$\int P_G(\mathrm{d}\psi)\, e^{\bar{\boldsymbol{\eta}}\cdot\boldsymbol{\psi}+\bar{\boldsymbol{\psi}}\cdot\boldsymbol{\eta}} = e^{\bar{\boldsymbol{\eta}}\cdot G\boldsymbol{\eta}}$$

Before defining expectations, we need one more operator on a Grassmann algebra to mimic the behavior of these same integrals done in the complex variables case: derivatives of Grassmann functions. If we are given a polynomial of degree N in N variables, then deriving this polynomial in all of the variables should return the maximal degree term. We will take this as the defining property of derivation, as we defined functions to be nothing more than Taylor series. Now notice that integrating a Grassmann function in all variables (i.e., in all the basis elements) returns exactly the same thing, the coefficient of the maximal degree term.

Definition 3.2.9 (Grassmann derivatives)**.** Let $f(\boldsymbol{\psi}, \bar{\boldsymbol{\psi}})$ be a Grassmann function, the *Grassmann derivatives* of f with respect to the basis elements are the linear operators defined as

$$\frac{\partial}{\partial \psi_i} f := \int \mathrm{d}\psi_i \, f(\boldsymbol{\psi}, \bar{\boldsymbol{\psi}})$$

Proof (Lemma 3.2.6). Recalling the "anticommutativity property" of Grassmann integrals from Equation (3.4), it is clear that the same property holds for derivatives too. We can make use of Grassmann derivatives to prove this lemma in a way that is closer to the complex variables case. Given a vector of Grassmann variables $\boldsymbol{\eta}$, we can write something similar to (3.2):

$$\psi_i \bar{\psi}_j e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi}} = -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{-\bar{\boldsymbol{\psi}} \cdot G^{-1} \boldsymbol{\psi} + \bar{\boldsymbol{\eta}} \cdot \boldsymbol{\psi} + \bar{\boldsymbol{\psi}} \cdot \boldsymbol{\eta}} \Big|_{\boldsymbol{\eta} = \bar{\boldsymbol{\eta}} = 0} .$$

We can substitute this in the integral to get that

$$\begin{aligned}
\int P_G(\mathrm{d}\psi) \, \psi_i \bar{\psi}_j &= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \int P_G(\mathrm{d}\psi) e^{\bar{\boldsymbol{\eta}} \cdot \boldsymbol{\psi} + \bar{\boldsymbol{\psi}} \cdot \boldsymbol{\eta}} \Big|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{\bar{\boldsymbol{\eta}} \cdot G \boldsymbol{\eta}} \Big|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} (\bar{\eta}_i G_{ij} \eta_j) \Big|_0 \\
&= G_{ij}
\end{aligned}$$

which is exactly the initial claim. □

We are finally ready to prove that Gaussian Grassmann integrals satisfy Wick's theorem.

Theorem 3.2.10 (Wick's theorem)**.** *Let $M \in \mathbb{N}$ be the degree of a monomial in Grassmann variables, and let $A = \{a_1, \dots, a_M\}$ and $B = \{b_1, \dots, b_M\}$ be two sets of M indices. The integral of a monomial is*

$$\int P_G(\mathrm{d}\psi) \left(\prod_{\alpha \in A, \beta \in B} \psi_\alpha \bar{\psi}_\beta \right) = \sum_{\sigma \in \mathrm{S}_N} (-1)^{\mathrm{sgn}(\sigma)} \prod_{j=1}^{M} G_{a_j, b_{\sigma(j)}}$$

Proof. The proof is almost identical to the one for Theorem (3.1.6), with the only difference being that now we have to be careful with the sign of the permutations. In particular, deriving the M-th term of the exponential carries an additional factor of $(-1)^M$:

$$\text{l.h.s.} = (-1)^M \left(\prod_{\alpha \in A, \beta \in B} \frac{\partial}{\partial \bar{J}_\alpha} \frac{\partial}{\partial J_\beta} \right) e^{\bar{\boldsymbol{J}} \cdot G \boldsymbol{J}} \Big|_{\boldsymbol{J} = \bar{\boldsymbol{J}} = 0} .$$

The rest of the proof follows exactly as in Theorem (3.1.6). □

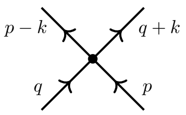
At this point, we can take this notion of Gaussian integrals to write the fermionic version of Section 3.1.

3.3 Perturbative series and Feynman diagrams

In this section we are finally going to look at how the order-by-order corrections introduced by a term like $e^{-\beta\lambda V}$ in Equation (3.1) can, in fact, be written in terms of Feynman diagrams, which are nothing more than a graphical way to use Wick's theorem and produce valid contractions in n-point functions. To do so, we introduce the concept of a truncated expectation. This is the object that ultimately will contain the important diagrams. Both bosonic and fermionic cases follow the same procedure but present some slight differences here and there, which we shall point out when needed.

Suppose we have an interaction potential V, formally written as a homogeneous polynomial of complex or Grassmann variables. We can represent such a potential as a *vertex*, i.e., a node in a graph, with as many labeled outgoing lines as there are conjugate variables in the expression of V, and as many labeled incoming lines as there are plain variables. For example, our quartic correction of Section 3.1 would be the formal way of writing a two-body interaction potential $V = \sum_{p,q,k} \hat{V}(k) a^*_{q+k} a^*_{p-k} a_p a_q$ and its vertex would be

$V =$



.

The *value* of a vertex is defined exactly as the specific expression of V. In a diagram, vertices can be connected by lines called *propagators* or even *contractions*, and as the name suggests the value of such propagators is given by the 2-point function. In particular, if a line labeled i exits a vertex and enters another as a line labeled j, then to such a line we associate the value G_{ij}. A vertex may have an outgoing leg connecting back to an incoming line in the same vertex, and in this work we will call this *a tadpole*. The *value of a diagram* Γ, denoted $\mathrm{val}(\Gamma)$, is obtained by multiplying the values of vertices and propagators following the labeling and contractions dictated by the diagram. The definitions of the vertices and the propagators involved in the diagrams are called *Feynman rules*. Finally, even though we will not make use of such a definition in our work, a Feynman rule may contain many different "types" of incoming/outgoing lines and associated propagators. This means that in the vertex there might be both fermionic and bosonic variables, and some of them may belong to different sets of variables (for example different Grassmann algebras to represent different families of fermions). Each type of line gets its own propagator, and they should be distinguished graphically.
We will compute values of Feynman diagrams explicitly in the next chapter, specifically in Section 4.1, so we refer to those computations as examples of the general rules we just laid out.

We use diagrams to compute expectations. If V is our interaction potential, its expectation $\mathbb{E}_G(V)$ is the sum of all possible Wick contractions, as we proved, i.e., all possible diagrams with one vertex and all lines contracted. Adding more copies of V inside the expectation means adding vertices, while multiplying by other expectations means multiplying the values of the respective sums of diagrams. We will prove that the most important diagrams are actually the connected ones.

Definition 3.3.1 (Truncated expectation)**.** In the bosonic case, let $P_G(\mathrm{d}z)$ be a complex Gaussian measure, and let $V_1, \dots, V_s$ be s polynomials complex variables. We define the *truncated expectation* of $V_1, \dots, V_s$ as

$$\mathbb{E}^{\mathrm{T}}_G(V_1; \dots; V_s) \coloneqq \lim_{R\to\infty} \frac{\partial^s}{\partial t_1 \cdots \partial t_s} \log\left(\int P_G(\mathrm{d}z) e^{\sum_{j=1}^s t_j V_j(\boldsymbol{z},\boldsymbol{z}^*)} \chi(|z| < R) \right)\Bigg|_{\boldsymbol{t}=0}.$$

In the fermionic case, let $P_G(\mathrm{d}\psi)$ be a Grassmann Gaussian measure, and let $V_1, \dots, V_s$ be s polynomials of Grassmann variables. We define the *truncated expectation* of $V_1, \dots, V_s$ as

$$\mathbb{E}^{\mathrm{T}}_G(V_1; \dots; V_s) \coloneqq \frac{\partial^s}{\partial t_1 \cdots \partial t_s} \log\left(\int P_G(\mathrm{d}\psi) e^{\sum_{j=1}^s t_j V_j(\boldsymbol{\psi},\bar{\boldsymbol{\psi}})} \right)\Bigg|_{\boldsymbol{t}=0}.$$

We denote $\mathbb{E}_G^{\mathrm{T}}(V;r)$ the truncated expectation of r copies of the same polynomial V. Given a multiindex $I=(I_1,\dots,I_n)$, we will denote $\mathbb{E}_G^{\mathrm{T}}(V_I)=\mathbb{E}_G^{\mathrm{T}}(V_{I_1},\dots,V_{I_n})$

Remark 23. Note that the special case where $s=1$ reads as $\mathbb{E}_G^{\mathrm{T}}(V;1)=\mathbb{E}_G(V)$.

Proposition 3.3.2 (Invariance of exponentials)**.** *Let V be a polynomial of complex (or Grassmann) variables, and $P_G(\mathrm{d}z)$ (or $P_G(\mathrm{d}\psi)$) be a complex (or Grassmann) Gaussian measure. Then*

$$\int P_G(\mathrm{d}z)e^{V(\boldsymbol{z},\boldsymbol{z}^*)}=e^{\sum_{j=0}^{\infty}\frac{1}{j!}\mathbb{E}_G^{\mathrm{T}}(V;j)}.$$

Proof. We will compute the logarithm of the left hand side in the complex case, and the thesis will follow immediately after. The Grassmann case is completely analogous and doesn't require any modification. To compute the logarithm we take a Taylor expansion around zero, and note that in the Grassmann case this just coincides with the definition of the integrand.

$$\begin{aligned}\log\int P_G(\mathrm{d}z)e^{V(\boldsymbol{z},\boldsymbol{z}^*)}&=\log\int P_G(\mathrm{d}z)e^{tV(\boldsymbol{z},\boldsymbol{z}^*)}\big|_{t=1}\\&=\sum_{j=0}^{\infty}\frac{1}{j!}\frac{\mathrm{d}^j}{\mathrm{d}s^j}\log\left(\log\int P_G(\mathrm{d}z)e^{tV(\boldsymbol{z},\boldsymbol{z}^*)}\right)^j\Bigg|_{s=0}t^j\Bigg|_{t=1}=\star.\end{aligned}$$

But now we notice that this contains exactly the definition of $\mathbb{E}_G^{\mathrm{T}}(V;j)$, thus:

$$\star=\sum_{j=0}^{\infty}\frac{1}{j!}\mathbb{E}_G^{\mathrm{T}}(V;j)t^j\big|_{t=1}=\sum_{j=0}^{\infty}\frac{1}{j!}\mathbb{E}_G^{\mathrm{T}}(V;j)$$

□

Theorem 3.3.3 (Linked cluster theorem)**.** *Let $V_1,\dots,V_s$ be polynomials of complex (or Grassmann) variables. Then*

$$\mathbb{E}_G^{\mathrm{T}}(V_1;\dots;V_s)=\mathbb{E}_G(V_1V_2\cdots V_s)-\sum_{p=2}^{s}\sum_{\{I_1,\dots,I_p\}\in P_p}\prod_{j=1}^{p}\mathbb{E}_G^{\mathrm{T}}(V_{I_j})$$

where P_p is the set of all possible partitions of the set of indices $\{1,\dots,s\}$ into p disjoint subsets.

Proof. We consider the complex case to outline the proof, but the Grassmann case is even simpler, as the characteristic functions that appear below are not needed (we put them in the complex case to manage convergence of the integrals). Consider $r\le s$ indices $i_1,\dots,i_r$, and define the following quantity for $t,R\in\mathbb{R}$:

$$\begin{aligned}\mathbb{E}_{G,t,R}(V_{i_1}\cdots V_{i_r})&:=\frac{\mathbb{E}_G(V_{i_1}\cdots V_{i_r}e^{\sum_{j=1}^{s}t_jV_j(z,z^*)}\chi(|z|<R))}{\mathbb{E}_G(e^{\sum_{j=1}^{s}t_jV_j(z,z^*)}\chi(|z|<R))}\\&=\frac{\int P_G(\mathrm{d}z)\left(\prod_{j=1}^{r}V_{i_j}\right)e^{\sum_{j=1}^{s}t_jV_j(z,z^*)}\chi(|z|<R)}{\int P_G(\mathrm{d}z)e^{\sum_{j=1}^{s}t_jV_j(z,z^*)}\chi(|z|<R)}\end{aligned}$$

And its truncated version

$$\mathbb{E}_{G,t,R}^{\mathrm{T}}(V_{i_1}\cdots V_{i_r})=\frac{\partial^r}{\partial t_1\cdots\partial t_r}\left(\log\int P_G(\mathrm{d}z)e^{\sum_{j=1}^{s}t_jV_j(z,z^*)}\chi(|z|<R)\right).$$

Then $\mathbb{E}_G = \mathbb{E}_{G,0,\infty}$ and $\mathbb{E}_G^{\mathrm{T}} = \mathbb{E}_{G,0,\infty}^{\mathrm{T}}$. The aim is to prove the more general formula in which we use $\mathbb{E}_{G,t,R}$ instead of the regular expectation, and prove the claim by setting $t = 0$ and taking $R \to \infty$. We will proceed by induction, starting from the base case $r = 1$:

$$\begin{aligned}\mathbb{E}_{G,t,R}^{\mathrm{T}}(V_k) &= \frac{\partial}{\partial t_j}\left(\log \int P_G(\mathrm{d}z)e^{\sum_{j=1}^s t_j V_j(z,z^*)}\chi(|z|<R)\right)\\ &= \frac{\int P_G(\mathrm{d}z)V_j(z,z^*)e^{\sum_{j=1}^s t_s V_s(z,z^*)}\chi(|z|<R)}{\int P_G(\mathrm{d}z)e^{\sum_{j=1}^s t_s V_s(z,z^*)}\chi(|z|<R)}\\ &= \mathbb{E}_{G,t,R}(V_k).\end{aligned}$$

We now assume the modified version of the claim holds for $r-1$ and prove that the case r follows from this. The inductive hypothesis reads

$$\mathbb{E}_{G,t,R}^{\mathrm{T}}(V_1;\dots;V_{r-1}) = \mathbb{E}_G(V_1V_2\cdots V_{r-1}) - \sum_{p=2}^{r-1}\sum_{\{I_1,\dots,I_p\}\in P_p}\prod_{j=1}^p \mathbb{E}_{G,t,R}^{\mathrm{T}}(V_{I_j}).$$

Deriving both sides of the equation by t_{i_r} gives us, by definition,

$$\frac{\partial}{\partial t_{i_r}}(\text{l.h.s.}) = \mathbb{E}_{G,t,R}^{\mathrm{T}}(V_{i_1};\dots;V_{i_r})$$

and

$$\frac{\partial}{\partial t_{i_r}}\mathbb{E}_{G,t,R}(V_{i_1},\dots,V_{i_{r-1}}) = (\dots) = \mathbb{E}_{G,t,R}(V_{i_1},\dots,V_{i_r}) - \mathbb{E}_{G,t,R}(V_{i_1},\dots,V_{i_{r-1}})\mathbb{E}_{G,t,R}(V_{i_r}).$$

Then, using the inductive hypothesis, we can rewrite the factor $\mathbb{E}_{G,t,R}(V_{i_1},\dots,V_{i_{r-1}})$ and iteratively make all possible partitions in which i_r is left alone in the expression, concluding the proof. □

Definition 3.3.4 (Connected Feynman diagrams)**.** A Feynman diagram is said to be *connected* if it is a connected graph.

In other words, in a connected Feynman diagrams, every vertex must be accessible to every other by following propagator lines.

Theorem 3.3.5 (Connected Feynman diagrams and trunacted expectations)**.**

$$\mathbb{E}_G^{\mathrm{T}}(V_1;\dots;V_s) = \sum_{\Gamma\in\mathcal{G}_{s,\mathrm{C}}}\mathrm{val}(\Gamma)$$

where $\mathcal{G}_{s,\mathrm{C}}$ is the set of all connected diagrams with s vertices.

Proof. We proceed by induction, with the base case $s = 1$ being trivially true, by construction of $\mathbb{E}_G^{\mathrm{T}}(V;1)$. We will call $\mathcal{G}_{s,p,\mathrm{CC}}$ the set of all connected diagrams with s vertices divided into p connected components. Assuming now the claim holds for the case $s-1$, we can use the linked cluster theorem to see that

$$\begin{aligned}\mathbb{E}_G^{\mathrm{T}}(V_1;\dots;V_s) &= \mathbb{E}_G(V_1;\dots;V_s) - \sum_{p=2}^{s}\sum_{\{I_1,\dots,I_p\}\in P_p}\prod_{j=1}^p \mathbb{E}_G^{\mathrm{T}}(V_{I_j})\\ &= \mathbb{E}_G(V_1;\dots;V_s) - \sum_{p=2}^{s}\sum_{\Gamma\in\mathcal{G}_{s,p,\mathrm{CC}}}\mathrm{val}(\Gamma)\\ &= \sum_{\Gamma\in\mathcal{G}_{s,1,\mathrm{CC}}}\mathrm{val}(\Gamma).\end{aligned}$$

□

In the next chapter we will discuss how we can rewrite the expectations on Gibbs states as a series of expectations of the powers of the interaction potential, and we will see how the linked cluster theorem tells us those expectations are actually *truncated*, with a number of copies of V given by the order at which we decide to truncate the expression. Ultimately what this means is that our computations will involve, order by order, a sum of all connected diagrams with a number of vertices equal to the chosen order, as is expected in diagrammatic perturbation theory.

Chapter 4

Ground State Energy in the Mean Field Limit

CHECK MISSING LINKS IN REFERENCES. Our objective is to choose a mean field scaling regime in the number of particles for both the kinetic and the interaction terms in the Hamiltonian, for a system that has a smooth, two-body interaction potential.

First, let us consider a system whose physical parameters, such as the mass of the fermions, are such that the kinetic energy is simply given by $H_0 = \sum_{i=1}^{N} -\hbar^2 \Delta_i$ and the interaction by $\lambda \sum_{i<j} V(x_i - x_j)$. Second, because of Pauli's exclusion principle, we know that – at least for free fermions – the ground state in momentum space will be a ball centered around the origin whose radius depends on the number of particles, as each momentum state can be occupied by at most one fermion (ignoring the degeneracy introduced by other degrees of freedom, which are not relevant to our discussions). This is the Fermi ball, and in three spatial dimensions the energy of a completely filled Fermi ball scales with the number of particles as $N^{5/3}$ (the Fermi energy ϵ_F scales as $N^{2/3}$, while the energy of the entire systems scales as $\epsilon_\mathrm{F}^{2/5}$ [7, p. 132]). Introducing weak interactions between particles we get that the potential energy scales with N^2, because the number of pair-wise interactions will be $N(N-1)/2 \sim N^2$ [1]. Here, we are assuming that the potential itself V is of order $\mathcal{O}(N^0)$[6], an assumption that can be realized by considering potentials such that the support of their Fourier transform is compact and doesn't depend on N.

To retrieve a sensible, non-trivial limit for the energy as $N \to \infty$, we choose to rescale the physical parameters [1] as

$$\hbar = N^{-1/3}, \quad \lambda = N^{-1} \tag{4.1}$$

where the redefinition of $\hbar$ is to be understood not as a redefinition of a fundamental constant, but rather of a parameter of the system that ultimately represents the semi-classical behavior of this scaling limit. Naming this parameter $\hbar$ is, at least in our setting, just a matter of convenience. With these prescriptions, the RPA ground state energy reproduces the scaling in the number of particles N that has been derived via exact methods in [1] and [6]. In particular, these results show the ground state energy is given by

$$E_N^{\mathrm{GS}} = E_N^{\mathrm{HF}} + E_N^{\mathrm{RPA}} + \mathcal{O}(N^{-\frac{1}{3}-\alpha}) \tag{4.2}$$

where

- E_N^{HF} is the Hartree-Fock approximation for the ground state energy, containing the unperturbed energy of order $\mathcal{O}(N)$, a direct excitation contribution of order $\mathcal{O}(N)$ and a momentum exchange contribution of order $\mathcal{O}(N^0)$;

- E_N^{RPA} is the direct excitation contribution in the RPA and of order $\mathcal{O}(N^{-1/3})$;
- α is a positive real number that signals how the exchange correction to the RPA is subleading in N with respect to the RPA itself.

In Theorem 4.2.3 – the main result of this thesis – we give an original proof that second order perturbation theory reproduces the expected scaling of each term in Equation (4.2), within the aforementioned scaling regime.

4.1 Explicit calculation at second order

The state space for a system of N identical fermions in a box of side length 2π (for convenience) with periodic boundary conditions is the space of completely antisymmetric L^2-functions on the torus $\mathbb{T}^{3N} = (\mathbb{R}^3/(2\pi\mathbb{Z}^3))^N$:

$$L^2_a(\mathbb{T}^{3N}) := \{\psi \in L^2(\mathbb{T}^{3N}) : \psi(x_{\sigma(1)}, ..., x_{\sigma(N)}) = \mathrm{sgn}(\sigma)\psi(x_1, ..., x_N)\, \forall\sigma \in S_N\}$$

The Hamiltonian for weakly interacting particles can be written as a sum of the kinetic energy terms plus a sum of two-body interaction terms:

$$H = \hbar^2 \sum_{i=1}^{N} (-\Delta_{x_i}) + \lambda \sum_{1\leq i<j\leq N} V(x_i - x_j) \tag{4.3}$$

where we assume $V : \mathbb{R}^3 \to \mathbb{R}$ to be a Schwartz function and its Fourier transform to have compact support. This is motivated by the need to control the convergence of the perturbative series, as we will discuss in detail in the rest of our work. Of course, V has period 2π in all variables, and we assume moreover that $V(x_i - x_j) = V(|x_i - x_j|)$, which implies that $\hat{V}(-k) = \hat{V}(k)$.
Working in the grand canonical picture requires lifting this operator to the fermionic Fock space, and requiring that the ground state energy is attained for a fixed number of particles that is greater than zero means that the Hamiltonian picks up an additional term $-\mu\mathcal{N}$, where μ is the *chemical potential*. We absorb the chemical potential in the new definition of the free Hamiltonian as $H_0 = \sum_k(\hbar^2|k|^2 - \mu)a^*_k a_k$. Putting this all together, the Hamiltonian for this system is written as

$$H = H_0 + \lambda V = \sum_{k\in\mathbb{Z}^3} (\hbar^2|k|^2 - \mu)a^*_k a_k + \sum_{k,p,q\in\mathbb{Z}^3} \lambda\hat{V}(k)a^*_{p-k}a^*_{q+k}a_q a_p$$

where $a^{\#}_k = a^{\#}_-(\psi_k)$ are creation/annihilation operators on the fermionic Fock space that create/annihilate states of definite momentum $k \in \mathbb{Z}^3$:

$$\psi_k(x) = (2\pi)^{-\frac{3}{2}} e^{ik\cdot x} \quad \text{for } x \in \mathbb{T}^3.$$

In the limit of zero temperature for a system of identical fermions, the chemical potential is also a value between the highest energy level occupied and the first excited level. We can therefore express it in terms of the number of particles in the ground state, as this parameter serves as a constraint for such a system through the relation

$$N = \sum_{k\in\mathbb{Z}^3} n_{\mathrm{F}}(\epsilon_k) \xrightarrow[\beta\to\infty]{} \sum_{k\in\mathbb{Z}^3} \chi(\epsilon_k \leq \mu) = |B_{\mathrm{F}}| \tag{4.4}$$

where k_{F} is the Fermi momentum, defined by $\epsilon_{k_{\mathrm{F}}} = \mu$, and $B_{\mathrm{F}} = \{p \in \mathbb{Z}^3 : |p| \leq k_{\mathrm{F}}\}$ is the Fermi ball. We re-derive the Fermi-Dirac distribution in Lemma 4.1.2. Equation (4.4) reduces to a counting problem for which we can use an argument by Gauss, which gives a lower and an upper bound to the number of points on a lattice inside a sphere. Namely, they are the number of unit cubes that are completely and at least partially inside the sphere, respectively. The result of such bounds can be summed up into the estimate given by the volume of the sphere plus an error proportional to the surface CITATION NEEDED:

$$N = |B_{\mathrm{F}}| = \frac{4\pi}{3} k_{\mathrm{F}}^3 + \mathcal{O}(k_{\mathrm{F}}^2)$$

which implies that

$$k_{\mathrm{F}} = \left(\frac{3}{4\pi} N\right)^{\frac{1}{3}} + \mathcal{O}(N^0). \tag{4.5}$$

Since we want to compute the second order contribution to the ground state energy, we will calculate the free energy of the system and take its limit as the temperature goes to zero. The free energy is given by

$$F = -\frac{\log(Z)}{\beta}$$

where $Z = \mathrm{Tr}(e^{-\beta H})$ is the partition function, and we will compute this logarithm perturbatively using Duhamel's formula, the subject of the following proposition. Recall that at thermal equilibrium the expectation value of a bounded operator is given by the Gibbs state functional, which is a quasi-free state, as stated in Theorem 2.4.7. We will take expectations with respect to H_0, so the Gibbs state is given by

$$\langle \cdot \rangle_\beta = \frac{\mathrm{Tr}(\,\cdot\, e^{-\beta H_0})}{Z}.$$

Proposition 4.1.1 (Duhamel's formula)**.** *Given $H = H_0 + \lambda V$ as defined above, the following approximation for the free energy holds formally at second order in λ, up to an overall factor of $-1/\beta$:*

$$\log \mathrm{Tr}\, e^{-\beta H} = \log \mathrm{Tr}\, e^{-\beta H_0} - \beta\lambda \langle V\rangle_\beta + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \left(\langle V(\tau')V(\tau)\rangle_\beta - \langle V(\tau')\rangle_\beta \langle V(\tau)\rangle_\beta\right). \tag{4.6}$$

Proof. Start by rewriting the operator $e^{-\beta H} e^{\beta H_0}$ using the fundamental theorem of calculus:

$$\begin{aligned}
e^{-\beta H} e^{\beta H_0} &= \mathbb{1} + \int_0^\beta \mathrm{d}\tau \big(\frac{\mathrm{d}}{\mathrm{d}t} e^{-tH} e^{tH_0}\big)(\tau) \\
&= \mathbb{1} + \int_0^\beta \mathrm{d}\tau e^{-\tau H}(-H + H_0) e^{\tau H_0} \\
&= \mathbb{1} + \int_0^\beta \mathrm{d}\tau e^{-\tau H} e^{\tau H_0} e^{-\tau H_0}(-\lambda V) e^{\tau H_0} \\
&= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau e^{-\tau H} e^{\tau H_0} V(\tau) = \star
\end{aligned}$$

where we defined $V(\tau) = e^{-\tau H_0} V e^{\tau H_0}$. The integral is well defined as $\tau \mapsto \|e^{-\tau H} e^{\tau H_0} V(\tau)\|$ is a continuous function on the compact domain $[0,\beta]$ and is therefore Lebesgue integrable, which implies that $\tau \mapsto e^{-\tau H} e^{\tau H_0} V(\tau)$ is Bochner integrable. We briefly discuss integration of functions $\mathbb{R}^n \to \mathfrak{B}(\mathcal{H})$ in Appendix B. Now we can see that in the integral we have the same term that appears in the l.h.s. but at a generic temperature τ. Iterating this procedure just once gives us

$$\begin{aligned}
\star &= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau \left(\mathbb{1} + \lambda \int_0^\tau \mathrm{d}\tau' e^{-\tau' H} e^{\tau' H_0} V(\tau')\right) V(\tau) \\
&= \mathbb{1} - \lambda \int_0^\beta \mathrm{d}\tau V(\tau) + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' V(\tau')V(\tau) + \mathrm{R}(\lambda, V, \beta).
\end{aligned}$$

where $\mathrm{R}(\lambda, V, \beta)$ is the remainder of this expression. Giving an estimate for the remainder that also allows to prove the convergence of this iterative process is a whole problem of its own, so we will simply truncate the expression and drop the remainder. Now we can invert this relation to isolate $e^{-\beta H}$ and take the trace. Since the trace is a continuous operator from trace class operators to $\mathbb{R}$ and $Ve^{-\beta H_0}$ is trace class, we can move it inside the integral (see Appendix A.1 and Appendix B) to get:

$$\begin{aligned}
\operatorname{Tr} e^{-\beta H} &= \operatorname{Tr} e^{-\beta H_0} - \lambda \int_0^\beta \mathrm{d}\tau \operatorname{Tr}(V(\tau) e^{-\beta H_0}) + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \operatorname{Tr}(V(\tau')V(\tau)e^{-\beta H_0}) \\
&= (\operatorname{Tr} e^{-\beta H_0}) \left(1 - \lambda \int_0^\beta \mathrm{d}\tau \, \langle V(\tau) \rangle_\beta + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \, \langle (V(\tau')V(\tau) \rangle_\beta \right).
\end{aligned}$$

We can now take the logarithm on both sides. Recall the Taylor expansion $\log(1 + x) \sim x - x^2/2$ as $x \to 0$ and apply it to the r.h.s., but note that the first integral gives two contributions (one at first order and one at second order) while the second integral only contributes to the second order. In the end, we are left with:

$$\begin{aligned}
\log \operatorname{Tr} e^{-\beta H} &= \log \operatorname{Tr} e^{-\beta H_0} - \lambda \int_0^\beta \mathrm{d}\tau \, \langle V(\tau) \rangle_\beta - \frac{1}{2} \left(\lambda \int_0^\beta \mathrm{d}\tau \langle V(\tau) \rangle_\beta \right)^2 \\
&\quad + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \, \langle (V(\tau')V(\tau) \rangle_\beta \\
&= \log \operatorname{Tr} e^{-\beta H_0} - \beta\lambda \langle V \rangle_\beta \\
&\quad + \lambda^2 \int_0^\beta \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau' \left(\langle V(\tau')V(\tau) \rangle_\beta - \langle V(\tau') \rangle_\beta \langle V(\tau) \rangle_\beta \right)
\end{aligned} \tag{4.7}$$

where in the last step we performed the first integral and rewrote the square as a double integral, then taking into account the symmetry under $\tau \longleftrightarrow \tau'$ we could remove the factor $1/2$ by integrating only over the lower triangle $\{(\tau, \tau') \in [0, \beta]^2 : \tau' \leq \tau\}$. □

It is possible to see that iterating Duhamel's formula leads exactly to the definition of the τ-ordered exponential $\mathcal{T}[e^{-\beta H}]$, and at every order the integrand looks exactly like the content of Theorem 3.3.3. To prove that the perturbative series in the operator formalism and in Grassmann variables are formally identical, we need to identify operator expectations $\langle \cdot \rangle_\beta$ with Grassmann expectations $\mathbb{E}_G(\,\cdot\,)$, but also identify the propagator $\langle a_k^*(\tau) a_{k'}(\tau') \rangle_\beta$ with the covariance $G_{k,k'}(\tau, \tau')$. To introduce Grassmann variables we need to have a finite number of physical modes in our system. Momentum modes can be made finite by introducing a cutoff Λ:

$$\mathcal{D}_\Lambda := \{k \in \mathbb{Z}^3 : |k| < \Lambda\} = \mathbb{Z}^3 \cap \mathbb{B}_\Lambda(0), \quad \text{with } |\mathcal{D}_\Lambda| = M,$$

but the propagator depends on the temperature of the operators, which is a continuous parameter. To fix this issue we should resort to using *Matsubara frequencies* and introduce their own cutoff Ω. We point to Appendix C for a brief discussion on this topic.
Usually this procedure doesn't cause any issues, and in the following calculations we do not concern ourselves with it, instead we will assume that we can remove the cutoff Ω and work directly with continuous temperature.

Let then $\boldsymbol{\psi} = (\psi_1, ..., \psi_M)$ and $\bar{\boldsymbol{\psi}} = (\bar{\psi}_1, ..., \bar{\psi}_M)$ be the Grassmann variables associated to the families of operators a and a^* respectively. Having introduced the cutoff, the kinetic part of the Hamiltonian can now be written as a simple finite linear combination of fermionic number density operators, which makes it a bounded operator:

$$H_0 = \sum_{k \in \mathcal{D}_\Lambda} (\epsilon_k - \mu) a_k^* a_k$$

The full physical theory will then be recovered in the limit $\Lambda \to \infty$ (and $\Omega \to \infty$).

We begin the calculation by building the Gaussian measure for the expectations $\langle \cdot \rangle_\beta$. Note that expectations of the form $\langle V(\tau)V(\tau')\rangle_\beta$ contain operators at different temperatures.

Lemma 4.1.2. *The two-point function for this system has the following form:*

$$G_{k,k'}(\tau,\tau') = \langle a_k^*(\tau)a_{k'}(\tau')\rangle_\beta = \delta_{k,k'} n_{\mathrm{F}}(\epsilon_k) e^{(\tau'-\tau)(\epsilon_k-\mu)} \tag{4.8}$$

where $n_{\mathrm{F}}(\epsilon) = (e^{\beta(\epsilon-\mu)}+1)^{-1}$ *is the Fermi-Dirac distribution at temperature* β.

Proof. Let us first focus on finding an expression for field operators at a given temperature $a^\#(\tau)$. Recall from the proof of Theorem 4.1.1 that for an operator A its evolution at temperature τ is given by $A(\tau) = e^{-\tau H_0} A e^{\tau H_0}$. Once again we will make use of the fundamental theorem of calculus to write

$$\begin{aligned}
e^{-\tau H_0} a_k e^{\tau H_0} &= a_k + \int_0^\tau \mathrm{d}\lambda \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(e^{-\lambda H_0} a_k e^{\lambda H_0}\right) \\
&= a_k + \int_0^\tau \mathrm{d}\lambda\, e^{-\lambda H_0}(-H_0 a_k + a_k H_0) e^{\lambda H_0} \\
&= a_k + \int_0^\tau \mathrm{d}\lambda\, e^{-\lambda H_0}[a_k, H_0] e^{\lambda H_0} \\
&= a_k + \int_0^\tau \mathrm{d}\lambda\, e^{-\lambda H_0} \sum_{k'\in\mathcal{D}_\Lambda} (\epsilon_{k'}-\mu)[a_k, a_{k'}^* a_{k'}] e^{\lambda H_0} = \star.
\end{aligned}$$

Recall that, for operators A, B and C defined on a common domain, it holds that

$$[A, BC] = \{A, B\}C - B\{A, C\}$$

as can be shown by explicitly writing the expressions for the anticommutators. Using this identity we rewrite the commutator that appears in the last expression in terms of the canonical anticommutator as follows:

$$\begin{aligned}
\star &= a_k + \int_0^\tau \mathrm{d}\lambda\, e^{-\lambda H_0} \sum_{k'\in\mathcal{D}_\Lambda} (\epsilon_{k'}-\mu)(\{a_k, a_{k'}^*\}a_{k'} - a_{k'}^*\{a_k, a_{k'}\}) e^{\lambda H_0} \\
&= a_k + \int_0^\tau \mathrm{d}\lambda\, e^{-\lambda H_0}(\epsilon_k-\mu) a_k e^{\lambda H_0} \\
&= a_k + (\epsilon_k-\mu)\int_0^\tau \mathrm{d}\lambda\, a_k(\lambda)
\end{aligned}$$

where we used that $\{a_k, a_{k'}^*\} = \delta_{k,k'}$ and that $\{a_k, a_{k'}\} = 0$. We now have an expression for $a_k(\tau)$ in terms of its integral, and if we take the derivative with respect to τ we get the following Cauchy problem:

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}\tau} a_k(\tau) = (\epsilon_k-\mu) a_k(\tau) \\ a_k(0) = a_k \end{cases}.$$

Using Theorem B.0.3, it suffices to find one such solution explicitly, which is:

$$a_k(\tau) = e^{\tau(\epsilon_k-\mu)} a_k(0) = e^{\tau(\epsilon_k-\mu)} a_k.$$

The procedure is completely analogous in the case of $a_k^*(\tau)$ and the result differs only by a sign in the exponent:

$$a_k^*(\tau) = e^{-\tau(\epsilon_k-\mu)} a_k^*(0) = e^{-\tau(\epsilon_k-\mu)} a_k^*.$$

By linearity, we can take the scalar exponential factors out of the expectation value to write the two-point function in terms of the quantity $\langle a_k^* a_{k'}\rangle_\beta$:

$$\langle a_k^*(\tau) a_{k'}(\tau')\rangle_\beta = e^{-\tau(\epsilon_k-\mu)} e^{\tau'(\epsilon_{k'}-\mu)} \langle a_k^* a_{k'}\rangle_\beta.$$

It remains to determine the expectation $\langle a_k^* a_{k'}\rangle_\beta$. Recall that the spectrum of the number density operators $a_k^* a_k$ is $\mathbb{N}_0$, and in the fermionic case the occupation number basis decomposes the Fock space like we discussed in Proposition (2.2.6). Using the cutoff on high momentum, the trace in the Gibbs state functional becomes a sum over a finite amount of states. Let $\{|\boldsymbol{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0,1\}\ \forall i = 1,\dots,M\}$ be the complete orthonormal set of eigenvectors of M independent number density operators (in our case, $M = |\mathcal{D}_\Lambda|$). Using the momenta $k \in \mathcal{D}_\Lambda$ as indices for the basis states labels n_k, we write

$$\begin{aligned}
\langle a_k^* a_{k'}\rangle_\beta = \frac{\operatorname{Tr}(a_k^* a_{k'} e^{-\beta H_0})}{\operatorname{Tr} e^{-\beta H_0}} &= \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} \langle \boldsymbol{n}| e^{-\beta H_0} a_k^* a_{k'} |\boldsymbol{n}\rangle \\
&= \delta_{k,k'} \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} \langle \boldsymbol{n}| a_k^* a_k |\boldsymbol{n}\rangle \prod_{p\in\mathcal{D}_\Lambda} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} \sum_{\boldsymbol{n}\in\{0,1\}^M} n_k e^{-\beta(\epsilon_k-\mu)n_k} \prod_{p\neq k} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} \left(\sum_{n_k=0}^{1} n_k e^{-\beta(\epsilon_k-\mu)n_k} \right) \prod_{p\neq k} \sum_{n_p=0}^{1} e^{-\beta(\epsilon_p-\mu)n_p} \\
&= \delta_{k,k'} \frac{1}{Z} e^{-\beta(\epsilon_k-\mu)} \prod_{p\neq k} (1 + e^{-\beta(\epsilon_p-\mu)})
\end{aligned}$$

where we used the fact that H_0 is bounded and diagonal in the occupation number basis $\{|\boldsymbol{n}\rangle\}$, and that $a_k^* a_k |\boldsymbol{n}\rangle = n_k |\boldsymbol{n}\rangle$. Using many of the same steps for the denominator, we find that $Z = \prod_{p\in\mathcal{D}_\Lambda}(1 + e^{-\beta(\epsilon_p-\mu)})$, and thus the expression simplifies to

$$\begin{aligned}
\langle a_k^* a_{k'}\rangle_\beta &= \delta_{k,k'} \frac{e^{-\beta(\epsilon_k-\mu)} \prod_{p\neq k}(1 + e^{-\beta(\epsilon_p-\mu)})}{\prod_p (1 + e^{-\beta(\epsilon_p-\mu)})} \\
&= \delta_{k,k'} \frac{e^{-\beta(\epsilon_k-\mu)}}{1 + e^{-\beta(\epsilon_k-\mu)}} \\
&= \delta_{k,k'} \frac{1}{e^{\beta(\epsilon_k-\mu)} + 1}.
\end{aligned}$$

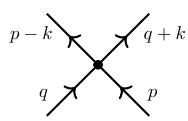
This is exactly the Fermi-Dirac distribution, and the thesis follows immediately. □

With the 2-point function determined we can go over to the computation via Feynman diagrams. Recall the definition for the expectation of a function of Grassmann variables

$$\mathbb{E}_G(\,\cdot\,) = \int P_G(\mathrm{d}\psi)(\,\cdot\,), \quad \text{where } P_G(\mathrm{d}\psi) = \frac{1}{\det G} \left(\prod_{j=1}^{N} \mathrm{d}\psi_j \mathrm{d}\bar{\psi}_j \right) e^{-\bar{\boldsymbol{\psi}}\cdot G^{-1}\boldsymbol{\psi}}$$

and the graphical representation of the interaction term via a vertex, which in our case is exactly the quartic term we discussed in Section 3.1:

$$V = \sum_{k,p,q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p =$$



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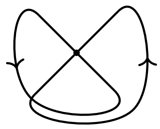
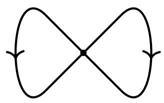
We can evaluate the expectations that show up in Duhamel's formula separately by following the diagrammatic rules we discussed in the previous chapter. Let's start with $\langle V\rangle_\beta$. Omitting the temperatures when writing two-point functions (as they are evaluated at the same temperatures) and calling $n_k = n_{\mathrm{F}}(\epsilon_k)$ the calculation is as follows:

$$\langle V\rangle_\beta = \mathbb{E}_G\Bigg(\sum_{k,p,q}\hat{V}(k)\bar{\psi}_{p-k}\bar{\psi}_{q+k}\psi_q\psi_p\Bigg) = \sum_{\Gamma\in\mathcal{G}_{\mathrm{C},1}}\mathrm{val}(\Gamma)$$

=  + 

$$\begin{aligned}
&= \sum_{k,p,q}\hat{V}(k)G_{p-k,p}G_{q+k,q} - \sum_{k,p,q}\hat{V}(k)G_{p-k,q}G_{q+k,p}\\
&= \sum_{k,p,q}\big(\hat{V}(k)\delta_{k,0}n_pn_q - \hat{V}(k)\delta_{k,p-q}n_pn_q\big)\\
&= \sum_{p,q}(\hat{V}(0) - \hat{V}(p-q))n_pn_q.
\end{aligned}$$

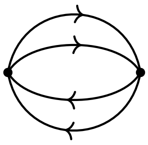
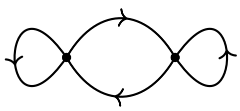
Let us now consider the term $\langle V(\tau')V(\tau)\rangle_\beta - \langle V(\tau')\rangle_\beta\langle V(\tau)\rangle_\beta$ in the double integral, and explain exactly where the problems with the second order arise. This time we have terms with two vertices, and using Theorem 3.3.3 the expression factorizes as follows, recalling the identification of Gibbs expectations $\langle\,\cdot\,\rangle_\beta$ with Grassmann expectations $\mathbb{E}_G(\,\cdot\,)$:

$$\begin{aligned}
\langle V(\tau')V(\tau)\rangle_\beta - \langle V(\tau')\rangle_\beta\langle V(\tau)\rangle_\beta &= \mathbb{E}_G(V(\tau')V(\tau)) - \mathbb{E}_G(V(\tau'))\mathbb{E}_G(V(\tau))\\
&= \mathbb{E}_G^{\mathrm{T}}(V(\tau');V(\tau))\\
&= \sum_{\Gamma\in\mathcal{G}_{\mathrm{C},2}}\mathrm{val}(\Gamma).
\end{aligned}$$

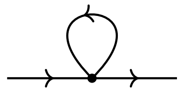
There are exactly two distinct topologies for the second order connected diagrams: those with no tadpoles, and those with two tadpoles. To see this, let us call A and B the two vertices of a connected diagram, and analyze the two topologies separately.

- **No tadpoles.** Consider one propagator exiting A (the other exiting A will be fixed by this choice): it has two possible destinations in B, both of which leave only two possible ways to complete the diagram. If, at first, we choose the other leg exiting A we would produce exact duplicates of the diagrams we obtained so far. We conclude that this topology admits 4 diagrams.

- **Two tadpoles.** There are 4 ways to put a tadpole on a vertex, up to a total of 16 possible ways of having a diagram with two tadpoles. Each of this configurations has to a unique way to complete the diagram, thus we conclude that this topology admits 16 diagrams.

No tadpoles:  Two tadpoles: 

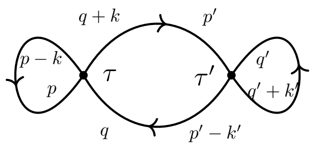
The second topology contains only divergent diagrams, and to see this we shall discuss the effect of a tadpole on the value of a diagram. Consider the tadpole diagram shown below.



Using Lemma 4.1.2 we can see that the Kronecker delta introduced by the tadpole propagator, connecting the vertex to itself, can only be either $\delta_{q+k,q}$, which implies $k=0$ (similarly for $\delta_{p-k,p}$), or $\delta_{q+k,p}$, which implies $k=p-q$ (similarly for $\delta_{p-k,q}$). Thus, adding a tadpole on each vertex in a second order connected diagram will lead to one of the four following cases, calling p,q,k the momenta that appear in the first vertex and p',q',k' the ones that appear in the second:

$$\begin{cases} k=0 \\ k'=0 \end{cases} \qquad \begin{cases} k=0 \\ k'=p'-q' \end{cases} \qquad \begin{cases} k=p-q \\ k'=0 \end{cases} \qquad \begin{cases} k=p-q \\ k'=p'-q' \end{cases} .$$

Note that each one of these conditions identifies 4 diagrams, up to a total of 16, as expected. Explicit calculation – of which we will give an example shortly – shows that in each of the above cases, the exponential will have exponent equal to zero, meaning the value of the entire diagram is a sum of two potentials $\hat{V}$ times four distributions n_{F}. Moreover, for a sum of this form the limit $\Lambda\to\infty$ is well defined and the limit $\beta\to\infty$ can be taken inside the sum: the proofs for these facts are completely analogous to the ones presented in Theorem 4.1.4 and Lemma 4.2.1, so we defer to those. We now show in one of the above cases that the contribution to the free energy is indeed zero, the other cases follow the same steps and lead to the same result. Let us evaluate the following diagram, where the vertices carry a label indicating the temperature of the vertex:



$$= \sum_{\substack{p,q,k \\ p',q',k'\in\mathcal{D}_\Lambda}} \hat{V}(k)\hat{V}(k')G_{p-k,p}(\tau,\tau)G_{q'+k',q'}(\tau',\tau')G_{q+k,p'}(\tau,\tau')G_{p'-k',q}(\tau',\tau)$$
$$= \sum_{\substack{p,q,k \\ p',q',k'\in\mathcal{D}_\Lambda}} \hat{V}(k)\hat{V}(k')\delta_{p-k,p}\delta_{q'+k',q'}\delta_{q+k,p'}\delta_{p'-k',q}n_p n_{q'} n_{p'} n_q e^{(\tau'-\tau)(\epsilon_{p'}-\epsilon_q)} = \star,$$

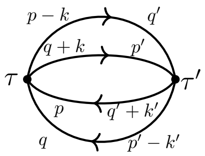
but now notice that the Kronecker deltas impose $k=k'=0$, which in turn implies $p'=q$. Thus, the exponential has a vanishing exponent, and we are left with

$$\star = \sum_{p,q,q'\in\mathcal{D}_\Lambda} \hat{V}(0)^2 n_q^2 n_p n_{q'} .$$

To retrieve the free energy contribution we integrate the above, as per Equation (4.7), in τ and τ', but since there is no dependence on these two variables, the integrals reduce to a factor of $\beta^2/2$ in the final expression. We mentioned we can take the limits $\Lambda\to\infty$ and $\beta\to\infty$: integrating, diving by $-\beta$, and carrying out these limits gets us to NOOOOO WHYYYYY, IT LOOKED SO PERFECT

$$\lim_{\substack{\beta\to\infty \\ \Lambda\to\infty}} \sum_{p,q,q'\in\mathcal{D}_\Lambda} \hat{V}(0)^2\left(-\frac{\beta}{2}n_q^2 n_p n_q\right) = \lim_{\beta\to\infty} \sum_{p,q,q'\in\mathbb{Z}^3} \hat{V}(0)^2\left(-\frac{\beta}{2}n_q^2 n_p n_q\right) = \infty.$$

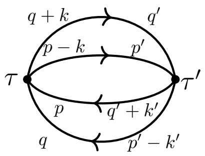
We may now look at the first topology, which is non-trivial even at zero temperature. The four diagrams evaluate to just two distinct terms up to renaming of some indices, and each with multiplicity two. We will see that these two terms correspond to the leading direct contribution and the subleading exchange contribution to the ground state energy. Let us evaluate them explicitly: WAIT A SECOND... CHECK SINGLE TERM $k = 0$ IN BOTH CONTRIBUTIONS. IT'S INFINITE.



$$= \sum_{\substack{p,q,k\\ p',q',k'}} \hat{V}(k)\hat{V}(k')G_{p-k,q'}(\tau,\tau')G_{q+k,p'}(\tau,\tau')G_{q'+k',p}(\tau',\tau)G_{p'-k',q}(\tau',\tau)$$

alignat you say?
never heard of it

$$= \sum_{p,q,k} \hat{V}(k)^2 n_q n_p n_{q+k} n_{p-k} e^{(\tau'-\tau)(\epsilon_{p-k}+\epsilon_{q+k}-\epsilon_p-\epsilon_q)}$$



$$= -\sum_{\substack{p,q,k\\ p',q',k'}} \hat{V}(k)\hat{V}(k')G_{q+k,q'}(\tau,\tau')G_{p-k,p'}(\tau,\tau')G_{q'+k',p}(\tau',\tau)G_{p'-k',q}(\tau',\tau)$$

c'mon now...

$$= -\sum_{p,q,k} \hat{V}(k)\hat{V}(p-q-k) n_q n_p n_{p-k} n_{q+k} e^{(\tau'-\tau)(\epsilon_{p-k}+\epsilon_{q+k}-\epsilon_p-\epsilon_q)}.$$

We then integrate these expressions and put the results back in Duhamel's formula. But note that the term $k = 0$ in both diagrams doesn't depend on τ, which means integrating in τ and τ' gives an overall factor of $\beta^2/2$ just like in the tadpole diagrams. We group the terms that contribute to zero (WELL, NOT SO SURE NOW) in a term $Q(\lambda, V, \beta)$:

$$\begin{aligned}
&\log \operatorname{Tr} e^{-\beta H}\\
&= Q(\lambda,V,\beta) + \log \operatorname{Tr} e^{-\beta H_0} - \beta\lambda \sum_{p,k\in\mathcal{D}_\Lambda} \left(\hat{V}(0)-\hat{V}(k)\right) n_p n_{p-k}\\
&\quad + 2\lambda^2 \sum_{\substack{p,q\in\mathcal{D}_\Lambda\\ k\in\mathcal{D}_\Lambda\setminus\{0\}}} \left(\frac{(\hat{V}(k)^2-\hat{V}(k)\hat{V}(p-q-k))n_p n_q n_{q+k} n_{p-k}}{\epsilon_{p-k}+\epsilon_{q+k}-\epsilon_q-\epsilon_p} \left(\beta + \frac{1-e^{-\beta(\epsilon_{p-k}+\epsilon_{q+k}-\epsilon_q-\epsilon_p)}}{\epsilon_{p-k}+\epsilon_{q+k}-\epsilon_q-\epsilon_p}\right)\right).
\end{aligned} \tag{4.9}$$

The next step is to show that removing the UV cutoff, i.e., taking the limit $\Lambda \to \infty$, leads to a well-defined expression. To prove this we do not need to use the specific expression $\epsilon_k = \hbar^2|k|^2$ for the kinetic energy, instead we only assume a condition on its asymptotic behavior for high momentum. A slight generalization which costs us nothing. In fact, we may also assume a weaker condition on the potential and drop the condition that $\hat{V}$ has compact support, as we can control its growth quite effectively – at least until we get to the zero temperature limit, when we will restore the initial condition.

A quick comment on a slightly less standard notation we will use. We say that two real or complex sequences $\{x_n\}_n$ and $\{y_n\}_n$ are asymptotic if, and only if, $x_n/y_n \longrightarrow 1$. We will indicate that two quantities are asymptotic only up to a proportionality constant by adopting the following notation:

$$x_n \asymp y_n \iff \exists C \neq 0 : \frac{x_n}{y_n} \longrightarrow C.$$

To control terms that contain the Fourier transform of the interaction potential we will make use of the following property:

Lemma 4.1.3. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a Schwartz function, then for all $m \in \mathbb{N}$ it holds that*

$$|\hat{f}(k)| \leq \frac{1}{|k|^m} \int_{\mathbb{R}^n} \mathrm{d}x \, |\partial^\alpha f(x)|.$$

Proof. Recall the expression for $\hat{f}(k)$ and note that we may rewrite the exponential in it as a derivative, then integrate by parts (the boundary term vanishes as f is a Schwartz function):

$$\hat{f}(k) = \int_{\mathbb{R}^n} \mathrm{d}^n x \, e^{ik\cdot x} f(x) = \int_{\mathbb{R}^n} \mathrm{d}^n x \left(\frac{1}{ik_j} \frac{\partial}{\partial x_j} e^{ik\cdot x} \right) f(x) = \frac{-1}{ik_j} \int_{\mathbb{R}^n} \mathrm{d}^n x \, e^{ik\cdot x} \frac{\partial}{\partial x_j} f(x).$$

We iterate this procedure m times to get get a derivative of f of order m. Note that every time we integrate by parts the boundary term vanishes because all derivatives must vanish at infinity. Lastly we bound $|\hat{f}(k)|$ by taking the modulus inside the integral:

$$|\hat{f}(k)| \leq \frac{1}{|k|^m} \int_{\mathbb{R}^n} \mathrm{d}^n x \left| \frac{\partial^m}{\partial x_{j_1} \cdots \partial x_{j_m}} f(x) \right|,$$

where $j_1, \ldots, j_m$ are used to indicate any derivative of order m, and we also used that $|k_j| \leq \sqrt{(k_1)^2 + \cdots + (k_n)^2} = |k|$. This concludes the proof. □

We now prove that we can remove the UV cutoff.

==================== PROVE NEW VERSION ====================

Theorem 4.1.4. *Asuume that $\hat{V}$ is a Schwartz function, and assume moreover that ϵ_k depends only on $|k|$ and that it satisfies $\lim_{|k|\to\infty} \epsilon_k/|k| > 0$. Then the expression in Equation (4.9) converges in the limit $\Lambda \to \infty$.*

Proof. The zeroth order is trivial, as it doesn't depend on Λ and is, in fact, finite: $e^{-\beta H_0}$ is trace-class and strictly positive definite, which implies $|\log \mathrm{Tr}\, e^{-\beta H_0}| < \infty$. Before moving on let's discuss the estimates we will use. As a consequence of Lemma 4.1.3 we have that $|\hat{V}(k)| \leq C_{1,m}/|k|^m$ for some $C_{1,m} > 0$ that does not depend on Λ, and for all $m \in \mathbb{N}$. As for the energy terms, we are assuming that $\epsilon_k \geq C_{2,m}|k|$ for all k outside of the ball of radius K for some $K, C_{2,m} > 0$ that again do not depend on Λ. Finally, the Fermi-Dirac distribution is controlled by an exponential decay in ϵ_k, which is bounded at least by an exponential decay in $|k|$.

Looking at the first order, we can write

$$\begin{aligned}\sum_{p,k\in\mathcal{D}_\Lambda} \left| \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k} \right| &\leq |\hat{V}(0)| \sum_{p,k\in\mathcal{D}_\Lambda} n_p n_{p-k} + \sum_{p,k\in\mathcal{D}_\Lambda} |\hat{V}(k)| n_p n_{p-k} \\ &= |\hat{V}(0)| \sum_{p\in\mathcal{D}_\Lambda} n_p \sum_{k\in\mathcal{D}_\Lambda} n_{p-k} + \sum_{p\in\mathcal{D}_\Lambda} n_p \sum_{k\in\mathcal{D}_\Lambda} |\hat{V}(k)| n_{p-k}.\end{aligned}$$

By the positivity of the general terms for all $k \in \mathbb{Z}^3$ we can bound the sums by the respective series over $\mathbb{Z}^3$, and because the general terms only depend on the modulus of the momentum we can employ the following strategy. Define the spherical shell $A_n = \{k \in \mathbb{Z}^3 : n \leq |k| < n+1\}$ and write a series with general term $b_{|k|}$ as

$$\sum_{k\in\mathbb{Z}^3} b_{|k|} = \sum_{n=0}^{\infty} \sum_{k\in A_n} b_{|k|}.$$

Because $c_1|A_n|b_{n+1} \leq \sum_{k\in A_n} b_{|k|} \leq c_2|A_{n+1}|b_n$ for all n if a_n is decreasing, and because $|A_n| \asymp n^2$, we can conclude that the behavior of $\sum_{k\in\mathbb{Z}^3} b_{|k|}$ and $\sum_{n=0}^{\infty} n^2 b_n$ is the same. Note that $\sum_{k\in\mathbb{Z}^3} n_k$

converges if, and only if, the series $\sum_{\ell\in\mathbb{N}} \ell^2 e^{-\beta\epsilon_\ell}$ converges, by the asymptotic comparison test. The latter does converge absolutely as shown by the root test:

$$\left|\ell^2 e^{-\beta\epsilon_\ell}\right|^{1/\ell} = e^{\frac{2\log(\ell)-\beta\epsilon_\ell}{\ell}} \xrightarrow[\ell\to\infty]{} Q \in [0,1).$$

Similarly, $\sum_{k\in\mathbb{Z}^3} |\hat{V}(k)|n_k$ converges too. We split the series into a finite sum over a ball D centered at the origin and of finite radius plus the sum over the complement, and the latter converges, as

$$\sum_{k\in D^{\mathrm{C}}} |\hat{V}(k)|n_k \leq \sum_{k\in D^{\mathrm{C}}} \frac{C_1 n_k}{|k|^m} \leq \sum_{k\in D^{\mathrm{C}}} C_1 n_k.$$

Thus, in the limit in $\Lambda\to\infty$, the first order is an absolutely convergent series.

Very similar arguments can be used to show that the second order has an absolutely convergent series as a limit. Indeed, we can bound the first term in the sum (the second order direct term) by

$$\sum_{p,q,k\in\mathcal{D}_\Lambda} \hat{V}(k)^2(\dots) \leq \sum_p \frac{4n_p}{\epsilon_p-\mu}\sum_q n_q \sum_k \hat{V}(k)^2 n_{q+k}n_{p-k}\left(\beta+\frac{1}{\epsilon_p-\mu}\right),$$

and turning once more sums over $\mathbb{Z}^3$ into series over spherical shells, this is absolutely convergent, because the additional factor $(\epsilon_p-\mu)^{-1}$ does not change the result of any of the comparison tests done before. The exact same argument can be applied to the other term (second order exchange term), and this concludes the proof. □

We can then essentially replace the sums over $\mathcal{D}_\Lambda$ in Equation (4.9) with series over $\mathbb{Z}^3$. Inserting the factor of $-1/\beta$ to get the free energy leads us to its complete expression:

$$\begin{aligned}
F &=: F^{(0)} + F^{(1)}_{\mathrm{D}} + F^{(1)}_{\mathrm{X}} + F^{(2)}_{\mathrm{D}} + F^{(2)}_{\mathrm{X}} \\
&= F^{(0)} + \lambda \sum_{p,k\in\mathbb{Z}^3} \hat{V}(0)n_p n_{p-k} - \lambda \sum_{p,k\in\mathbb{Z}^3} \hat{V}(k)n_p n_{p-k} \\
&\quad + \lambda^2 \sum_{p,q,k\in\mathbb{Z}^3} \frac{4\hat{V}(k)^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p+\epsilon_q+\epsilon_{q+k}+\epsilon_{p-k}-4\mu}\left(1+\beta^{-1}\frac{1-e^{-\beta(\epsilon_p+\epsilon_q+\epsilon_{q+k}+\epsilon_{p-k}-4\mu)}}{\epsilon_p+\epsilon_q+\epsilon_{q+k}+\epsilon_{p-k}-4\mu}\right) \\
&\quad + \lambda^2 \sum_{p,q,k\in\mathbb{Z}^3} \frac{16\hat{V}(k-q)\hat{V}(p-q)n_q^2 n_p n_{q-k}}{2\epsilon_q+\epsilon_p+\epsilon_{q-k}-4\mu}\left(1+\beta^{-1}\frac{1-e^{-\beta(2\epsilon_q+\epsilon_p+\epsilon_{q-k}-4\mu)}}{2\epsilon_q+\epsilon_p+\epsilon_{q-k}-4\mu}\right)
\end{aligned} \tag{4.10}$$

where each term has been named appropriately following the order in which they appear: $F^{(0)}$ is the free energy of the unperturbed system (no interactions), $F^{(i)}_{\mathrm{D}}$ is the direct term at order i and $F^{(i)}_{\mathrm{X}}$ is the exchange term at order i.

4.2 Mean field scaling

With the convergence of the free energy under control, the next step is to look at the limit $\beta\to\infty$ to retrieve the ground state energy and complete our perturbative proof of the mean field scaling of Equation (4.2).

Lemma 4.2.1. *With the same hypotheses as in Theorem 4.1.4, all the series in Equation (4.10) converge uniformly in β over $[\delta,\infty)$ for some $\delta>0$, and the limit $\beta\to\infty$ can thus be taken inside the sums.*

Proof. The only troubling factor is the Fermi-Dirac distribution, as it has a discontinuity at $\epsilon_k = \mu$ in the limit $\beta \to \infty$. By splitting each sum into a finite part up to, say, $\epsilon_k = 2\mu$, and a series over the remaining values of k, we can avoid this discontinuity. Moreover, we are restricting the domain on which to check for uniform convergence to $[\delta, \infty)$ for some $\delta > 0$ because we are considering a limit away from $\beta = 0$. With these things in mind we can see that the series $\sum_k n_k$ converges uniformly, as

$$\sum_{k:\epsilon_k \geq 2\mu} n_k \leq \sum_k \sup_{\beta\in[\delta,\infty)} n_k \leq \sum_k e^{-\delta(\epsilon_k - \mu)}$$

and the upper bound converges absolutely by the same arguments used in the previous theorem. Modifying this series with a prefactor of $1/(\epsilon_k - \mu)$, $\hat{V}(k)$ or $\hat{V}^2(k)$ in the general term does not change the result, as these factors do not depend on β and still lead to an absolutely convergent series as an upper bound. We can see that, by rewriting sums of the form $\sum_{p,q,k}(\ldots)$ as $\sum_q(\ldots\sum_p(\ldots\sum_k(\ldots)))$, we only need to check the convergence of the innermost series in all the terms of Equation (4.10), and the rest will follow from arguments that we already covered. Consider the term $F_{\mathrm{D}}^{(1)} + F_{\mathrm{X}}^{(1)}$:

$$\begin{aligned}\sum_{p,k}\left|\hat{V}(0) - \hat{V}(k)\right| n_p n_{p-k} &= \sum_p n_p \sum_k \left|\hat{V}(0) - \hat{V}(k)\right| n_{p-k} \\ &\leq \sum_p n_p \sum_k \left|\hat{V}(0) - \hat{V}(k)\right| \sup_{\beta\in[\delta,\infty)} n_{p-k} = \star\end{aligned}$$

but the series containing $\sup_\beta(\ldots)$ converges to some $C > 0$ as we discussed above, and we can therefore write $\star = \sum_p C n_p$, which again converges. For the term $F_{\mathrm{D}}^{(2)}$, we can identify two terms that need our attention, up to trivial bounds:

$$A := \sum_k \frac{\hat{V}(k)^2 n_{q+k} n_{p-k}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu} \quad \text{and} \quad B := \sum_k \left| \frac{\hat{V}(k)^2 n_{q+k} n_{p-k}(1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu)})}{\beta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)^2} \right|.$$

The exchange term $F_{\mathrm{X}}^{(2)}$ will exhibit the same behavior, both in terms of dependence on β and in terms of point-wise convergence. The series A is very simple: we bound the Fermi-Dirac distributions with their supremum and notice that

$$A \leq \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu}.$$

Let's now discuss B. Distributing the product at the numerator, the first term is a product of Fermi-Dirac distributions divided by β, which means it is monotonically decreasing in β for all q, p, k outside the Fermi ball. As for the exponential term one can check that the resulting general term is monotonically decreasing in β far enough from the origin, as the only extremal point is a maximum that moves towards the origin as k grows. Therefore, once again, the supremum is found at $\beta = \delta$ and we are left with

$$B \leq \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2} + \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_p + \epsilon_q + 2\epsilon_{q+k} + 2\epsilon_{p-k} - 6\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2},$$

which finally shows that the second order is uniformly convergent too, completing the proof. □

Now that we have proved that we can take the limit of zero temperature inside the series, we are left with the task of computing it. The zero temperature Fermi-Dirac distribution is just the characteristic function of the Fermi ball B_{F}:

$$n_{\mathrm{F}}(\epsilon) \xrightarrow[\beta\to\infty]{} \chi(\epsilon \leq \mu).$$

This turns every sum in Equation (4.10) into a finite sum over intersecting Fermi balls. We present a useful lemma that will help us control the growth of these sums as N approaches infinity.

Lemma 4.2.2 (Riemann sum approximation)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a compactly supported, Lipschitz continuous function, with Lipschitz constant L. Then for $N \in \mathbb{N}$ we have the following approximation by integration, with $m \in \mathbb{N}$ and $h = N^{-1/m}$ the lattice spacing:*

$$N^{-\frac{d}{m}} \sum_{k \in h\mathbb{Z}^d} f(k) = \int_{\mathbb{R}^d} \mathrm{d}^d x\, f(x) + \mathcal{E}(h),$$

where $\mathcal{E}(h) \leq hLK_f\sqrt{d} = \mathcal{O}(h)$, and K_f a constant that depends on the volume of the support of f.

Proof. Let $h = N^{-1/m}$ be the lattice spacing in $N^{-1/m}\mathbb{Z}^d$. Then define, at each point in the lattice, a cell of side length h as

$$C_k := [hk_1, h(k_1+1)] \times \cdots \times [hk_d, h(k_d+1)].$$

Note that the volume of a cell is $\int_{C_k} \mathrm{d}^d x = h^d = N^{-d/m}$. We can estimate the error of integration through the difference between the sum and the integral. The domain of integration is split into cells that intersect on sets of measure zero, and the integrand has compact support, so the integral can be split as a sum of integrals over the individual cells. Changing variables with $k \mapsto k' = k/h$ we can write

$$\begin{aligned}\left| N^{-\frac{d}{m}} \sum_{k \in h\mathbb{Z}^d} f(k) - \int_{\mathbb{R}^d} \mathrm{d}^d x\, f(x) \right| &= \left| N^{-\frac{d}{m}} \sum_{k \in \mathbb{Z}^d} f(hk) - \sum_{k \in \mathbb{Z}^d} \int_{C_k} \mathrm{d}^d x\, f(x) \right| \\ &= \left| \sum_{k \in \mathbb{Z}^d} \int_{C_k} \mathrm{d}^d x\, \chi_{\mathrm{supp}(f)}(x)(f(hk) - f(x)) \right| \\ &\leq L \sum_{k \in \mathbb{Z}^d} \chi_{\mathrm{supp}(f)}(hk) \int_{C_k} \mathrm{d}^d x\, |hk - x| = \star,\end{aligned}$$

where in the last step we used Lipschitz continuity. Now we can bound the integral of $|hk - x|$ on C_k by $h\sqrt{d} = \max_{C_k}(|hk - x|)$ times the volume of a cell C_k, to get that

$$\star \leq L \sum_{k \in \mathbb{Z}^d} \chi_{\mathrm{supp}(f)}(hk) h\sqrt{d} \left(\int_{C_k} \mathrm{d}^d x \right) = hL\sqrt{d} N^{-\frac{d}{m}} \mathcal{O}(N^{\frac{d}{m}}) = \mathcal{O}(h).$$

where we used that $\sum_{k \in \mathbb{Z}^d} \chi_{\mathrm{supp}(f)}(hk)$ is of order $N^{d/m}$ because f is compactly supported. The thesis follows immediately from this bound, as $h = N^{-1/m}$. $\square$

ADD NOTE TO INCLUDE CASES WHERE THE COMPACT SUPPORT IS DETERMINED BY AN INDICATOR FUNCTION, WHICH IS NOT LIPSCHITZ BUT STILL ALLOWS FOR THIS APPROXIMATION.

For each term in Equation (4.10), we write E instead of F to indicate the same term in the limit $\beta \to \infty$. That is, $E_{\mathrm{gs}} = \lim_{\beta \to \infty} F = E^{(0)} + E^{(1)}_{\mathrm{D}} + E^{(1)}_{\mathrm{X}} + E^{(2)}_{\mathrm{D}} + E^{(2)}_{\mathrm{X}}$.
Recall that the chemical potential μ is chosen to sit inside the spectral gap between the Fermi energy (highest occupied energy level in the ground state) and the first excited energy level. Any value inside the spectral gap can be chosen, but since this gap closes as N grows, we may choose the

specific scaling IS IT NEEDED NOW?

$$\begin{aligned}\mu &:= \left(\max_{p\in B_{\mathrm{F}}} \epsilon_p\right) + \frac{\hbar^2}{2} \\ &= \hbar^2 \left(\frac{3}{4\pi}N\right)^{\frac{2}{3}} + \frac{\hbar^2}{2} \\ &= C^2 + \frac{1}{2}N^{-\frac{2}{3}},\end{aligned} \tag{4.11}$$

with $C = 3/4\pi$. Such a definition avoids, for all N, the singularity of the form $(\epsilon - \mu)^{-1}$ in the denominators of Equation (4.10) and also allows to control the growth of the domains of the sums over momenta that appear in the limit $\beta \to \infty$. Note, however, that summing over a Fermi ball with this new definition of the chemical potential does not introduce new terms. In fact, $\hbar^2/2$ being smaller than the spectral gap between the Fermi energy and the first excited level means that no new momenta will lie in the Fermi ball by enlarging its radius by such amount.

Theorem 4.2.3. *The ground state energy of the Hamiltonian in (4.3) exhibits the following scalings in the number of particles N at second order in perturbation theory:*

- *WHAT ABOUT $-\frac{Q(\lambda,V,\beta)}{\beta}$ AND $k = k' = 0$?*
- $E^{(0)} = \mathcal{O}(N)$,
- $E^{(1)}_{\mathrm{D}} = \mathcal{O}(N)$ *and* $E^{(1)}_{\mathrm{X}} = \mathcal{O}(N^0)$,
- $E^{(2)}_{\mathrm{D}} = \mathcal{O}(N^{-\frac{1}{3}})$ *and* $E^{(2)}_{\mathrm{X}} =$ *CHECK.*

Proof. Of course we proceed order by order, starting from the unperturbed free energy to make sure that everything is in order. Recall that $B_{\mathrm{F}} = \{p \in \mathbb{Z}^3 : |p| < k_{\mathrm{F}}\}$ and expand the expression for $F^{(0)}$ as

$$F_0 = -\frac{\log(Z)}{\beta} = -\sum_{p\in\mathbb{Z}^3} \frac{1}{\beta}\log(1+e^{-\beta(\epsilon_p-\mu)}) = \sum_{p\in B_{\mathrm{F}}} (\epsilon_p - \mu) + \sum_{p\notin B_{\mathrm{F}}} \log(1+e^{-\beta(\epsilon_p-\mu)}),$$

where in the limit the second term vanishes. From here we can change the index summed over in $\sum_{|p|\le CN^{1/3}}$ to approximate the result with an integral. Recall the mean field regime defined in Equation (4.1) and that $k_{\mathrm{F}} = \mathcal{O}(N^{1/3})$. We take $p \mapsto p' = N^{-1/3}p$, which means that in the Fermi ball it holds $|p'| \le C$. Using Lemma 4.2.2 we come to the approximation

$$\begin{aligned}\sum_{p\in B_{\mathrm{F}}} (\hbar^2 p^2 - \mu) &= N^{-\frac{2}{3}} \sum_{\substack{p\in\mathbb{Z}^3:\\ |k|\le CN^{1/3}}} p^2 - \mu \sum_{p\in B_{\mathrm{F}}} 1 \\ &= N^{-\frac{2}{3}} N \left(\frac{1}{N} \sum_{\substack{p'\in N^{-1/3}\mathbb{Z}^3:\\ |p'|\le C}} (N^{\frac{1}{3}}p')^2 \right) - N\mu \\ &= N\left(\int_0^C 4\pi \mathrm{d}p\, p^4 + \mathcal{O}(N^{-\frac{1}{3}})\right) - N\mu \\ &= \mathcal{O}(N)\end{aligned}$$

where we used that, from Equation (4.11), $\mu = \mathcal{O}(N^{-2/3})$. This shows that indeed $E^{(0)}$ is of order N. Onto the first order, recall the scaling limit $\lambda = N^{-1}$ to see that

$$E_{\mathrm{D}}^{(1)} + E_{\mathrm{X}}^{(1)} = \lambda \sum_{\substack{k\in\mathbb{Z}^3, p\in B_{\mathrm{F}}:\\ p-k\in B_{\mathrm{F}}}} \left(\hat{V}(0) - \hat{V}(k)\right) = N^{-1}\hat{V}(0)\left(\sum_{p,k\in B_{\mathrm{F}}} 1\right) - N^{-1} \sum_{\substack{k\in\mathbb{Z}^3, p\in B_{\mathrm{F}}:\\ p-k\in B_{\mathrm{F}}}} \hat{V}(k)$$
$$= \mathcal{O}(N) + N^{-1}\mathcal{O}(N) = \mathcal{O}(N) + \mathcal{O}(N^0).$$

where in the last steps we used the condition that $\operatorname{supp}(\hat{V})$ is compact, so that $\sum_k \hat{V}(k) = \mathcal{O}(1)$ and $\sum_{p,k} \hat{V}(k) = \mathcal{O}(N)$. With this expression we can conclude that the first order indeed contributes to order $\mathcal{O}(N)$ with the direct term and to order $\mathcal{O}(1)$ with the exchange term.
We now tackle the second order. Consider the expression for $E_{\mathrm{D}}^{(2)}$, it follows from the definition in Equation (4.11) that we may write

$$E_{\mathrm{D}}^{(2)} = -2\lambda^2 \sum_{k\in\mathbb{Z}^3} \sum_{\substack{p,q\in B_{\mathrm{F}}:\\ q+k, p-k\in B_{\mathrm{F}}}} \frac{\hat{V}(k)^2}{\hbar^2((p-k)^2 + (q+k)^2 - q^2 - p^2)}$$
$$= N^{-2} \sum_{k\in\mathbb{Z}^3} \sum_{\substack{p,q\in B_{\mathrm{F}}:\\ q+k, p-k\in B_{\mathrm{F}}}} \frac{\hat{V}(k)^2}{\hbar^2(k^2 + k\cdot(q-p))},$$

and changing variables like before with ($p \mapsto p' = \hbar p$, same for q and k) we get

$$E_{\mathrm{D}}^{(2)} = \sum_{k\in\hbar\mathbb{Z}^3} \frac{1}{N^2} \sum_{\substack{p\in\hbar\mathbb{Z}^3:\\ |p|\le C\\ |p-k|\le C}} \sum_{\substack{q\in\hbar\mathbb{Z}^3:\\ |q|\le C\\ |q+k|\le C}} \frac{\hat{V}(\hbar^{-1}k)^2}{k^2 + k\cdot(q-p)}.$$

WHAT NOW?

$\square$

Appendices

Appendix A

More about Hilbert spaces

We group in this appendix a series of definitions and theorems that are not necessarily part of the basic foundations of quantum mechanics, but are still crucial in understanding how exactly some computations are carried out and why they hold true. Some of the results presented here go beyond the scope of our work as far as their proof is concerned, and only serve the purpose of painting a clearer picture of what some tools actually are.

The following definitions and propositions are taken from [2].

A.1 Special operators

Definition A.1.1 (Self-adjoint operators)**.** An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \to \mathcal{H}$ is called *self-adjoint* if $A = A^*$.

Note that this condition is stronger that being symmetric.

Definition A.1.2 (Compact operators)**.** An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \to \mathcal{H}$ is called *compact* if $D(A) = \mathcal{H}$ and there exist complete orthonormal systems $\{u_i : i \in \mathbb{N}\}$ and $\{v_i : i \in \mathbb{N}\}$ for $\mathcal{H}$ and a sequence $\{\lambda_i : i \in \mathbb{N}\}$ such that $\lambda_i \to 0$ such that

$$A\phi = \sum_{n=0}^{\infty} \lambda_n \langle v_n, \phi \rangle u_n$$

for all $\phi \in \mathcal{H}$.

Definition A.1.3 (Trace class and Hilbert-Schmidt operators)**.** A compact operator A is said to be *trace class* if $\sum_{n=1}^{\infty} |\lambda_n| < \infty$ and is said to be *Hilbert-Schmidt* if $\sum_{n=1}^{\infty} |\lambda_n|^2 < \infty$. For a trace class operator A we define its *trace* as

$$\mathrm{Tr}(A) := \sum_{n=1}^{\infty} \lambda_n \langle v_n, u_n \rangle.$$

Proposition A.1.4. *The trace of a trace class operator A is finite, and given an orthonormal basis $\{\phi_n\}_{n \in \mathbb{N}}$ for $\mathcal{H}$ the trace is*

$$\mathrm{Tr}(A) = \sum_{n=1}^{\infty} \langle \phi_n, A\phi_n \rangle.$$

Remark 24. The rigorous derivation of the quantities used in this work relies heavily on the properties of the operator $e^{-\beta H}$, given H a self-adjoint operator (in our case H is also bounded, but spectral theory can extend this to the unbounded case). This operator is trace-class, and it is strictly positive definite. Moreover, if V is a bounded operator defined on the image of $e^{-\beta H}$ then it also holds that $Ve^{-\beta H}$ is trace class.

Definition A.1.5 (Projection)**.** A linear operator P on a Hilbert space $\mathcal{H}$ that is idempotent ($P^2 = P$) is called a *projection operator* or *projector*. If P is also self-adjoint it is called *orthogonal projection.*

Proposition A.1.6. *Given a Hilbert space $\mathcal{H}$, projectors defined on $\mathcal{H}$ have the following properties:*

a) every projector P is such that $P \neq 0$ and $\|P\| = 1$;

b) P is a projector if, and only if, $\mathbb{1} - P$ is a projector;

c) if P is a projector, then

$$\mathcal{H} = \operatorname{ran} P \oplus \ker P, \quad P|_{\operatorname{ran} P} = \mathbb{1}, \quad P|_{\ker P} = 0.$$

In particular, the spectrum of P is $\sigma(P) = \{0, 1\}$;

d) there is a bijection between projectors defined on $\mathcal{H}$ and closed subspaces of $\mathcal{H}$.

e) if $\{e_1, \ldots, e_N\}$ for $N \in \mathbb{N}$ or $N = \infty$ is an orthonormal system in $\mathcal{H}$, then the projection P_M on $M = \operatorname{span}\{e_1, \ldots, e_N\}$ is

$$P_M x = \sum_{n=1}^{N} \langle e_n, x\rangle e_n, \quad \forall x \in \mathcal{H}.$$

Definition A.1.7 (Isometries and Unitaries)**.** Given two Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, a linear map $U : \mathcal{H} \to \mathcal{K}$ with the property that

$$\|Ux\|_{\mathcal{K}} = \|x\|_{\mathcal{H}} \quad \forall x \in \mathcal{H}$$

is called an *isometry* between $\mathcal{H}$ and $\mathcal{K}$. A surjective isometry is called a *unitary operator.*

Unitary operators are of particular interest in quantum mechanics, as the property of preserving the inner product means that these are the natural objects to describe operations that need to preserve probability amplitudes.

Proposition A.1.8. *For a bounded linear operator between Hilbert spaces $U : \mathcal{H} \to \mathcal{K}$, the following statements are equivalent:*

a) U is a unitary operator;

b) $U^*U = \mathbb{1}_{\mathcal{K}}$ and $UU^* = \mathbb{1}_{\mathcal{H}}$;

c) $U\mathcal{H} = \mathcal{K}$ and $\langle x, y\rangle_{\mathcal{H}} = \langle Ux, Uy\rangle_{\mathcal{K}}$, for all $x, y \in \mathcal{H}$.

Recall that the dual of a complex Hilbert space $\mathcal{H}$ is the Hilbert space $\mathcal{H}^* = \{\varphi : \mathcal{H} \to \mathbb{C} : \varphi \text{ is a continuous linear functional}\}$, where continuity is with respect to the norm topology on $\mathcal{H}$. To conclude this part of our discussion on the basics of operators on a Hilbert space, we give three more results whose importance we will not discuss further, as it goes beyond the scope of our work.

Theorem A.1.9 (Riesz representation theorem)**.** *Let $\mathcal{H}$ be a complex Hilbert space. For every $x \in \mathcal{H}$ the map*

$$\begin{aligned} F : \mathcal{H} &\longrightarrow \mathbb{C} \\ y &\longmapsto F(y) := \langle x, y\rangle \end{aligned}$$

defines a continuous functional, and thus $F \in \mathcal{H}^*$. Conversely, for every $F \in \mathcal{H}^*$ there exists a unique $x_F \in \mathcal{H}$ such that $F(y) = \langle x_F, y\rangle$ for all $y \in \mathcal{H}$. Moreover, it holds that $\|x_F\|_{\mathcal{H}} = \|F\|_{\mathcal{H}^*}$.

Remark 25. The map $J : \mathcal{H} \to \mathcal{H}^*$ that sends a vector to its uniquely identified continuous functional on $\mathcal{H}$ defines a canonical "isomorphism" between a Hilbert space and its dual, which means Hilbert spaces are self-dual. Technically J is not even a homomorphism since it is conjugate-linear rather than linear, and calling it an isomorphism is a bit imprecise, but this does not pose any problem to us.

Theorem A.1.10 (Spectral theorem for self-adjoint operators)**.** *Let $A : \mathcal{H} \to \mathcal{H}$ be a self-adjoint operator on the Hilbert space $\mathcal{H}$ over the field $\mathbb{K}$. Then there exists a measure space (X, Σ, μ), a unitary operator $U : L^2(X, \mathrm{d}\mu) \to \mathcal{H}$ and a function $f : X \to \mathbb{K}$ such that*

$$A = UM_fU^*$$

where M_f is the multiplication by f, i.e., the operator defined by $(M_f\psi)(x) = f(x)\psi(x)$.

Remark 26. In the case where A is also compact, then the spectrum $\sigma(A)$ is discrete and the function f could be seen as mapping an index over $\sigma(A)$ to an eigenvalue, and the claim would read as $(M_f\psi)_i = f_i\psi_i$, which tells us that ψ_i is an eigenvector with eigenvalue f_i. This special case makes it easier to understand this version of the spectral theorem as a generalization to the case of self-adjoint operators with possibly continuous spectrum.

Theorem A.1.11 (C*-algebra of bounded operators)**.** *Let $\mathcal{H}$ be a Hilbert space. The set*

$$\mathfrak{B}(\mathcal{H}) = \{A : \mathcal{H} \to \mathcal{H} : A \text{ is bounded } \}$$

is a Banach space, i.e., a complete normed space, via the usual definition of addition between operators (assuming the domain is the entire space $\mathcal{H}$) and multiplication by a scalar. It is also a C-algebra, i.e., a normed algebra with an involution $*$, via the adjoint of bounded operators.*

A.2 Direct sums and tensor products of Hilbert spaces

Recall that, given two vector spaces V and W over a field $\mathbb{K}$, their *direct sum* is the set $V \times W$ endowed with a vector space structure defined via the following operations:

$$\begin{aligned}
&(v_1, w_1) + (v_2, w_2) \coloneqq (v_1 + v_2, w_1 + w_2), \\
&\lambda(v, w) \coloneqq (\lambda v, \lambda w)
\end{aligned}$$

for all $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$ and $\lambda \in \mathbb{K}$, and is denoted $V \oplus W$. Suppose now $\mathcal{H}_1$ and $\mathcal{H}_2$ are Hilbert spaces, then we have the natural result

Proposition A.2.1. *If $\langle \cdot, \cdot \rangle_i$ denotes the inner product on $\mathcal{H}_i$ for $i = 1, 2$, then the bilinear $\langle \cdot, \cdot \rangle$ given by*

$$\langle (x_1, x_2), (y_1, y_2) \rangle \coloneqq \langle x_1, y_1 \rangle_1 + \langle x_2, y_2 \rangle_2$$

defines an inner product with respect to which the space $\mathcal{H}_1 \oplus \mathcal{H}_2$ is complete. Thus $(\mathcal{H}_1 \oplus \mathcal{H}_2, \langle \cdot, \cdot \rangle)$ is a Hilbert space.

We can extend this definition of direct sum to the case where we consider a countable set of Hilbert spaces $\{\mathcal{H}_i : i \in \mathbb{N}\}$ over the same field $\mathbb{K}$.

Proposition A.2.2. *The set $\mathcal{H}$ of sequences $x = \{x_i : x_i \in \mathcal{H}_i,\ i \in \mathbb{N}\}$ that satisfy*

$$\sum_{i=1}^{\infty} \|x_i\|_i^2 < \infty$$

is a Hilbert space with the operations $\{x_i\} + \{y_i\} \coloneqq \{x_i + y_i\}$, $\lambda\{x_i\} \coloneqq \{\lambda x_i\}$ *and* $\langle \{x_i\}, \{y_i\} \rangle \coloneqq \sum_{i=1}^{\infty} \langle x_i, y_i \rangle_i$. *This Hilbert space is denoted*

$$\mathcal{H} = \bigoplus_{i=1}^{\infty} \mathcal{H}_i.$$

To build the tensor product we will follow an algebraic approach which involves a similar procedure but with additional steps. Given two vector spaces V and W over a field $\mathbb{K}$, define the vector space of finite linear combinations of ordered pairs in $V \times W$ as

$$\Lambda(V, W) \coloneqq \left\{ \sum_{i=1}^{N} a_i (x_i, y_i) : a_i \in \mathbb{K},\ x_i \in V,\ y_i \in W\ \forall i = 1, \dots, N \right\}.$$

On this space we impose algebraic relations through the quotient space

$$V \otimes W \coloneqq \Lambda(V, W)/\Lambda_0$$

where Λ_0 is the subspace generated by elements

$$\begin{aligned}
&(x, y_1 + y_2) - (x, y_1) - (x, y_2),\\
&(x_1 + x_2, y) - (x_1, y) - (x_2, y),\\
&(\lambda x, y) - \lambda(x, y),\\
&(x, \lambda y) - \lambda(x, y)
\end{aligned}$$

for all $x, x_1, x_2 \in V$ and $y, y_1, y_2 \in W$. If $\chi : \Lambda(V, W) \to \Lambda(V, W)/\Lambda_0$ is the projection to the quotient space, then we denote the elements in the quotient as $x \otimes y = \chi((x, y))$. Once again we can now extend this definition to the case of Hilbert spaces.

Proposition A.2.3. *Let* $\mathcal{H}_1$ *and* $\mathcal{H}_2$ *be two Hilbert spaces over the same field. If* $\langle \cdot, \cdot \rangle_i$ *denotes the inner product on* $\mathcal{H}_i$ *for* $i = 1, 2$, *then the bilinear* $\langle \cdot, \cdot \rangle$ *given by*

$$\langle x_1 \otimes x_2, y_1 \otimes y_2 \rangle \coloneqq \langle x_1, y_1 \rangle_1 \langle x_2, y_2 \rangle_2$$

defines an inner product on $\mathcal{H}_1 \otimes \mathcal{H}_2$. *The metric completion* $(\mathcal{H}_1 \tilde{\otimes} \mathcal{H}_2, \langle \cdot, \cdot \rangle)$, *unique up to isomorphism, is a Hilbert space, called tensor product of Hilbert spaces.*

Remark 27. Note that, in general, the metric completion is required, as the algebraic tensor product by itself may not be complete with respect to the norm induced by the inner product defined in the proposition. However, with a slight abuse of notation we still denote $\mathcal{H}_1 \otimes \mathcal{H}_2$ the tensor product of the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$.

In the case where $\mathcal{H}_1$ and $\mathcal{H}_2$ are separable Hilbert spaces, we can give an explicit construction of their tensor product. Indeed, let $\{u_i : i \in \mathbb{N}_1\}$ and $\{v_i : i \in \mathbb{N}\}$ be orthonormal systems for $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively. Then it holds that

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \overline{\operatorname{span}\{u_i \otimes v_j : i, j \in \mathbb{N}\}},$$

which means that, in this scenario, the tensor product can be naturally identified with the closure of the space of finite linear combinations of elements $u \otimes v$, where $u \in \mathcal{H}_1$ and $v \in \mathcal{H}_2$, which are called *simple tensors* or *pure tensors*. Moreover, the set of simple tensors with factors taken from the respective complete orthonormal systems forms a complete orthonormal system for the tensor product.

Remark 28. With a simple inductive process one can conclude that the N-fold tensor product of Hilbert spaces is a Hilbert space, and the closure of the set of finite linear combinations of simple tensors still coincides with that tensor product.

Remark 29. A useful note for the type of computations we want to tackle in this work is that the space $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^m)$ can be naturally identified with the space $L^2(\mathbb{R}^{n+m})$, and given $f \in L^2(\mathbb{R}^n)$ and $g \in L^2(\mathbb{R}^m)$ one has that

$$f \otimes g(x, y) = f(x)g(y).$$

Lastly, we may define operators on a tensor product starting from operators on the factors. Indeed, let $A : D(A) \subseteq \mathcal{H}_1 \to \mathcal{H}_1$ and $B : D(B) \subseteq \mathcal{H}_2 \to \mathcal{H}_2$ be operators on the respective Hilbert space, then we can define their tensor product as

$$(A \otimes B)(\phi \otimes \psi) \coloneqq (A\phi) \otimes (B\psi)$$

on the set of pure pure tensors, i.e.,

$$D(A \otimes B) = \{\phi \otimes \psi : \phi \in D(A),\ \psi \in D(B)\}.$$

Appendix B

Integration of operator-valued functions

We want to define integration for functions of the type $f : X \to \mathcal{B}$, where (X, Σ, μ) is a measure space and $\mathcal{B}$ is a Banach space (while differentiation is automatic if the measure space is $\mathbb{R}^n$ via the usual definition as limit of the incremental ratio). First we define *simple functions*: let $\{E_1, \ldots, E_n\}$ be elements of the σ-algebra Σ and $\{b_1, \ldots, b_n\}$ elements of $\mathcal{B}$, then the function

$$s(x) \coloneqq \sum_{i=1}^{n} \chi_{E_i} b_i$$

is called a simple function and its integral is defined as

$$\int_X s \, \mathrm{d}\mu \coloneqq \sum_{i=1}^{n} \mu(E_i) b_i.$$

Definition B.0.1 (Bochner integral)**.** A measurable function $f : X \to \mathcal{B}$ is called *Bochner integrable* if there exists a sequence of simple functions $\{s_n : n\mathbb{N}\}$ such that

$$\lim_{n\to\infty} \int_X \|f - s_n\|_{\mathcal{B}} \, \mathrm{d}\mu = 0,$$

i.e., if its point-wise norm is Lebesgue integrable. In this case, the *integral of* f is defined as

$$\int_X f \, \mathrm{d}\mu \coloneqq \lim_{n\to\infty} \int s_n \, \mathrm{d}\mu.$$

It can be shown that $\{\int_X s_n \mathrm{d}\mu : n \in \mathbb{N}\}$ is a Cauchy sequence, hence the integral is well-defined.

Remark 30. The integration rules for "nice" functions like powers and exponentials are identical to the case of scalar functions, just like for differentiation, but we will not prove this.

Proposition B.0.2. *Bochner integration satisfies the following properties:*

1. *a function $f : X \to \mathcal{B}$ is Bochner integrable if, and only if, $\int_X \|f\|_{\mathcal{B}} \, \mathrm{d}\mu < \infty$;*
2. *if $T : \mathcal{B} \to \mathcal{B}'$ is a continuous linear operator between Banach spaces and $f : X \to \mathcal{B}$ is Bochner integrable, then Tf is also Bochner integrable and it holds that*

$$T \int_E f \, \mathrm{d}\mu = \int_X Tf \, \mathrm{d}\mu$$

for all $E \in \Sigma$;

3. *(Dominated convergence theorem) if $\{f_n\}$ is a sequence of Bochner integrable functions that converges almost everywhere to a limit function f, and if there exists an integrable function g such that $\|f_n(x)\|_{\mathcal{B}} \leq g(x)$ almost everywhere on X, then it holds that*

$$\int_E \|f - f_n\|_{\mathcal{B}} \,\mathrm{d}\mu \to 0, \quad \textit{and} \quad \int_E f_n \,\mathrm{d}\mu \to \int_E f \,\mathrm{d}\mu$$

for all $E \in \Sigma$;

4. *(Fundamental theorem of calculus) if $f : X \to \mathcal{B}$ is Bochner integrable over $[a, b] \subseteq \mathbb{R}$ and $F : X \to \mathcal{B}$ is an antiderivative for f, then*

$$\int_{[a,b]} f \,\mathrm{d}\mu = F(b) - F(a).$$

Remark 31. In particular, since functionals in the dual of a Hilbert space are also continuous operators between Banach spaces, given $f : X \to \mathcal{B}(\mathcal{H})$ a Bochner integrable function, it holds that

$$\langle \phi, \left(\int_X f \,\mathrm{d}\mu \right) \psi \rangle_{\mathcal{H}} = \int_X \langle \phi, f\psi \rangle_{\mathcal{H}} \mathrm{d}\mu.$$

Remark 32. An important property, which we make use of in our work, is that the set of trace-class operators is a Banach space with the trace norm, and the trace is a continuous operator from this space to the base field. This means that, because of the above proposition, whenever we take the trace of an integral of an operator-valued function, we can take the trace inside the integral.

With this notion of integration (and with the usual notion of differentiation) we can give an important theorem on ordinary differential equations that we used in our work. The proof of this theorem is completely analogous to the case where the Banach space is just $\mathbb{R}^n$, with the only difference being that the point-wise norm of a function is given by the norm on $\mathcal{B}$.

Theorem B.0.3 (Existence and uniqueness of a solution to an ODE)**.** *Let $f : [a, b] \times \mathcal{B} \to \mathcal{B}$ be a Lipschitz continuous function on $[a, b]$ uniformly on $\mathcal{B}$, and let it define the following Cauchy problem for $(x_0, y_0) \in [a, b] \times \mathcal{B}$:*

$$\begin{cases} y'(x) = f(x, y(x)) \\ y(x_0) = y_0 \end{cases}$$

where $y : [a, b] \to \mathcal{B}$. Then there exists a unique solution for the Cauchy problem over $[a, b]$.

Appendix C

Matsubara Frequencies

We reference the discussion on finite temperature perturbation theory from [5, pp. 319–324].

Following the standard notation, we call $\rho = Z^{-1}e^{-\beta H}$ the Gibbs state at temperature β, where $Z = \mathrm{Tr}(e^{-\beta(H-\mu\mathcal{N})}) = \mathrm{Tr}(e^{-\beta K})$ is the grand canonical partition function. Recall that the temperature evolution (i.e., imaginary time evolution) of an operator A in Heisenberg's picture is given by

$$A(\tau) = e^{\tau K} A e^{-\tau K}.$$

Then, given $A(\tau)$ a polynomial of creation/annihilation operators at temperature τ and B a polynomial of creation/annihilation operators at temperature β, we ask how can we write the correlation function

$$G_{AB}(\tau) = \langle \mathcal{T}[A(\tau)B] \rangle = \mathrm{Tr}(\rho \mathcal{T}[A(\tau)B]),$$

also called the *temperature propagator*. We have the following remarkable identity:

$$G_{AB}(\tau) = (-1)^f G_{AB}(\tau - \beta\mathrm{sign}(\tau)),$$

where f is the number of fermionic interchanges that occur when A and B are interchanged. This identity states that $G_{AB}(\tau)$ is actually periodic in τ (with period β) if f is even, and antiperiodic if f is odd. Indeed, if we consider – as an example – the case where $\beta > \tau > 0$ (so that $\tau - \beta > 0$), we can make use of the cyclic property of the trace and the form of the Gibbs state to get that

$$\begin{aligned} G_{AB}(\tau - \beta) &= \mathrm{Tr}(\rho \mathcal{T}[A(\tau - \beta)B]) \\ &= (-1)^f Z^{-1} \mathrm{Tr}(e^{-\beta K} B e^{K(\tau-\beta)} A e^{-K(\tau-\beta)}) \\ &= (-1)^f Z^{-1} \mathrm{Tr}(e^{K\tau} A e^{-K\tau} e^{\beta K} e^{-\beta K} B e^{-\beta K}) \\ &= (-1)^f Z^{-1} \mathrm{Tr}(A(\tau) B e^{-\beta K}) \\ &= (-1)^f \mathrm{Tr}(\rho \mathcal{T}[A(\tau)B]). \end{aligned}$$

Similar steps prove the case where $\tau < 0$. Because $G_{AB}(\tau)$ is periodic, we may want to write it as a Fourier series:

$$G_{AB}(\tau) = \frac{1}{\beta} \sum_{k \in \mathbb{Z}} \tilde{G}_{AB}(i\omega_k) e^{-i\omega_k \tau}.$$

Here, ω_k are called *Matsubara frequencies*, and they are *even* integer multiples of $1/\beta$ if f is even, and *odd* multiples if f is odd:

$$\omega_k = \begin{cases} \frac{2k}{\beta} & \text{if } f \text{ is even} \\ \frac{2k+1}{\beta} & \text{if } f \text{ is odd} \end{cases}.$$

In formal perturbation theory with Grassmann integrals, we want the system to be composed of a finite number of modes. Then, to make the number of temperature modes finite, we can simply introduce a cutoff Ω on Matsubara frequencies

$$\mathcal{D}_\Omega = \mathbb{Z} \cap [-\Omega, \Omega],$$

and recover the original physical theory by taking the limit $\Omega \to \infty$.

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